## ABSTRACT

**PLARIZA, KEYVEN B.** Davao del Sur State College, College of Information and Digital Sciences, Matti, Digos City. October 2025. **EHIREMO: WEB-BASED RECRUITEMENT AND SELECTION MANAGEMENT FOR THE PROVINCIAL HUMAN RESOURCE MANAGEMENT OFFICE.** Undergraduate Thesis.

Adviser: **RAZEL ANN G. AGUILAR, MIT**

The primary objective of the project is the development of Web-Based Recruitment and Selection Management System designed to automate the process of Provincial Human Resource Management Office (PHRMO). The system seeks to mitigate existing inefficiency and limitations of the traditional manual recruitment process by introducing a digital platform equipped with key features such as online job posting, applicant registration, automated applicant ranking, report generation, and push/email notifications. These features collectively promote transparency, efficiency, and accuracy throughout the recruitment process. The system underwent evaluation using the ISO/IEC 2510 software quality framework, which assesses various software quality characteristics including functional suitability, compatibility, performance efficiency, usability, security, reliability, maintainability, and portability. Results revealed high levels of

user satisfaction, particularly in functional suitability (4.46), compatibility (4.46), compatibility (4.46), and usability (4.37). These findings confirm that the system meets user requirements and provides a satisfactory recruitment experience, with only minor recommendations for improving document management and performance sustainability. Overall, the project successfully digitized the PHRMO's recruitment process, demonstrating how information systems can enhance transparency and data-driven decision-making in public service. Furthermore, it contributes to Sustainable Development (SDG) 9: Industry, Innovation and Infrastructure, by fostering technological innovation and strengthening the digital infrastructure of public institutions toward responsive and efficient governance.

*Keywords: Recruitment System, Web-Based Application, ISO/IEC 25010,*

*SDG: 9 Automation, Human Resource Management,*

# eHiremo: A WEB-BASED RECRUITMENT AND SELECTION MANAGEMENT SYSTEM FOR THE PROVINCIAL HUMAN RESOURCE MANAGEMENT OFFICE

**PLARIZA, KEYVEN B.**

**CAPSTONE PROJECT SUBMITTED TO THE FACULTY OF THE COLLEGE OF INFORMATION AND DIGITAL SCIENCES, DAVAO DEL SUR STATE COLLEGE, MATTI, DIGOS CITY IN PARTIAL FULFILLMENT OF THE REQUIRMENTS FOR THE DEGREE OF**

**BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**

**JANUARY 2026**

## APPROVAL SHEET

This Capstone Project entitled "**eHiremo: WEB-BASED RECRUITMENT AND SELECTION MANAGEMENT SYSTEM OF THE PROVINCIAL HUMAN RESOURCE MANAGEMENT OFFICE"** prepared and submitted by **KEYVEN B. PLARIZA** in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, is hereby accepted.

**RHEA MAE L. PERITO, MSIS**
Member

_____
Date Signed

**EDUARDO F. AQUINO, MSME**
Member

_____
Date Signed

**RAZEL ANN G. AGUILAR, MIT**
Adviser

_____
Date Signed

**DOMINGO V. ORIGINES JR., IT.D.**
Chairperson

_____
Date Signed

Accepted and approved in partial fulfillment of the requirement for the degree of BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY (BSIT).

**DOMINGO V. ORIGINES JR., IT.D.**
Chairperson, Computing Department

_____
Date Signed

**FELOMINO P. ALBA ITD**
Dean, College of Information and Digital Sciences

_____
Date Signed

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# CHAPTER I

# THE PROBLEM AND ITS BACKGROUND

## Introduction

Human Resource Management (HRM) is essential to organizational effectiveness, as recruitment and selection form a core process in developing a skilled and high-performing workforce across institutions. Over time, the core principles of HRM have evolved, emphasizing the critical role of recruitment and selection in enhancing organizational effectiveness (Juicharoen et al., 2023).

In line with this development, Cappelli (2021) emphasized the importance of utilizing technology to optimize the hiring process, from job posting to candidate selection,by automating tasks that were traditionally performed manually. Similarly,Thawrani (2021) highlighted that online recruitment platforms enhance an organization's visibility and overall recruitment effectiveness. Ventura & Bringula (2021) also emphasized how e-recruitment bridges the gap between applicants and employers, making the hiring process more accessible and transparent.

Further advancements have shown that automation contributes significantly to improving HR efficiency. Sharma et al. (2022) found that automation in recruitment, such as scheduling interviews and examinations, improves efficiency and optimizes resource allocation. Vincentia (2023) further explained that with technological advancement, recruitment and selection in Human Resource Management have evolved into a more efficient and systematic process. The adoption of computerized systems reduces manual workload and allows HR personnel to streamline the participant selection process. Moreover, ALRawashdeh et al. (2024) found that electronic Human Resource Management (E-HRM) practices, including e-recruitment and e-training, enhance organizational performance by strengthening decision-making and communication processes.

Despite these technological advancements, the Provincial Human Resource Management Office (PHRMO) continues to rely on a manual recruitment and selection process, which poses several significant challenges. The preparation of essential documents, such as comparative assessment reports and lists of qualified applicants for examination remains time-consuming and tedious due to the lack of automated tools. This reliance on paper-based records leads to an excessive volume of documents that

are difficult to organize, store, and retrieve, thereby hindering workflow efficiency. In addition, applicant scores from examinations and interviews are calculated manually, making the process prone to human error and causing delays in producing accurate rankings. Communication with applicants is also hindered, as updates on application status, examination results, and interview schedules are not provided in real time, causing uncertainty and dissatisfaction among candidates. Interview scoring, which remains dependent on paper-based evaluation sheets, requires manual consolidation from multiple panel members, further prolonging the process and compromising consistency. The posting of job vacancies likewise presents challenges, as it is still done manually, limiting accessibility and visibility for potential applicants. As the number of applicants increases, tracking submitted applications becomes progressively more challenging, complicating the office's ability to monitor progress and maintain transparency. Taken together, these shortcomings underscore the inefficiency and lack of transparency in the current system, highlighting the urgent need for a streamlined, automated, and reliable approach to recruitment and selection.

Recognizing these gaps, the researcher developed a Web-Based Recruitment and Selection system for PHRMO. The system automates and digitalizes key processes such as the job postings, application submission, scheduling, application processing, and report generation to enhance efficiency. Through this system, the recruitment and selection process will shift from manual, paper-based methods to a digital, data-driven workflow that minimizes human error and improves turnaround time. Furthermore, this initiative aligns with Sustainable Development Goal (SDG) 9: Industry, Innovation, and Infrastructure, as it promotes technological innovations and strengthens the digital infrastructure of public institutions to support transparent and responsive governance.

## Objectives of the Study

The study's overall objective is to create a web-based recruitment and selection management system that will effectively enhance the PHRMO's hiring and selection procedure.

The study specifically seeks to accomplish the following goals:

1. Develop a page for super administrator that has the following functionalities:

1.1 View the ranking of applicants for a specific position with their ratings.

1.2 Select an Applicant for the appointment of the position;

1.3 View activity logs such as:

1.3.1 Posting, updating, or closing of vacant positions;

1.3.2 Updating applicant application status;

1.3.3 Selection of applicant for appointment;

2. Develop a page for administrator that has the following functionalities:

2.1 Post vacant position;

2.2 Result on Assessment and Evaluation of Applications;

2.3 Send Push notification and email for the following:

2.3.1 Application Status;

2.3.2 Schedule for the examination, Panel interview and Appointment;

2.4 Tabulation of the applicant's exam score and interview score;

2.5 Generate files and reports such as:

2.5.1 Comparative Assessment Report of Applicants;

2.5.2 HRMPSB list of examinees for examination;

2.5.3 Accession report;

3. Develop a page for applicant account that has the following functionalities:

3.1 View the published vacant position in the job portal;

3.2 Registration page for new account;

3.3 Submission portal for application documents such as:

3.3.1 Personal Data Sheets entry;

3.3.2 Scanned copy of School Credentials;

3.3.3 Scanned copy of certificate of Eligibility (CS, PRC License, LTO License);

3.3.4 Scanned Work Experience Sheet;

3.3.5 Scanned Certificate of Training; and

3.3.6 Scanned Performance Rating.

3.4 View and Track application status;

3.5 Receive Push Notification and Email for the following:

3.5.1 Application status;

3.5.2 Schedules of Exam, Interview and Appointment including date, time and venue;

3.5.3 Result of Examination and Selection.

4. Examine and evaluate the system using the following criteria:
    4.1 Functional Requirements;
    4.2 Performance efficiency;
    4.3 Compatibility;
    4.4 Usability;
    4.5 Reliability;
    4.6 Security;
    4.7 Maintainability; and
    4.8 Portability.

## Scope and Limitations

The study focuses on the development of a Web-Based Recruitment and Selection Management System designed to optimize the key stages of the recruitment and selection process of the Davao del Sur PHRMO. Specifically, the system covers the following major functions: job posting, submission and assessment of applications, and selection of qualified applicants for appointment. The system enables applicants to view and apply for published job vacancies, register an account, and submit the required application documents. Additionally, applicants can track the

status of their applications and receive timely updates regarding assessment results, interview schedules, and final outcomes.

For the administrator, the system facilitates the posting of job vacancies, evaluation of applications based on qualifications, and the generation of notifications related to application status, examinations, interviews, and appointments. It also supports the computation of exam and interview scores, the generation of comparative assessment reports. The super administrator role includes enhanced privileges such as viewing of top candidates, reviewing applicant ratings, confirming appointments, and tracking system activities related to job postings, applicant registrations and evaluation logs.

Despite its contributions, the study has several limitations. The system is currently used exclusively in the PHRMO office, and its automation functionality is limited in certain areas of the process. The accuracy of the system's outputs largely depends on the correctness and completeness of data entered by both users and administrators, and human intervention is still required to manage and execute certain processes. The study was conducted at the PHRMO, Matti, Digos City, Davao del Sur, and the system was developed for approximately five months.

## Significance of the Study

The primary objective of this study is to generate meaningful insights and make a necessary contribution to the current body of knowledge. The expected significance of the study lies in its potential to offer data-driven findings that can provide valuable benefits to a wide range of organizations and users.

**Civil Service Commission (CSC).** The outcome will help government agency responsible for overseeing and ensuring the transparency and efficiency of the recruitment and selection process, specifically in relation to the implementation of the web-based system at the Provincial Resource Management Office of Davao del Sur.

**Provincial Governor's Office (PGO). Provincial Governor's Office (PGO):** The outcome will help the governing body of Davao del Sur province. Within this study, the PGO is a key beneficiary of the web-based recruitment and selection system, as it aims to improve the efficiency and systematic approach to hiring for public service positions within the province.

**Provincial Human Resource Management Office (PHRMO).**

The outcome will help the department within the provincial government is responsible for managing human resources. The study anticipates that the PHRMO will benefit from the web-based recruitment and selection system, which aims to automate the hiring process and reduce reliance on manual paperwork.

**Applicants.** The outcome will help individuals seeking employment within the Davao del Sur Provincial Government. The study aims to provide applicants with a user-friendly, web-based platform for job applications, enabling remote access and application status tracking.

**Future researchers.** The outcome will help individuals or groups conducting subsequent studies or system development related to web-based recruitment and selection.

## Definition of Terms

The includes terminologies that are defined and used within the specific context of the study.

**Administrator.** It refers to a user account with the highest level of system access, granting full control for system configuration.

**Applicants.** It refers to the people who apply for a job opening. They are often considered potential employees or candidates for the position.

**Appointment.** It refers to a formal act of selecting and confirming an individual's employment for a specific position.

**Digitalize**. It refers to converting the operation of the process with the use of computers and the internet.

**Efficient.** It means reaching the highest level of productivity while spending the least amount of money or effort.

**Hiring.** It describes the procedure for evaluating applications, choosing the best applicants for interviews, testing applicants, selecting applicants to make a decision, and carrying out a variety of pre-employment examinations.

**Manual**. It refers to the processes involving one or more human's performing tasks, such as data entry and/or verification.

**Provincial Governor's Office (PGO).** It refers to the dynamic hub of administrative excellence and strategic governance at the heart of our province.

**Provincial Human Resource Management Office (PHRMO).** It refers to the exact HR office where the system will be deployed.

**Recruitment.** It refers to the entire process of finding, attractive, screening, interviewing, and hiring qualified candidates to fill open positions within an organization.

**Selection.** It refers to the process of choosing the most suitable candidate for a job position from a pool of applicants by evaluating them skills, experience, and qualifications through various methods like interviews, tests, and background checks.

**Streamline.** It refers to the process of enhancing a system's efficiency by simplifying its structure or operations.

**Super Administrator.** It defines a user who has total access to every item, folder, role template, and group within the system.

**Web-based.** It refers to the application or system that can be accessed online via a web browser.

# CHAPTER II

# REVIEW OF RELATED LITERATURE AND STUDIES

## Related Literature

This chapter discuss the relevant literature and related study that is relevant to this study, including the existing study and systems. The findings of previous studies will be examined and compared. The discussion will highlight the strengths and limitations of the previous works and how they relate to this study.

### Literature Map of the Study

A map of literature is a is a visual representation that outlines the various studies, theories, and concepts within a specific field of study, illustrating their interrelationships and hierarchical structure. It serves to highlight the connections between existing systems and relevant supporting research, thereby justifying the significance and relevance of the current study. By employing a literature map, the researcher is able to identify gaps in existing knowledge, pinpoint areas that require further exploration, and strategically enhance the study's focus and scope. This

tool enables a comprehensive understanding of the research landscape and informs the study's direction and methodology.

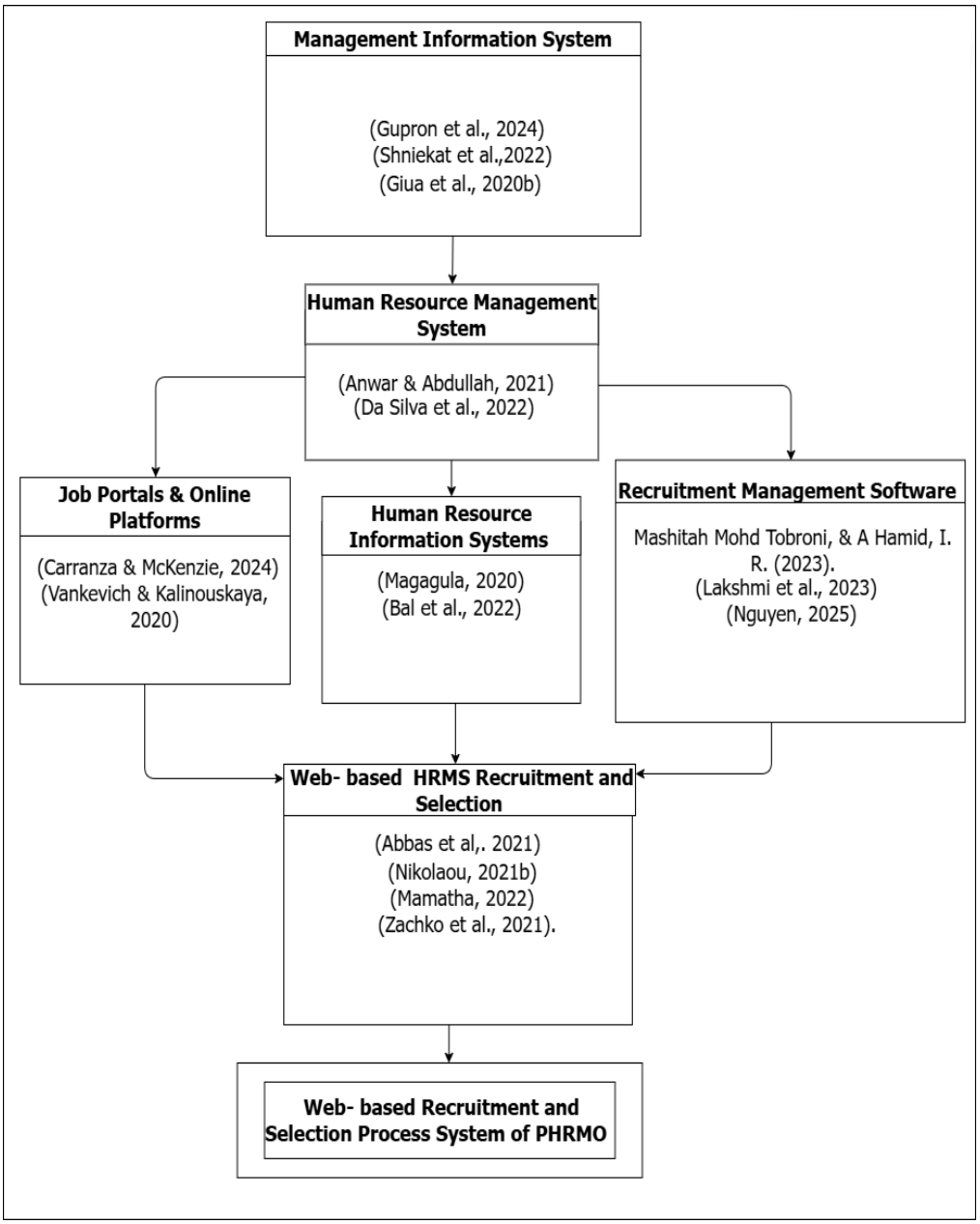



**Figure 1: Literature Map of the Study**

Figure 1 represents the relationship of essential components of the study about the eHiremo, it will guide the researcher about the focus of the studies and examine the significant relationships with this study.

**Management Information System (MIS)**

An organization's operations can be made more effective and efficient through the strategic use of Management Information Systems (MIS). More organized data and information management is made possible by MIS adoption, which facilitates quicker and more precise decision-making. In order to get the greatest employment outcomes, this study aims to investigate the concept and benefits of MIS in increasing productivity and effectiveness (Gupron et al., 2024).

According to Giua et al., (2020) highlight the technical issues such a lack of data standards, a lack of device compatibility, and High prices may prevent a certain technology from reaching its full potential, but another important one factor that appears to influence the adoption of MIS at the farm level and the desire to use it is farmer’s and farm worker’s personal opinions of innovative features like the relative benefit, usefulness, complexity, and potential customization of the technology.

## Human Resources Management System (HRMS)

The Human Resources Management System (HRMS) plays a vital role in different institutions to streamline and automate HR processes within an organization. According to the study of Da Silva et al., (2022) says that Human resources and the workplace are changing as a result of digitalization in a variety of industries and service sectors. Some Human Resource Management System have limitation in its capability of managing the HR process for instance in this study that focuses on Recruitment and Hiring stage. The process is very important because all processes of HR will always start in Recruitment of employees for the organization. One could argue that hiring is the main duty of the human resources division. Shortlisting, screening, and sourcing are all steps in the process (Shukur et al., 2021).

The study by Da Silva et al. (2022) and Shukur et al. (2021) highlight the broader impact of digitalization on HR functions emphasizing how technology is transforming workplaces and streamlining general HR tasks this study narrows its focus to one of the most critical and foundational phases of HR recruitment and selection. Unlike existing systems that often offer only limited capabilities in this area, the proposed HRMS in this study is specifically designed to enhance and automate every

step of the hiring process from job posting and application tracking to assessment, evaluation, and final appointment. By zeroing in on the recruitment stage, the study addresses a vital need in many institutions where manual hiring processes are still common and prone to delays and inconsistencies. In contrast to the general frameworks presented in earlier research, this system provides a more targeted and workflow-oriented solution that reflects the real operational demands of HR offices, particularly in government or public service sectors.

### Job Portals and Online Platforms

Nowadays, most of the transactions and processes are through online platforms due to advancement in technology and the increasing reliance in digital solutions. In the study of Vankevich and Kalinouskaya (2020), emphasized that the study of job openings and resumes gathered by online job portal scraping is therefore a more trustworthy source of information. It enables you to examine job openings and resumes in relation to the competencies mentioned and create a prediction about their dynamics. According to the study of Carranza and McKenzie (2024), explained that major issue in government in developing countries that they face pressure to intervene actively to help jobseekers find employment due

to lack of skills for employment. It also highlighted the two of the most popular policies are job training, and experience, it is predicated on the notion that many job seekers lack the skills that employers require, and job search assistance, which is predicated on the idea that even if workers possess the skills that employers require, search and matching frictions may make it difficult for them to be hired in positions that require them by the employees.

The studies by Vankevich and Kalinouskaya (2020) and Carranza and McKenzie (2024) highlight the growing role of online platforms and job portals in connecting job seekers with potential employers. While Vankevich and Kalinouskaya focus on the reliability of data gathered from online job portals for analyzing competencies and predicting employment trends, and Carranza and McKenzie explore the challenges in employment matching and skill gaps especially in developing countries this study takes a more hands-on approach by building an actual web-based platform tailored for the public sector's recruitment process. Unlike the general observations made in previous research, the eHiremo system goes beyond data analysis or policy implications and provides a practical tool for both applicants and HR personnel. It directly addresses the issues of accessibility, skills matching, and applicant tracking through its features

such as automated qualification checks, document uploads, and real-time status updates. In this way, the study not only aligns with but also builds upon existing literature by offering a concrete solution that integrates recruitment, selection, and communication in a centralized online platform.

### Human Resource Information Systems (HRIS)

A software program called HRIS aids companies in managing and automating a range of HR tasks. In the study of Magagula (2020) elaborate that the demand for companies to accomplish their goals has always been increasing and practicality in the workplace. Long-term research indicates that conventional Human Resource Management practices have been shifted to strategic human resource management with a significant information technology component. The human resource management offices nowadays should have HRIS to have an effective HR management. According to Bal et al., (2022) stated that human resource information systems facilitate tasks like finding new hires, keeping thorough records on current workers, and developing initiatives to enhance workers' abilities.

While studies like those of Magagula (2020) and Bal et al. (2022) emphasize the growing importance of Human Resource Information Systems (HRIS) in supporting strategic HR management, this study

focuses on a more specialized area within HRIS recruitment and selection. Magagula points out the shift from traditional HR practices to more technology driven strategies, and Bal et al. highlight how HRIS supports tasks such as recruitment, employee record management, and skill development. Building on these insights, this study presents eHiremo, a system designed not just to manage information but to actively support and automate the entire hiring process. Unlike general HRIS solutions that cover a wide range of HR functions, eHiremo zooms in on the initial and arguably most critical stage of HR—bringing the right people into the organization. Through features such as automated screening, centralized document submission, and real-time applicant tracking, this system demonstrates how HRIS principles can be applied in a focused and practical way to improve efficiency, accuracy, and transparency in public sector hiring.

### Recruitment Management System (HMS)

HMS is a tool designed to streamline and automate the hiring process for organizations. It helps HR manage job posting, track applicants, conduct interviews, and onboard new employees efficiently. Recruitment software helps organizations improve the efficiency of the

hiring process from sourcing all the way through to signing an employment contract.

It is highlighted by Lakshmi et., (2023) that the social recruiting, automated computerized job postings, candidate identification, identifying possible problems, assigning timeslots according to the supervisor's availability for job interviews, and automating the initial interview process are all features of the Human Resource Management System (HRMS).

### Web- based HRMS Recruitment and Selection

The project aims to come up with a Web-based Recruitment and Selection System that manage the hiring stage process of HR, and it also helps the applicants to lessen the hustle and difficulty of application process and optimize the workflow. The study of Abbas et al., (2021) emphasized that recruitment and selection techniques play a crucial role in improving organizational performance, with businesses increasingly adopting electronic and social media platforms for staffing, especially due to the impact of the COVID-19 pandemic. Furthermore, Mamatha (2022), also stated that E-Recruitment and E-Selection processes work together to enhance the efficiency of E-Human Resource Management (E-HRM) in IT

MNCs in Hyderabad, with advancements in AI further simplifying recruitment since 2020.

The studies of Abbas et al. (2021) and Mamatha (2022) highlight how recruitment and selection practices have evolved with the integration of web-based platforms and digital tools, especially accelerated by the COVID-19 pandemic. Abbas emphasizes the importance of digital recruitment in enhancing organizational performance, while Mamatha notes the role of E-Recruitment and E-Selection in improving HR operations, particularly in IT sectors. In contrast, this study focuses on the development of eHiremo, a web-based recruitment and selection system specifically designed for the structured and often rigid hiring workflows in public institutions. While earlier studies focus on general trends and corporate settings, eHiremo adapts these digital innovations to fit government hiring standards streamlining the application process for applicants while providing HR personnel with tools to evaluate, notify, and select candidates more efficiently. This project does not only follow the trend of web-based HRMS but grounds it in the practical realities and requirements of public sector recruitment, aiming to reduce applicant burden and improve overall transparency and accountability in hiring.

# Related Studies

A review of relevant research involves a comprehensive and critical evaluation of existing frameworks and literature pertinent to the subject matter under investigation. The primary objective is to offer an in-depth synthesis of prior scholarly efforts, identifying key findings and methodologies. Furthermore, the review aims to highlight any gaps or deficiencies in the current body of knowledge, thereby pinpointing areas that require further refinement or exploration to advance the field.

## List of System Related Studies

A related systems entails more than simply describing existing works; it involves a careful and critical assessment of their features, methodologies, and outcomes. This process highlights the strengths that can serve as benchmarks for good practices, while also uncovering limitations that may hinder effectiveness or long-term applicability. By identifying these research gaps, scholars and practitioners gain a clearer understanding of where current systems fall short and where innovation is most needed.

Table 1 presents findings from various studies that examine relevant literature, providing a comparative perspective of existing systems. These findings do not merely catalog functionalities but also reveal recurring challenges such as scalability, user accessibility, integration with emerging technologies, and sustainability. At the same time, the strengths noted—such as efficiency, accuracy, or adaptability—offer valuable insights into what has been achieved and what can be replicated or enhanced.

**Table 1. List of Systems Associated with the Study**

| Name of System | Author Year/ Cited | Description | Strength | Weaknesses | Application |
|---|---|---|---|---|---|
| Web-Based Recruitment Selection Application Development with Analytical Hierarchy Process (AHP) Method | Maruanaya et al. (2021) | It's a support mechanism for recruiting Integrated ICT Laboratory assistants at Budi Luhur University, utilizing predetermined criteria. | Through integrated score calculation and printing, the method increases efficiency and expedites assistant selection. | The Analytical Hierarchy Process (AHP) application does not automate the ultimate choice or make decisions on its own; rather, it merely supports decision-makers. | An online tool for hiring and choosing candidates using the analytical hierarchy process method |
| Applicant Tracking System: A Powerful Recruiters' Tool | (Novaković & Dražet., 2024) | ATS components are typically included in a more comprehensive Human | ATS systems effectively generate and disseminate job | Potential disadvantages include AI biases, an excessive reliance on automation, | Applicant Tracking System (ATS) |

*Continuation from Table 1…*

| | | | | | |
|---|---|---|---|---|---|
| | | Resources Information System (HRIS). Recruiters can accomplish the following tasks utilizing the user-friendly interface of a traditional applicant tracking system (ATS). | advertise ments, collect resumes, and screen candidates . | and difficulties for applicants who are not familiar with ATS optimization are not thoroughly discussed. . | |
| Graduate Students and Companie s Web Based E-Recruitme nt System | (Malki & Atlam, 2021) | This approach is to assist both job searchers and employers in replacing the current ways of job searching, application, and hiring. | It improved the recruitmen t experience , overcame current challenges , and serve HR departme nts in many sectors. | For both companies and job seekers, limited search and filtering capabilities impede speedy candidate matching. Additionally, user data may be compromised by insufficient security procedures. | Web-based application |
| Recruiting Models and Issues: Linkedin.c om as a Case Study | (Alqahta ni et al., 2023) | Recruitment has transformed, with the internet permitting global communication with employees, generating new opportunities and challenges . | Once only used for hiring, LinkedIn is now a profession al communic ation tool with an emphasis on personal branding. | The efficacy of recruitment platforms depends on user participation; inactive users or out-of-date profiles hinder their ability to match job prospects. . | Linkedin application |

*Continuation from Table 1…*

| | | | | | |
|---|---|---|---|---|---|
| Secured Recruitment Management System with Data Retention Approach | (Tobroni Hamid 2023) | The SRMS will help recruiters to manage their recruiting operations securely and efficiently. | The solution will automate candidate screening, helping staff find acceptable interview prospects. | The system's security features, such as its absence of SQL injection, brute force, and XSS protection, are not fully addressed. | Secured Recruitment Management System (SRMS) |
| Present-Day Human Resource Management (HRM) Hiring Processes | | Applicant tracking systems (ATS) handle applications through video interviews for remote assessment and AI-driven resume screening to identify eligible applicants. . | The report offers a comprehensive analysis of contemporary HRM hiring procedures, fusing strategic HR approaches with technological developments. | Potential Bias Toward Technology: An overemphasis on technical solutions can ignore the value of traditional, human-centric recruitment approaches . | Applicant tracking system |
| Sustainable HRM Goals in Selected IT Companies | Tiwari (2023) | The paper examines sustainable HRM practices in a few chosen IT firms, highlighting the necessity | Focuses on long-term human resource development | The suggested practices are not supported by actual data. | Web application |

Over the past few years, the development of this platform has improved each human resource office's hiring and selection process. The

effectiveness and dependability of the system demonstrated the continuous improvement and progress of this technology in systematic operations and digitization, such as applicant tracking, support systems, and online transactions. The enhancement of the recruitment and selection process through the incorporation of the most recent method and technique is highlighted by the following research and systems.

In order to improve the recruitment support system, Maruanaya et al. (2021) designed a system called Web-Based Recruitment Selection Application Development with Analytical Hierarchy Process (AHP) Method. The integration optimizes the acceptance of assistants by combining the calculation process and the development of final scores inside a single system, following to set terms and standards, hence enhancing effectiveness and efficiency. However, the system's accuracy and performance are not entirely optimized in terms of its user-friendliness, and the system's methodology was insufficiently precise to manage the operations. The study by Novaković & Dražet (2024) highlighted the significant influence of Applicant Tracking Systems (ATS) on contemporary hiring procedures in order to improve the candidate experience by facilitating communication and transparency throughout the hiring process. Additionally, it reduces the amount of time spent monitoring the applicants'

application status. The report emphasizes the advantages of Applicant Tracking Systems (ATS) for recruiters, although it is devoid of comparative analysis, empirical data, and information regarding AI integration. It misses the candidate experience and implementation issues, leaving potential for more detailed future research. However, it doesn't go into great detail about potential problems including screening applicants, over-reliance on automation, and biases in AI.

Malki and Atlam's (2021) study highlights how important computer systems are for improving a number of industries, especially hiring. Traditionally, recruiters engaged substantial time, effort, and financial resources in locating great prospects. The study on the Graduate Students and Companies' Web-Based E-Recruitment System underlines its foundational purpose in linking graduates with employers. However, the system falls short in embracing new technologies, advanced functionality, and a user-centric design. Additionally, it lacks empirical evaluation and ignores important factors like candidate experience and system security, indicating areas that should be improved in the future. However, it has restricted search and filtering feature hinder speedy candidate matching for both employers and job seekers and weak security measures may jeopardize use data.

Additionally, Alqahtani et al. (2023) looked at the effects of recruitment platforms, using LinkedIn as a case study to show how these platforms assist in overcoming conventional recruitment obstacles. The study investigates how recruiting and recruitment procedures for human resource management are streamlined by digital platforms such as LinkedIn. The study does not offer a thorough examination of LinkedIn's sophisticated AI-driven features, nor does it investigate the candidate experience or assess the platform's efficacy across different industries and geographical areas, even if it emphasizes the platform's role in resolving previous recruitment challenges. However, the efficacy of the recruitment platform depends on user involvement; inactive users or out-of-date profiles decrease the platform's ability to connect candidates with open positions.

In the study of Tobroni & Hamid (2023), addressed the difficulties of inadequate data security and poor data management in traditional recruitment systems. Sensitive application data is vulnerable to data breaches and illegal access because many current platforms do not protect it. Furthermore, the absence of a structured data retention strategy typically leads to non-compliance with legal and organizational norms. To handle these difficulties, the researchers offer a safe recruiting

management system that integrates comprehensive security mechanisms, including encryption, user authentication, and access restriction. The system also has a data retention mechanism that specifies how long applicant data will be kept on file and guarantees that it will be securely disposed of after that time.

By improving data security and compliance, this strategy promotes a more dependable and effective hiring process for both companies and candidates. Nevertheless, the system's security features—such as its absence of SQL injection, XSS, and brute force protection—are not addressed.

Furthermore, Pendse et al. (2024) claimed that traditional Human Resource Management (HRM) hiring practices are unsuitable for tackling the problems given by today’s quickly changing employment market. Organizations' capacity to draw in and hire top people is hampered by problems including ineffective screening, biased decision-making, subpar applicant experiences, and bad employer branding. The authors suggest integrating contemporary technology, such as data analytics, AI-driven resume screening, and applicant tracking systems (ATS), to address these issues.

The study comes to the conclusion that a strategic, tech-enabled approach to HRM combined with human insight can greatly increase the efficacy, efficiency, and fairness of hiring procedures, ultimately assisting businesses in creating more cohesive teams and preserving their competitiveness in the changing labor market. However, there may be a bias in favor of technology, and an excessive focus on technical solutions may obscure the need of conventional human-centered hiring practices.

According to Sadhana Tiwari's (2023) study, "Sustainable HRM Goals in Selected IT Companies," a major problem is that traditional HRM practices in IT companies are frequently short-term focused and unrelated to more general sustainability goals, which leads to low employee engagement and inadequate long-term human capital development. The study promotes the use of sustainable HRM methods, which match human resource strategies with social, ethical, and environmental goals, in order to address this. This strategy places a strong emphasis on corporate responsibility, long-term employee development, and well-being. According to the study's findings, including sustainability into HR procedures not only improves an organization's reputation and resilience but also cultivates a staff that is more moral and prepared for the future.

Nevertheless, there is little empirical evidence to back up the suggested procedures.

Using the Analytical Hierarchy Process (AHP), Maruanaya et al. (2021) created a web-based recruitment selection application that enhanced decision-making by automating the computation and ranking of candidates. Nevertheless, their technique was not very user-friendly or precise; the application simply helps the decision-makers; it is not automated for the final choice.

The current study, eHiremo: Web-Based Recruitment and Selection Process System, on the other hand, solves a number of issues found in the related system under examination. eHiremo offers a comprehensive end-to-end solution, whereas earlier systems frequently lacked either sophisticated security, full feature integration, or customized application to government organizations. It has multi-role dashboards for applicants, HR administrators, and super administrators, secure document submissions, application tracking, and automatic notifications. It is also built with workflows unique to the government, guaranteeing accessibility, transparency, and compliance. By combining best practices from previous systems and addressing their gaps, eHiremo aims to improve the overall

recruitment process for both applicants and HR staff, making it more efficient, secure, and user-friendly.

## Concept of the Study

The methodical approach of the "eHiremo: Web-based Recruitment and Selection Process System" put in place by the Provincial Human Resource Management Office (PHRMO) at the Davao del Sur Provincial Capitol is outlined in the conceptual framework shown in the image. This framework exhibits the seamless integration of numerous components within the recruitment and selection process, enabled by the web-based system. The PHRMO job portal is the main point of entry for candidates. Applicants use the portal to look through open positions and send in their applications. The technology streamlines the hiring process by processing candidates' data through an organized interface after they submit their information. The application is then a data survey instrument, which is a tool for gathering data or information in a standardized and organized way from a selected group of participants. The instrument was altered to meet project requirements for the assessment procedure once virtual development was finished. Respondents were requested to rate the system's efficiency using a questionnaire derived from the International

Standard Organization's (ISO/IEC) International standard for software quality characteristics 25010. Functional suitability, performance efficiency, usability, compatibility, reliability, maintainability, security, and portability are some of the criteria that the questionnaire focuses on.

The eHiremo system integrates multiple functional modules to manage candidate data. These include database administration, which guarantees effective applicant sorting and evaluation, automated creation of application files, and digital submission of application requirements.

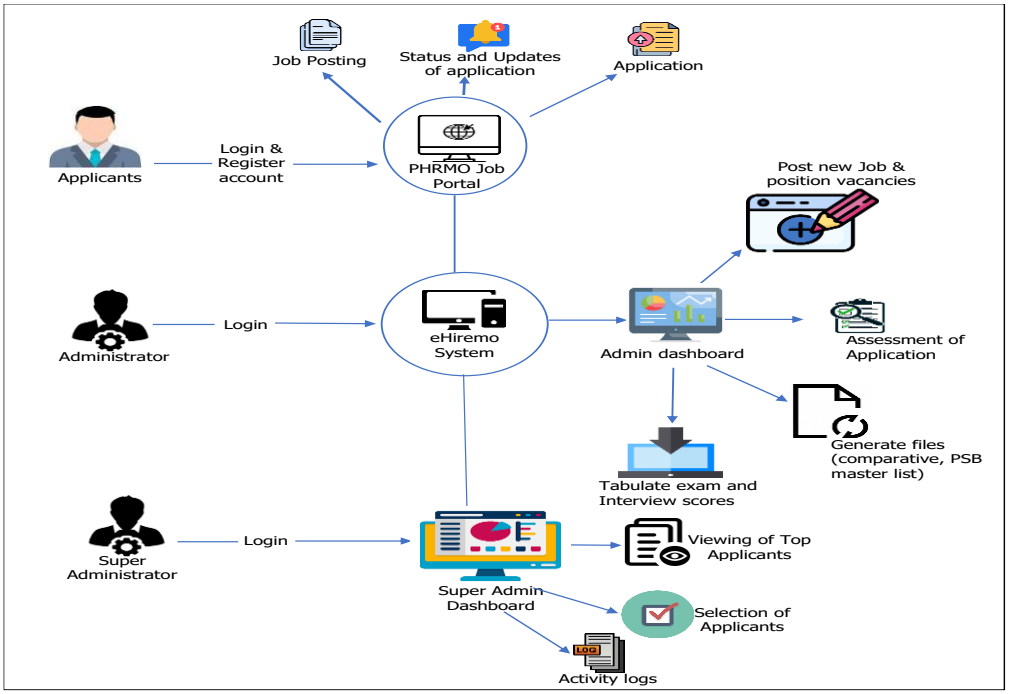



**Figure 2. Conceptual Framework**

The solution uses sophisticated data analytics to support decision-making, enabling HR staff to evaluate applicants impartially. The system is also built with an emphasis on improving data security and guaranteeing adherence to privacy laws, which raises the recruitment process's dependability even more.

From the applicant's first engagement with the PHRMO portal until the processing of their application within the eHiremo system, this conceptual model emphasizes the procedural flow of information. The framework promotes a more effective, transparent, and user-friendly hiring process that benefits both the company and the candidates by placing an emphasis on automation and data-driven strategies. It is in line with contemporary information technology trends that emphasize streamlining organizational procedures, improving applicant experiences, and increasing system engagement.

# CHAPTER III

## METHODOLOGY

This chapter provides a comprehensive overview of the capstone study, including the research locale, system design, data collection methods, analysis techniques, and diagrams used throughout the investigation, as well as the methodologies and procedures employed in conducting the study.

### Capstone Locale

The study was carried out at the Davao del Sur Provincial Capitol Provincial Human Resource Management Office, which is situated in Matti, Digos City. The study area's geographic coordinates were around 6.7662443466519955° N latitude and 125.30671243762794° E longitude.




. **Figure 3. Capstone Locale of the Study**

## Research Design of the Study

In order to satisfy certain needs and facilitate successful implementation, research design entails developing a comprehensive framework that describes the system's parts, interfaces, and data structures. Figure 4 illustrates the Rapid Application Development (RAD) paradigm, a flexible software development methodology that adapts to evolving requirements and user feedback throughout the development process. The researchers have chosen to employ the Rapid Application Development (RAD) technique for the medication reminder application due to its numerous advantages. The system may be continuously enhanced and changed in response to user requests through numerous feedback loops due to its iterative development technique.

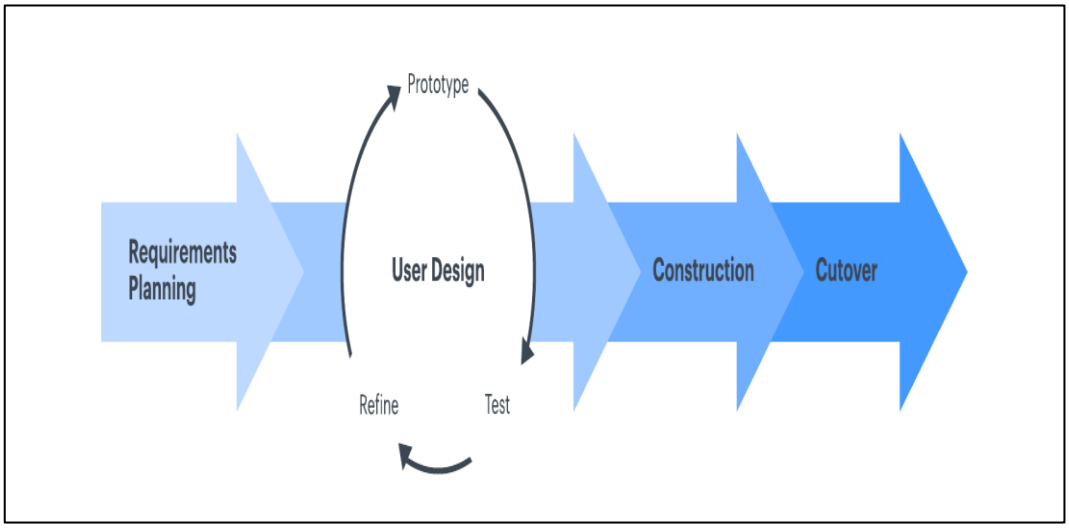



**Figure 4. Rapid Application Development of the System**

I selected Rapid Application Development (RAD) over other SDLCs because it prioritizes speed and flexibility while accelerating development through iterative prototyping and ongoing user feedback. RAD is well-suited for projects with stringent time restrictions, changing requirements, and the need for ongoing user input, enabling quick modifications and lowering the probability of delivering misaligned solutions. Its focus on quick delivery makes it a useful strategy for modern development demands, even though it might not be the best for more complicated or safety-critical systems.

Figure 4 shows how the eHiremo's development process follows the four-phase RAD paradigm. The first stage, "Requirements Planning," comprises gathering essential requirements and establishing the Provincial Human Resource Management Office's project scope. Determining what the HR department needs to recruit and select candidates is the main goal of this phase. The next stage, "User Design," focuses on creating wireframes and prototypes to demonstrate the functionality and interface of the system. In order to assure the effective organization and retrieval of applicant data, this phase also comprises developing the database design. During the construction phase, the designs are transformed into a fully operational web-based system. Last but not least, the Cutover phase

entails setting up the system, making it usable, and ensuring that it effectively supports the Recruitment and Selection process of the Provincial Human Resource Management Office. These stages guarantee that the system is effectively created and expands in response to user requirements, thereby enhancing recruitment and selection processes.

**Requirements Planning**

This crucial first stage describes the project's objectives, needs, and scope. Developers, stakeholders, and users must work closely together to gather pertinent data, evaluate present issues, and create a thorough plan to direct the project's advancement.

In order to obtain requirements directly from the HRMO staff, the researcher visited the Provincial Human Resource Management Office (PHRMO) several times throughout this phase. An initial interview was done to acquire a comprehensive understanding of the center's current operations, difficulties, and specific requirements for the proposed system. Subsequent follow-up visits were carried out to enhance user specifications, during which more discussions with staff members helped clarify important data requirements and system functionalities. It was achieved by conducting the following steps: (1) outlining specified

objectives associated with the planned application, and (2) finding suitable development processes and tools for executing the Web-based Recruitment and Selection Process System.

### System Used

The technological requirements required for the creation of a web-based recruitment and selection process system are covered in length in this section. The key components required to effectively create a system are referred to as system requirements. A deep comprehension of the relevant software and hardware guarantees that all resources are available and suitable for the system.

**Hardware Used.** The hardware listed in the table 2 is an important thing that keep the system work according to the user desired performance and optimize its capabilities. Therefore, if the system hardware does not meet the requirement will result unwanted poor system performance, crashes, vulnerability with the its security. The carefully selected and listed hardware utilized in the system's development is listed in Table 2 along with its specifications. This assisted the system to work smoothly and guarantee that the system could be constructed without failure.

**Table 2. Hardware Requirements for System Development**

| Hardware | Specifications |
|---|---|
| **Laptop**  | Intel(R) Core (TM) i5-6200U<br>CPU 2.30GHz  2.40 GHz<br>8gb RAM<br>256 ROM |

**Software Used.** Because these software tools are essential to the project's overall success, they were carefully selected by taking into account each tool's capabilities and compatibility.

**Table 3. Software Requirements for System Development**

| Software | Specification |
|---|---|
| Operating system | Windows 10 pro |
| Database Management | phpMyAdmin |
| Source Code Editor | Visual Studio Code 2022 |
| Programming Language | PHP 8.2.4 |
| Framework | Laravel 11 |

The software requirements needed to develop the system were listed in Table 3. The database, programming software, and operating system are all included in the software. The system needed to run Windows 10 or later in order to guarantee compatibility. Developers typically utilize MySQL as a database management system (DBMS) and Visual Studio Code as an integrated development environment (IDE) while producing programming software.

**Data Used**

The Provincial Human Resource Management Office (PHRMO) provided structured and structured data for this study, including the applicant's personal information, family background, educational background, civil service eligibility, work experience, volunteer work or civic involvement, training programs attended, other information, references, government IDs, PHRMO job details, and recruitment selection data. These data were acquired from existing printed and digital records, forms, and applicant entries, confirming their applicability to the real-world Recruitment and Selection activities of the Provincial Human Resource Office. To promote trustworthy analysis and aid in the creation of a significant and useful system, particular emphasis was paid to data quality, correctness, and confidentiality.

**Data tables.** The necessary data tables for the system are shown in the tables below. Within a DBMS, structured data is represented by data tables, which are arranged in grid-like rows and columns.

A thorough examination of the eHiremo database structure is provided within the phpMyAdmin environment, as seen in Figure 4. 25 unique, normalized tables that are essential for overseeing the Provincial Human Resource Office's (PHRMO) recruitment and selection process are included in the primary pane. These tables include Personal Data Sheet (PDS) data from the applicants and it was divided into different tables in order for the data to be manage in a clean way and the data base table is normalized with relationships and a foreign key. users table for the authentication and system's accounts with different roles type. examination_scores, interview_ratings table for the examination and interview stage, jobs table for the publication of job vacancies, notifications table for applicants' updates and activity_logs table to ensure the process happened in the system will be recorded.

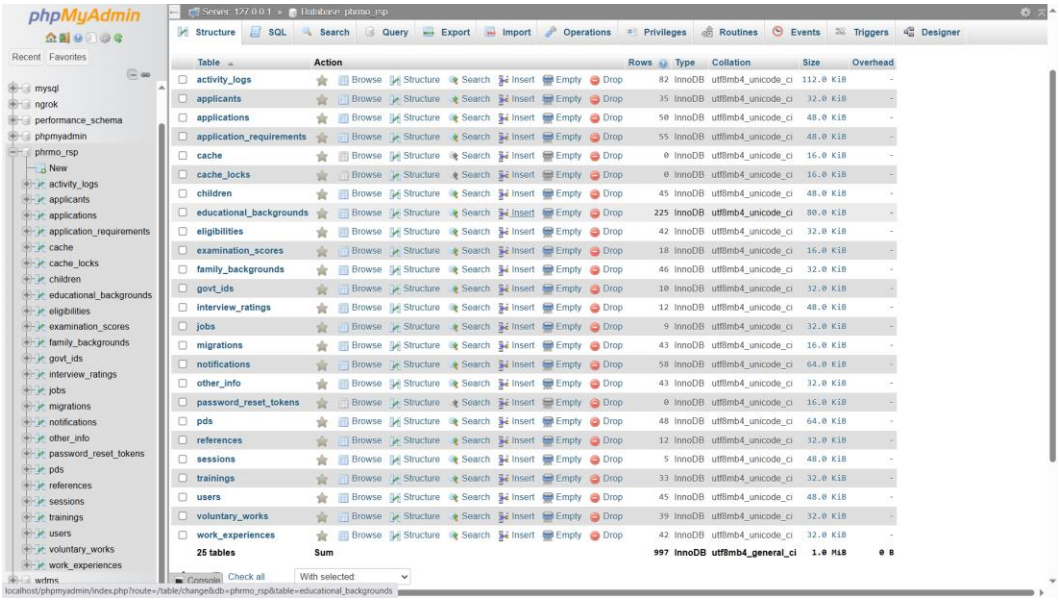



**Figure 5. Database Structure**

## User Design

User Design A system outline was built utilizing project parameters during the architectural stage of system design. Entity connection diagrams, data flow diagrams, use-case diagrams, flowcharts, operational frameworks, and system designs were among the numerous design aspects that made up this. The information acquired at this point had an impact on the subsequent stages of system architecture and design. While suggesting several system designs and architectural solutions, the project's goals and specifications were taken into account.

## Diagrams

Diagrams are visual tools that depict a range of elements of a system, such as its behavior, structure, processes, and component interconnections. They enhance stakeholder understanding, make difficult concepts easier to understand, and support system analysis, design, and documentation.

**Entity Relationship Diagram of the System.** A database system's entities and their connections are shown graphically via the entity relationship diagram (ERD). It provides insight into the logical structure of different objects and concepts by demonstrating how they interact with one another. The entity relationship diagram for the eHiremo: Web-based Recruitment and Selection Process System of PHRMO is shown in the following diagram, which stores entities and makes their logical relationships clear.

The recruitment and selection system's structured architecture is shown by the Entity-Relationship Diagram (ERD), which shows the orderly interactions between candidates, administrators, and job-related data. Every step of the hiring process is precisely tracked and meticulously documented because to the Applicants entity's central connection to vital documents like applications, test results, interview ratings, and

notifications. Supporting entities such as PDS_main, work_experience, educational_background, eligibility, training, voluntary_work, and family_background provide comprehensive details about each applicant's qualifications and personal information, ensuring fair evaluation based on both academic and experiential credentials.

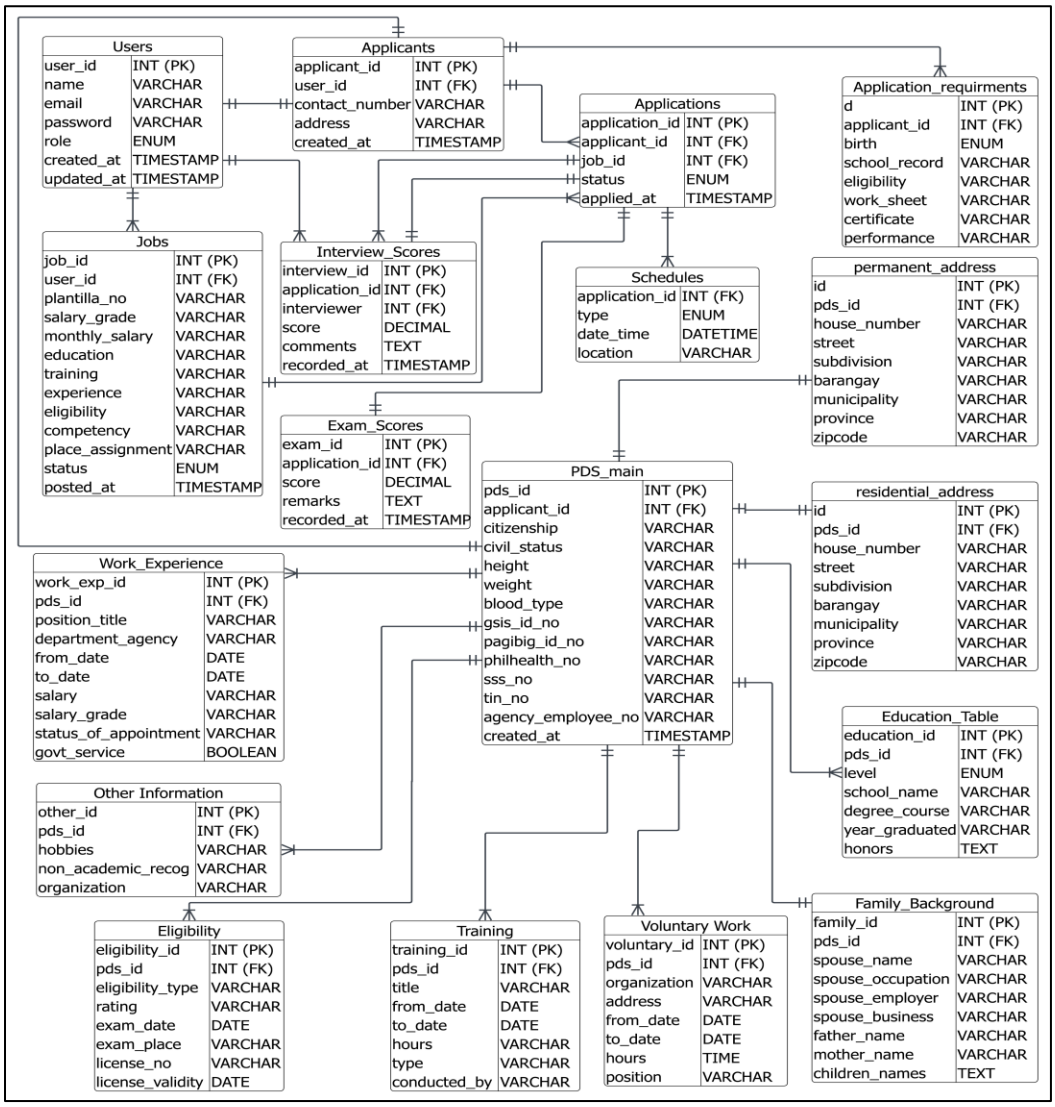



**Figure 6. Entity Relationship Diagram of the eHiremo**

**Data flow Diagram of the Study.**

Applications are connected to the Jobs entity, which enables the system to track open positions and match them with qualified applicants. Meanwhile, constraints such as primary keys and foreign keys guarantee data integrity, and features like timestamps ensure accuracy in capturing crucial events. All things considered, the ERD exhibits a well-organized database model that strikes a balance between effectiveness, equity, and transparency, facilitating a methodical and trustworthy hiring procedure.

An online recruitment and selection system is conceptualized at a high level in the Data Flow Diagram (DFD) below, which shows the systemic interaction between external entities and internal operations. The data flow for applicant registration, job application administration, evaluation, and selection is shown in the diagram. Through separate processes including Login/Registration, Application, and Selection and Appointment, external entities—Applicant, Administrator, and Super Administrator—interact with the system. Data stores that enable the long-term storage and retrieval of relevant information, such as Users_Table, Application_Table, and score repositories, mediate these relationships.

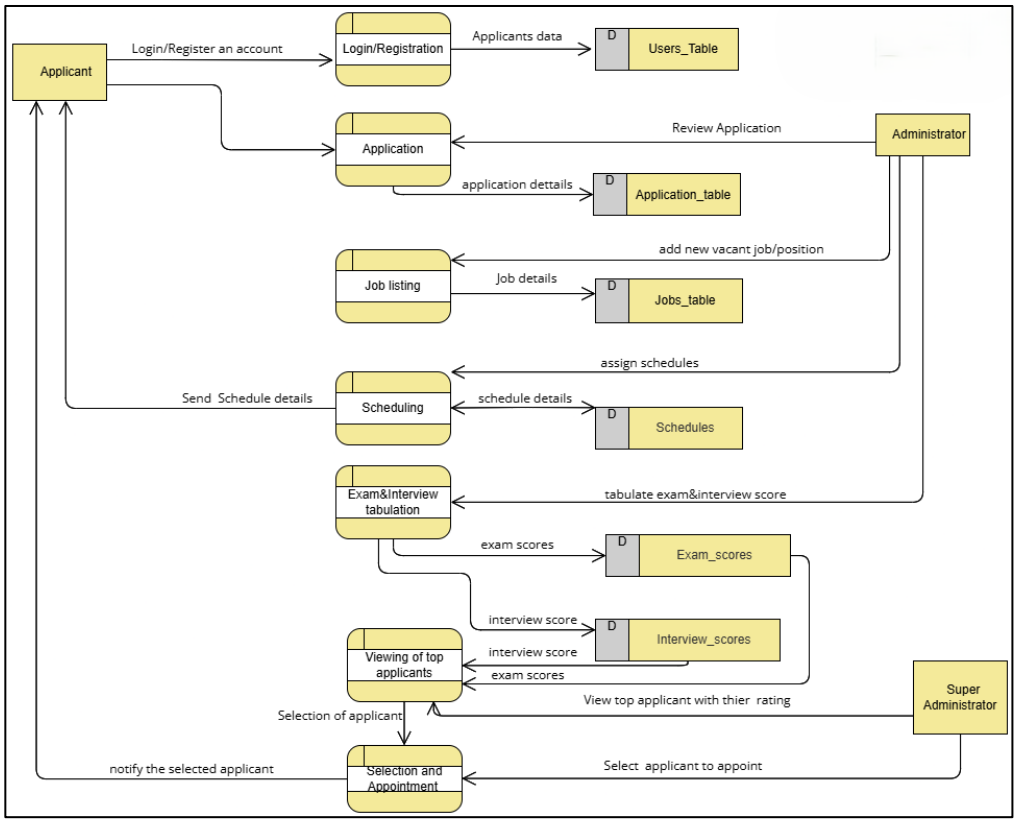



**Figure 7. Level 1 Data Flow Diagram of the System**

**Use Case Diagram of the System.** In system analysis, the case model—a kind of behavioral diagram—is used as a methodological tool to make the complex activities inside the system easier to understand. By showing how users (actors) and the system interact in predetermined scenarios (use cases), it outlines and organizes the system requirements. This model's main goal is to identify the system's internal and external influences.

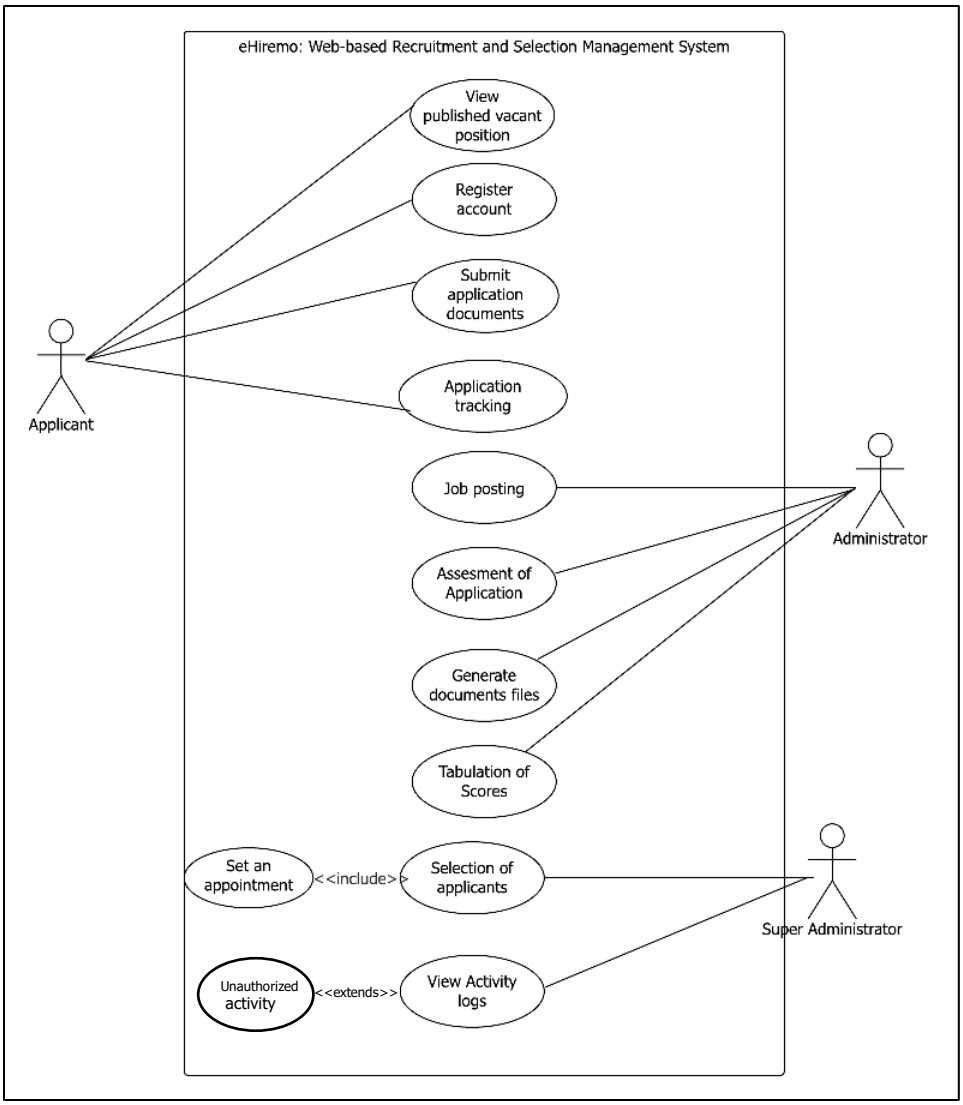



**Figure 8. Use Case Diagram of the System**

The figure offers a clear image of how the recruitment process works for the Provincial Human Resource Management Office at Davao del Sur Provincial Capitol. Applicants first register, log in to the system, look

for opportunities, and then apply for the positions that pique their interest. Once an application is filed, administrators step in to analyze it and determine whether to approve or reject it. Additionally, applicants are kept informed throughout the process by receiving notifications and updates. Finally, the super administrator plays a significant part by picking out the best candidate and officially appointing them. All things considered, the diagram shows a seamless, methodical process that guarantees each application is thoroughly examined and that the best applicant is selected.

**Flowchart of the System.** A flowchart is a diagram that sequentially displays the activities, steps, and process flow of the eHiremo. Diagrams are useful for planning, visualizing, documenting, and improving the system process. Figure 6 shows the flowchart for the research. The entire process of submitting a job application is depicted in the flowchart below. It all starts when a job posting is made public and candidates are invited to apply. Applications will go through a review procedure after they are submitted. The application procedure ends when an applicant is informed that their application does not meet the necessary requirements. An email or account notification is issued to those who are accepted. A planned examination and a panel interview are the next steps in the procedure. Upon completion of all evaluations, the top-performing

applicants are ranked based on their overall performance.The hiring team then selects the best candidate for the position.

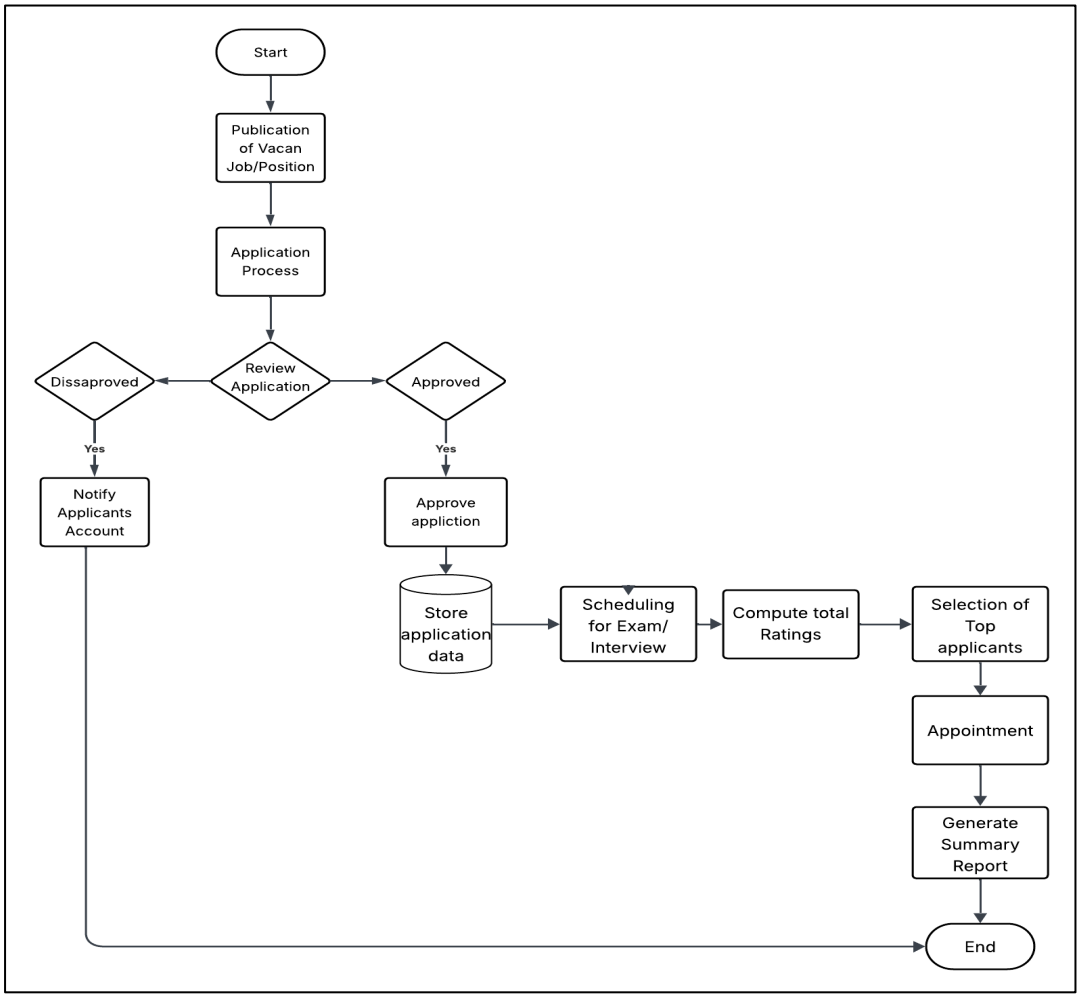



**Figure 9. Flow Chart of the System**

The voyage concludes with the creation of a summary of the complete application procedure. This methodical process guarantees that

each candidate is given an equal opportunity and that the most qualified candidate is carefully selected.

**Operational Framework.** The operational framework included the essential steps, tasks, and materials needed to carry out a project, program, or policy from start to finish.

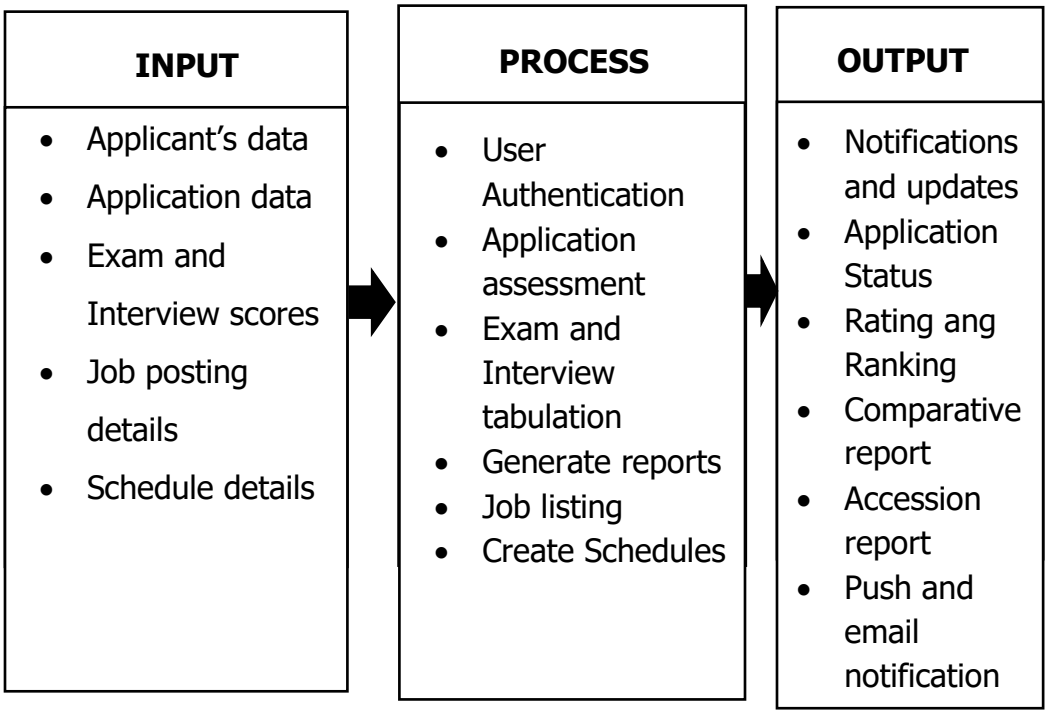



**Figure 10. Operational Framework of the eHiremo**

A comprehensive summary of the elements of the recruiting and selection system, their relationships, and the information flow was given, as seen in Figure 10. It describes the system's inputs, procedures, and results, such as authentication, managing applicant data, managing exams

and interviews, managing applications, managing alerts and changes, scheduling, and creating reports.

## Construction

Within the framework of rapid application development, the construction phase is a critical phase since it focuses on translating the defined design into a working software system. Its main purpose is to create a system that satisfies the needs and expectations of the client while staying in line with the project's objectives.

### Implementation Activities

This section explains how each user account may access and utilize the Web-Based Recruitment and Selection Management System. The Super Administrator process flow is depicted in Figure 11, which emphasizes the recruitment and selection system's authentication and access control methods. The administrator's function in overseeing candidate appointments is given special attention.

The process begins when the user enters their login credentials, after which the system validates the information to confirm its accuracy before granting access.

**How the Project Worked**

This project helped manage Recruitment and Selection in the Provincial Human Resource Office (PHRMO). The Super Admin could control the Selection of applicant to be appointed in the position, while the admin will manage the users account, application assessment, tabulation of exam and interview score, manage the applicant's notifications and updates for the applicant's application status, exam, interview, and appointment schedules, and reports. The applicant could access the Job portal for viewing of Job publication, register account for the job application, track application status, receives push and email notifications and updates and submit multiple application entry for different job. Each function has its own set of tools to perform its tasks precisely and effectively. Refer to the User's Manual for Admin and Super Admin in Appendix 4, pp. 126–136, for complete instructions on how to utilize the system.

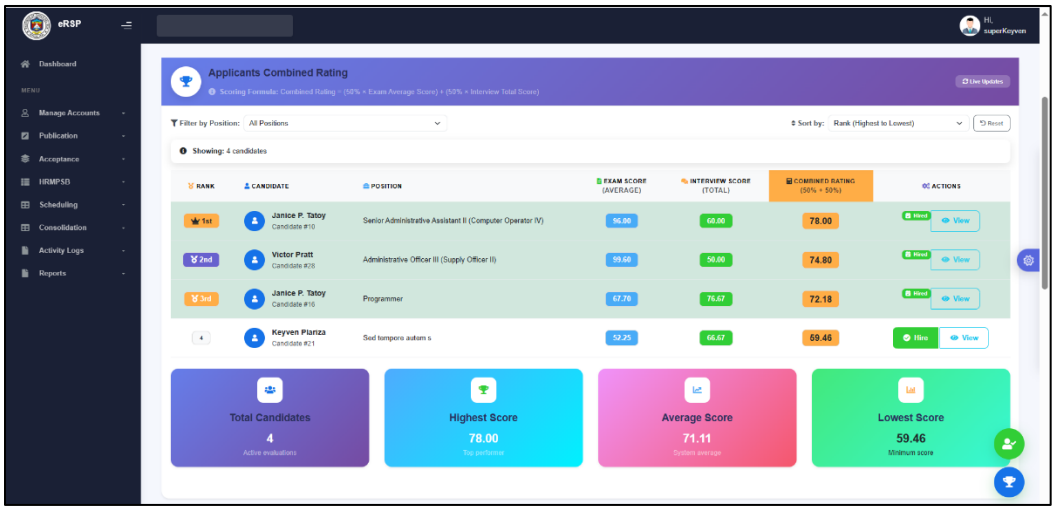



**Figure 11. GUI Dashboard Super Administrator**

As shown in the figure 11, the Graphical User Interface of the Super administrator dashboard displays the applicant's ranking candidate for appointment according to their rating it can also rank according to their position using the "Sort by" and the Super admin can select an applicant to be appointed using "Hire" button this will update the applicant and notify them. There is also a side bar for navigation to other process the same to the admin, however there is specific functions for super admin only which is the viewing of "Activity Log" and Hiring functionality which is only the super admin has the privilege to access. There is also cards that holds real time data of the system for "Total candidates", "Highest score", "Average score", and "Lowest score".

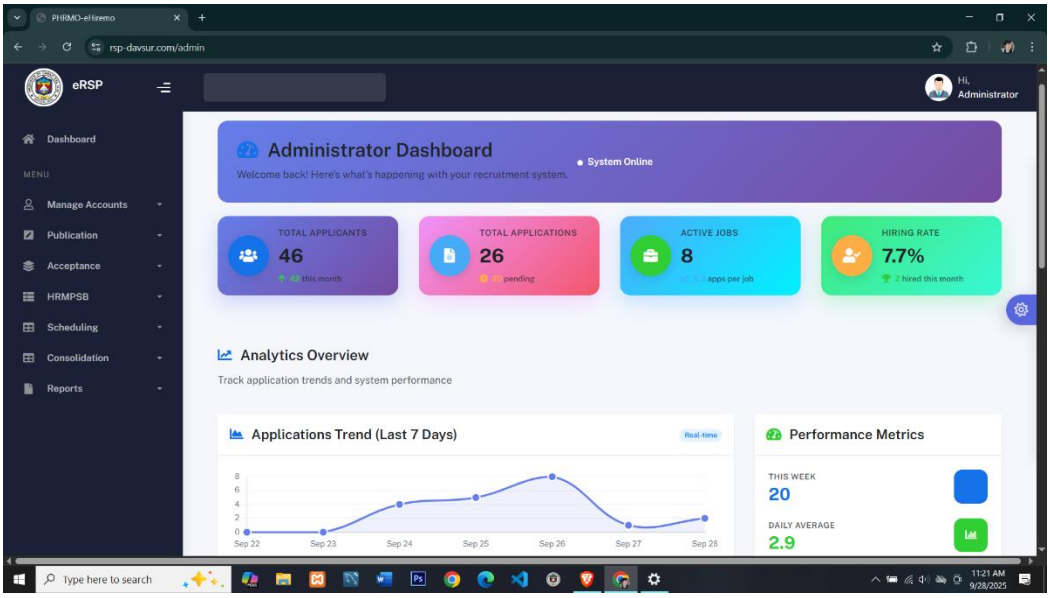



**Figure 12. GUI Dashboard Administrator**

As shown in the figure 12, the Administrator Dashboard Graphical User Interface it displays the administrator functions in the system there is a sidebar for navigating the pages for the specific functions, "Mange Account" enable the admin to manage users account in the system "Publication" is where the Job listing process happened the admin can manage the Job listed in the website " Acceptance" is where the application of the applicants undergo assessment and review their documents submitted "HRMPSB" let the admin to generate the list of applicant that will undergo PSB "Scheduling" it enable the admin to set schedules for the examination, interview and appointment to the applicant "Consolidation" this allow the Admin to Inpute the score of the applicant in their exam and

interview rating "Reports " enable the admin to generate reports included in the process and at the center there is a cards that holds and display the number of "Total Applicants", " Total Application", "Active jobs", and "Hiring rate" and it is Realtime data below of that is a simple analytics to give brief information to Administrator about the data in the application process.

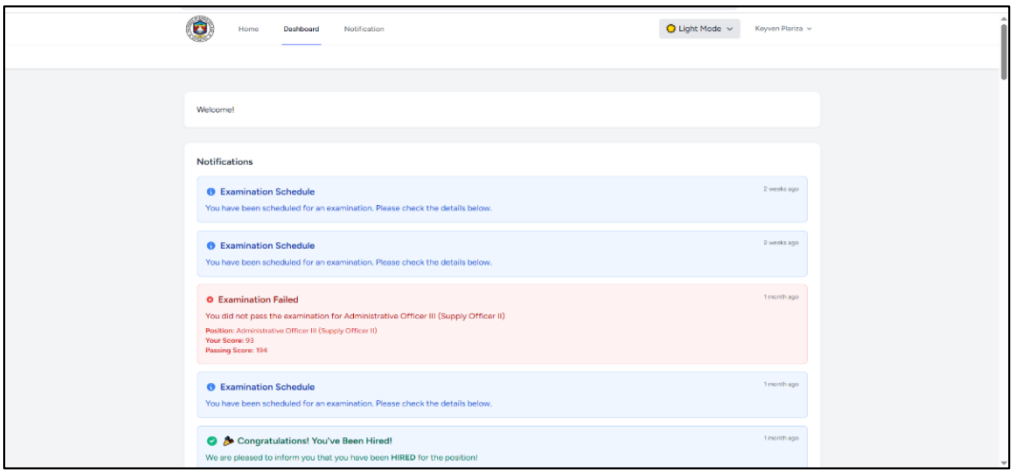



**Figure 13. GUI Dashboard Applicant**

As show in the figure 13, the Applicant Dashboard Graphical User Interface is displayed there three navigations at the top "Home", "Dashboard" and "Notification" which is use to navigate different page. And there is also a system configuration to enable user to choose the color theme of the GUI and a navigation for the profile of the account. In the

center of the GUI the real time notification will be displayed and the application progress will also display in that section.

**Deployment Activities**

In order to ensure that the system was correctly installed, configured, and prepared to effectively support the PHRMO activities, deployment was an essential stage in moving it from development to practical usage.

**Installation Activities.** This functioned as a manual, offering precise instructions to guarantee a seamless and effective installation procedure. The system was implemented in the manner described below: First. Configuring the server for hosting. The Hostinger was utilized for hosting, as seen in Figure 14.

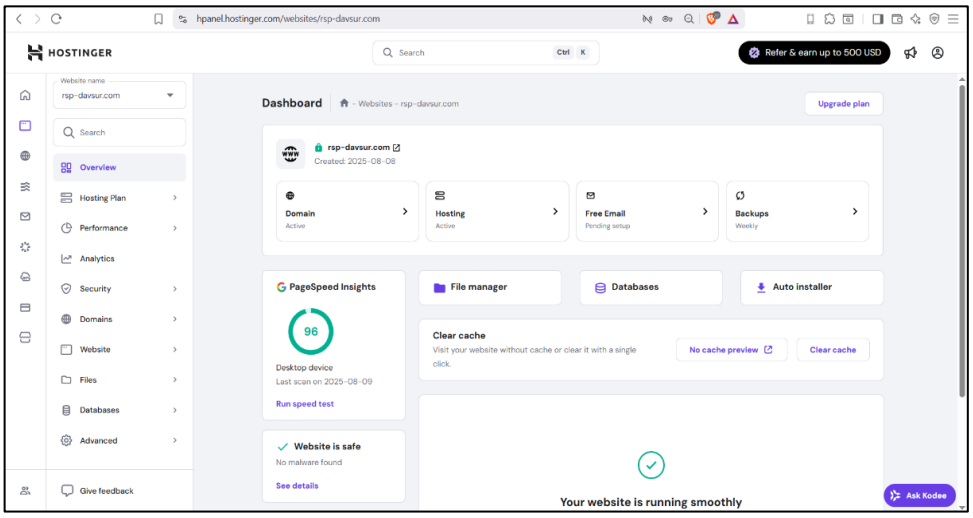



**Figure 14. Hosting Server**

2. Zip the project file in the WinRAR to upload in the web hosting.

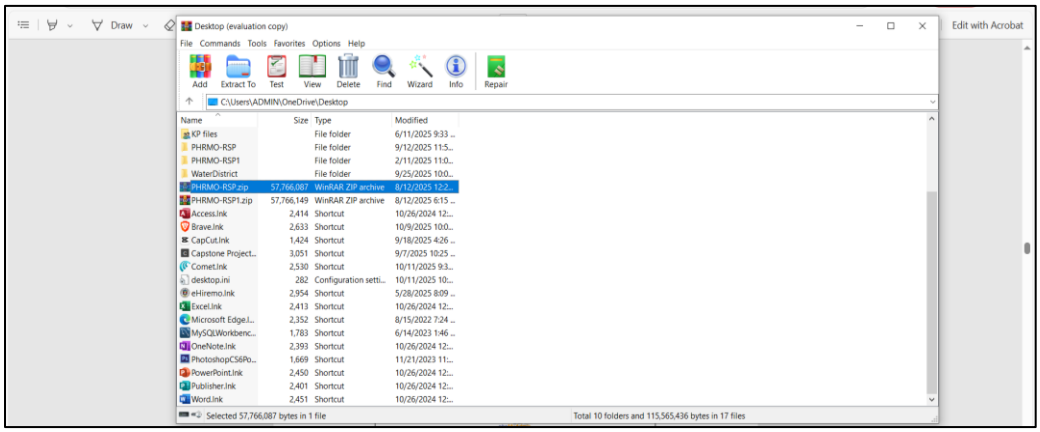



**Figure 15. Zipping the Project files**

3. Upload and extract the project file in the File manager of the server

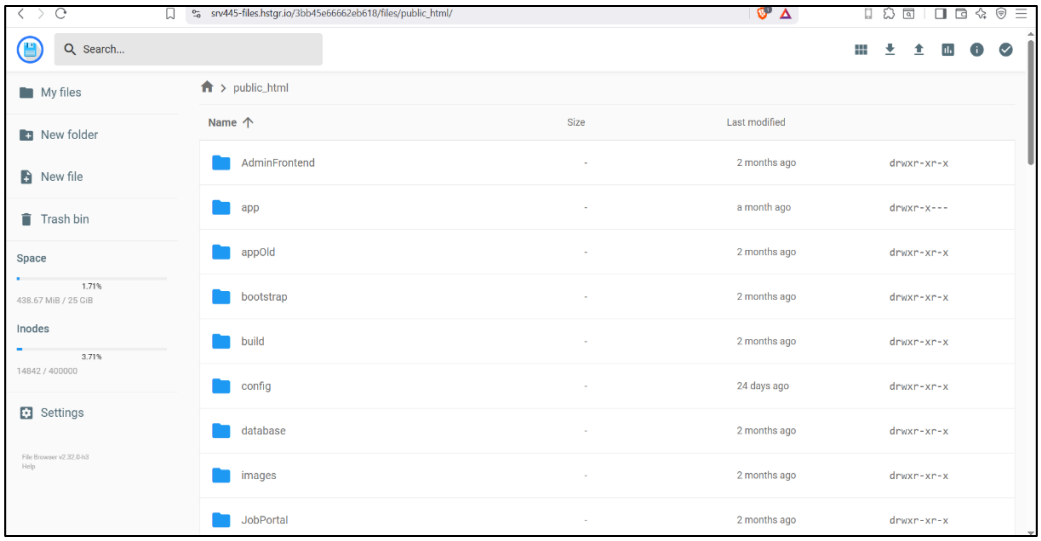



**Figure 16. File upload in the server**

4. Upload in the PhpMyadmin MySQL Database. The system's database was uploaded and managed using phpMyAdmin, as seen in Figure 17.

This made sure that every table and data structure required for the system's operation was set up correctly.

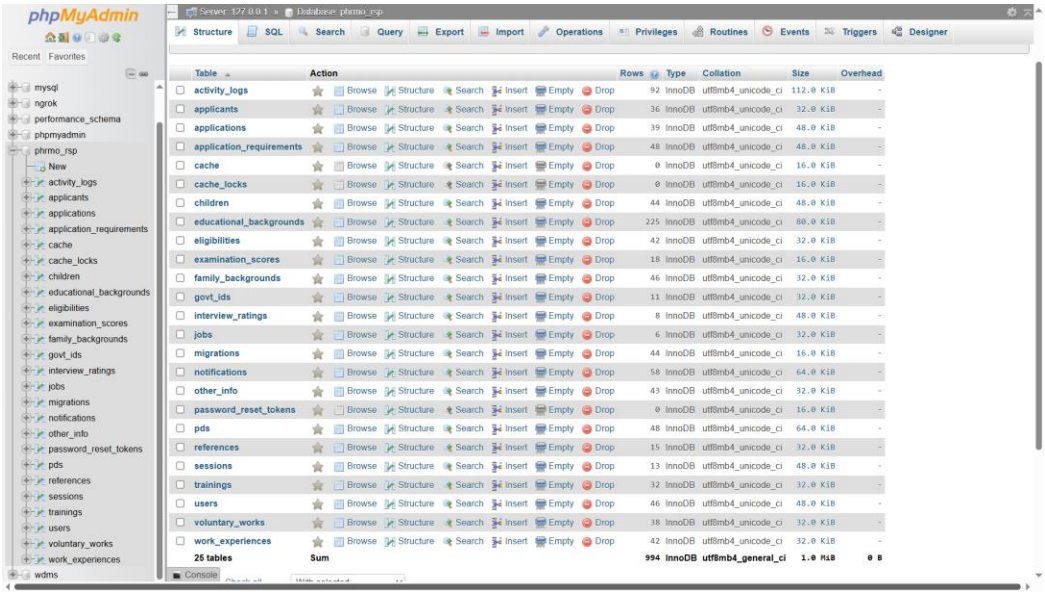



**Figure 17. Upload Database**

**Maintenance Activities**. To ensure efficacy, this required creating a thorough action plan that included all building, maintenance, and repair work. Preventive, corrective, and enhancing procedures were all part of the maintenance strategy.

1. Modifying the source code. Using a coding editor like Visual Studio coding, the developer accessed the project files. Here, essential adjustments were performed, such as bug corrections, feature

improvements, or optimizations. The code editor highlighted the changed lines or files to reflect the system modifications, as seen in Figure 18.

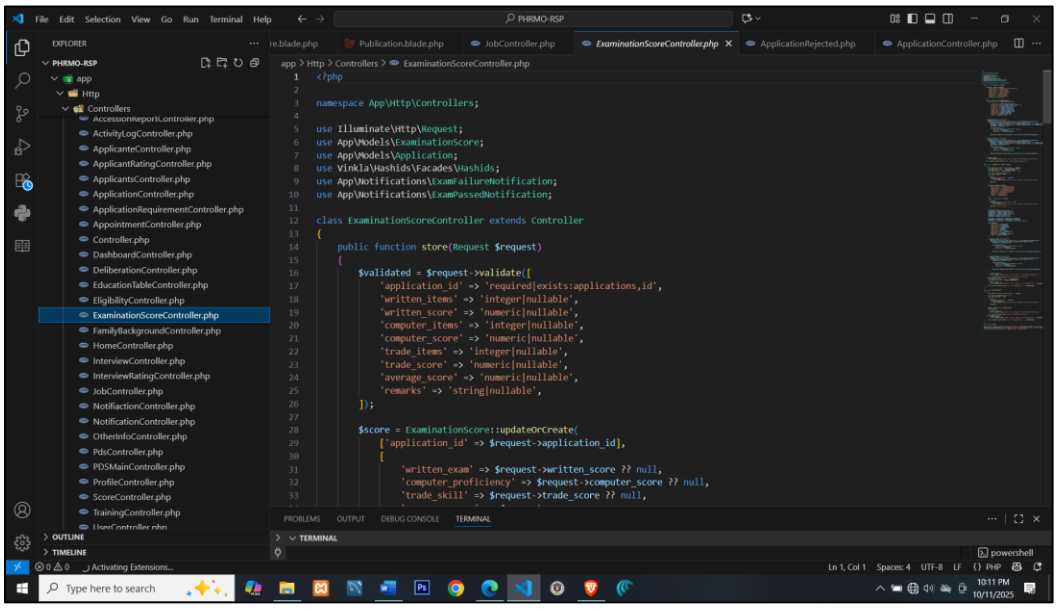



**Figure 18. Modifying the source code**

2. Update the project remotely with Putty so that the changes made in the local machine will implement in the server

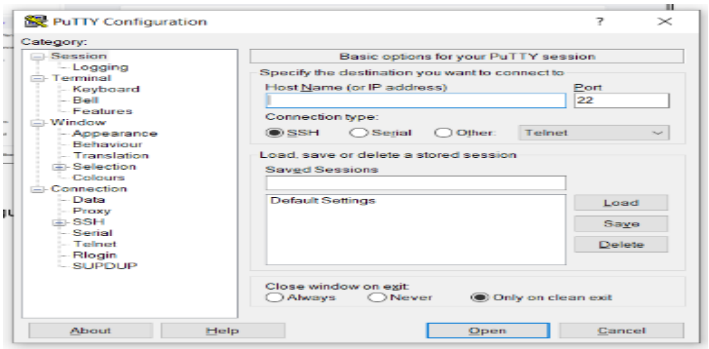



**Figure 19. Update the Project remotely**

**Backup and Recovery Activities.** This required copying and carefully storing vital data in order to generate a backup in case of data loss or damage.

1. Upload the entire project in the GitHub repository functions as a secure remote repository that automatically retains every committed version of the project. This ensures that in the event of hardware malfunction, accidental file deletion, or local data corruption, the complete system files can be easily restored from the cloud repository.

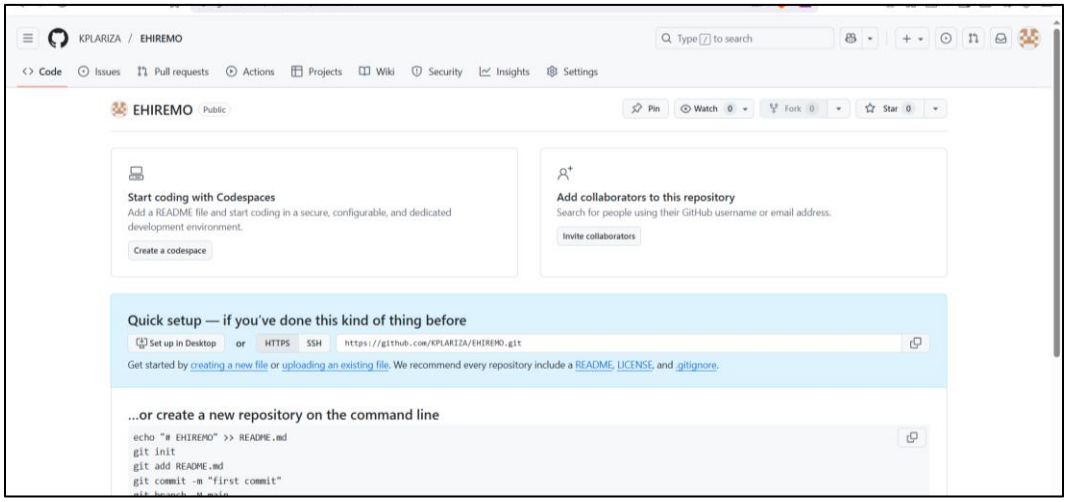



**Figure 20. GitHub Repository of the Project**

2. Hostinger automatically generates daily and weekly backups of all hosted files and databases. These backups are securely stored on

Hostinger's remote servers and can be restored directly through the hPanel (Hostinger Control Panel).

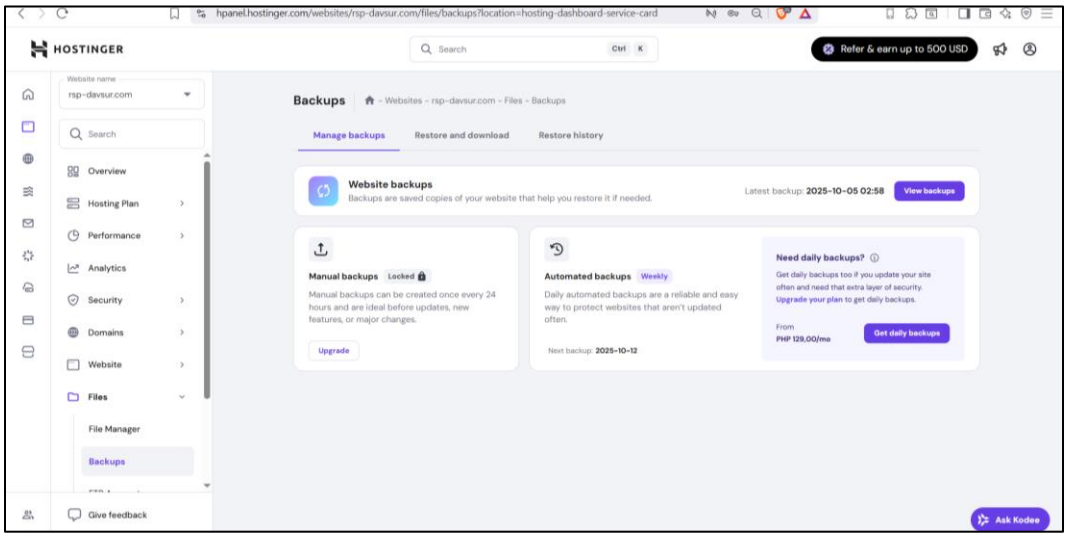



**Figure 21. Hostinger backup function**

## Cutover

The cutover phase signaled the end of the quick development of applications framework, which served as the study's system development life cycle model. The deployment procedure began once all of the objectives had been met and the system was judged ready for usage. To make sure the produced system reached acceptable levels of functionality, dependability, and usability, testing and user training were part of the process.

### Testing and Evaluation

This stage outlines the procedures and techniques for testing and examining the system. It includes the functionality, utility, and dependability of the system. The findings are obtained by computing the mean value.

**Respondents**. Respondents voluntarily and kindly provide vacant time to engage in systems evaluations, data gathering, or give an important information that is essential to the project's success, making them an essential element of any research effort. The project's responder distribution is displayed in Table 4.

**Table 4. Distribution of Respondent**

| EVALUATORS | SAMPLE SIZE |
|---|---|
| Advisory Committee | 4 |
| Applicants | 10 |
| PHRMO | 5 |
| **Total** | 19 |

Nineteen (19) respondents made up the whole sample size for examination, as shown in Table 4. These assessors came from a variety of backgrounds, including the PHRMO team, the advisory committee, and applicants. To evaluate the project's functional needs, performance efficiency, compatibility, usability, dependability, security, maintainability,

and portability, they were given descriptive survey questionnaires. These assessors' input helped with the system's overall assessment and enhancement.

**Sampling and Design Technique**. After the project was tested using Purposive Sampling, four members of the Advisory Committee, five PHRMO employees, and ten applicants assessed the system's functional requirements, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

**Evaluation Procedure.** This study's assessment process offers a methodical way to evaluate the produced system's efficacy, quality, and performance. The evaluation's findings will direct decision-making, point out areas where the system may be improved, and support the goals of the study. The researcher will get official clearance to enter the system and gather pertinent data before beginning the examination. This procedure makes it easier to get precise and representative assessment data while guaranteeing adherence to ethical research norms. Secure a permit to conduct from RDI (refer to appendix 9 page 162).

1. The HR officer and PHRMO staff signed and approved the permission letter requested by the researcher (refer to appendix 11 page 164).

2. The responders are briefed by the developer on how to utilize the system (see appendix 18, page 166).
3. To assess the system functionalities and features the developer will give the testers ample opportunity to utilize the system and document their feedback on the assessment form.
4. Gave participants a set amount of time (three to five days) to finish the survey, then collected the completed forms.
5. Arranged and examined the information gathered in order to make judgments and put the evaluation's results into practice (see appendix 2, page 143 for the data tabulation).

**Survey Instrument of the Study**

A data survey instrument is a tool used to collect information or data from a sample of individuals or groups in a systematic and consistent manner. The instrument was altered to meet project requirements for the assessment procedure once virtual development was finished. Respondents were asked to score the system's performance according to functional suitability, performance efficiency, usability, compatibility, reliability, maintainability, security, and portability using a questionnaire modified from the ISO-defined system and software quality framework and

model 25010. The questionnaire focuses on the eHiremo: Web-based Recruitment and Selection Management System's ISO-based software quality framework. **Likert Scale of Measurement for Functional Suitability, Performance Efficiency, Usability, Portability, Reliability, and Maintainability.** The system's ability to carry out its intended tasks is evaluated using Likert scales. The Likert scale used to evaluate the system is shown in Table 5.

**Table 5. Likert Scale of Measurement for Functional Suitability Performance Efficiency Usability, Maintainability, Portability, Reliability, and**

| Mean Range | Descriptive Equivalent | Interpretation |
|---|---|---|
| 4.51 – 5.00 | Strongly Agree | The measure demonstrates the respondents' confidence in the system's functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability, all of which have been successfully met with no errors. |
| 3.51 – 4.50 | Agree | Despite that all functional and non-functional objectives not being fully met and the possibility of small problems like bugs and malfunctions, it emphasizes that the participants acknowledge the results. |
| 2.51 – 3.50 | Fairly Agree | The measure assumes that the goals was achieved in some way and shows that the respondents accept the findings. However, some functional and non-functional. |
| 1.51 – 2.50 | Disagree | Based on the data, this statistic demonstrates that the respondents thought almost all of them were unmet. |

| | | |
|---|---|---|
| 1.00 – 1.50 | Strongly Disagree | The indication emphasizes the fact that the participants was not persuaded that the system achieved its goals. |

The scale ranges (1) are the lowest score on the scale, which goes between one (1) and five (5). This Likert scale provides an organized way to gauge how well the system is perceived by users, which aids in pinpointing areas that need work. "Strongly Agree" is defined as 4.51 to 5.00, "Agree" as 3.51 to 4.50, "Fairly Agree" as 2.51 to 3.50, "Disagree" as 1.51 to 2.50, and "Strongly Disagree" as 1.00 to 1.50.

**Statistical Tools**

It is the tool for statistical analysis evaluates and interprets research data using mathematical methods and strategies. It entails finding trends, connections, and knowledge that will be utilized to problem-solving or decision-making. The data will be compiled and recorded after respondents' responses to the survey's circulated questionnaire have been gathered. The total number of respondents divided by the sum of the evaluators' scores yields the mean. The mean that is weighted will show the impact of the recruitment and selection system.

Equation to calculate the mean value:

$$\bar{X} = \Sigma x / n$$

Where:

$\bar{X}$ = signifies the score

$\Sigma x$ = shows the sum of the scores from each assessor.

$n$ = indicates the total number of assessors.

# CHAPTER IV

# RESULTS AND DISCUSSIONS

The study "eHiremo: Web-based Recruitment and Selection Management System for the Provincial Human Resource Management Office" is presented in this chapter along with its findings, analysis, and debates. It covers the key findings and clarifies their significance. The results are arranged according to the objectives of the study, with an emphasis on the functionality, performance efficiency, compatibility, usability, security, dependability, maintainability, and portability of the system. Comparisons with earlier research and relevant insights are also highlighted.

.

## Super Admin module for Viewing Applicant Ranking and Rating

The Viewing of Applicant Ranking and Rating module was successfully developed and integrated into the Web-Based Recruitment and Selection Management System to automate the ranking of applicants based on their examination and interview results. The ranking mechanism

employs an equal-weight linear combination, wherein the total score is derived by assigning 50% to the examination score and 50% to the interview score.

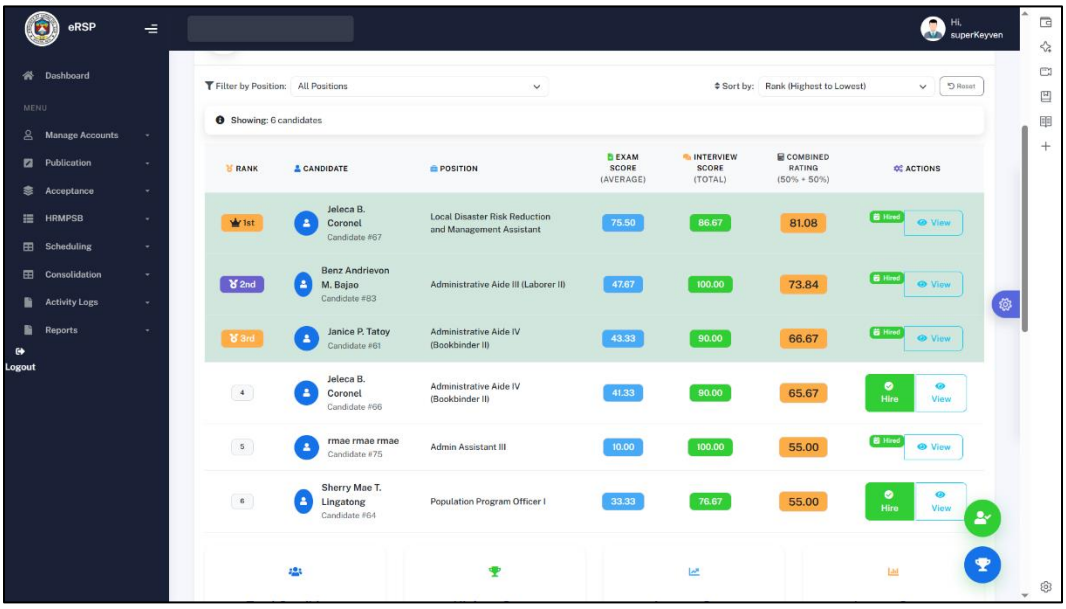



**Figure 22. GUI for Viewing of Applicant Ranking and Rating**

Figure 22 show the Graphical User Interface (GUI) for viewing of the applicant ranking and rating. It displays the list of applicants that are ranked according to their combine exam and interview rating. The Super Administrator can filter the list of applicants based on the available positions. In the Action column, two buttons are provided: "Hire" and "View." Clicking the "Hire" button selects the applicant for appointment to the corresponding position. Once the applicant is successfully appointed,

the "Hire" button changes to "Hired" and becomes non-clickable**,** preventing any conflicts or duplicate appointments within the system. The "View" button will only display the overview of the applicant and can also use the "Hire" function button directly in the view modal.

```
Function: Comparative Assessment and Hiring Process
Require: Ranked list of applicants with examination and
interview scores
Ensure: Applicant's combined rating and hiring decision are
accurately determined
   1. Start
   2. View the ranked list of applicants.
   3. For each applicant:
        a. Compute combined rating = (Examination Score × 0.50)
      + (Interview Score × 0.50).
        b. Record the computed rating.
   4. Display applicants based on their combined rating.
   5. Stop
```

**Figure 23. Pseudocode for Viewing and Selection of the Top Applicant**

The process begins by viewing the ranked list of applicants. Each applicant's combined rating is computed using 50% of the examination score and 50% of the interview score. After calculating the ratings, applicants are ranked accordingly. Those who meet the required qualification criteria are selected or hired, while others are not. The process then ends (see corresponding code snippet in Appendix Figure 5.1, p. 152).

### Super Admin module for selection of applicant for appointment

The Super Admin module for selection of applicant for appointment enables the super admin to perform the selection of applicant among the top five ranked applicant to be appointed to the position.

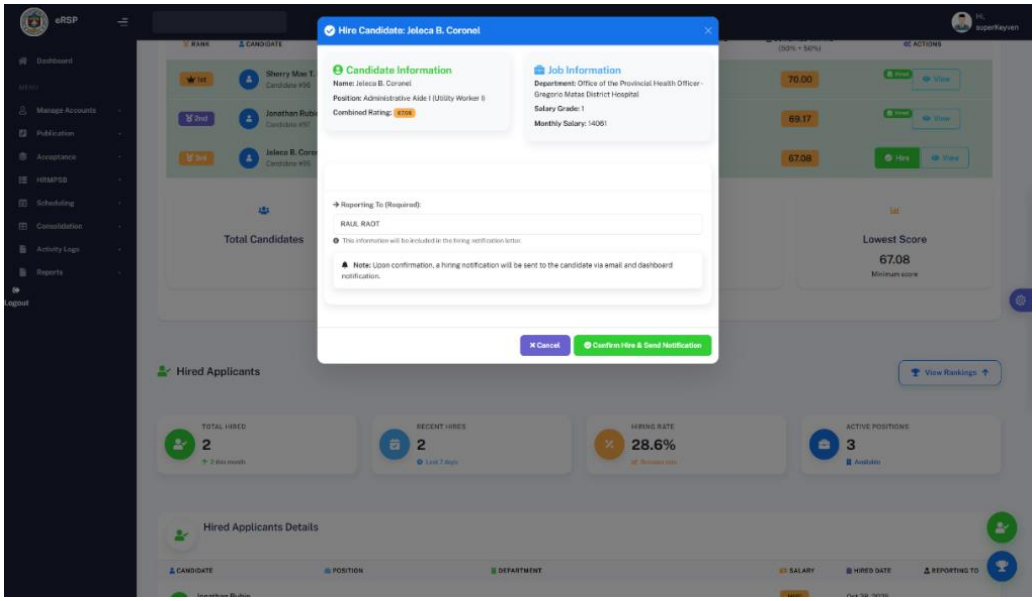



**Figure 24. GUI for Selection of Applicant for Appointment**

Figure 24 shows the Graphical user interface (GUI) for Selection of Applicant for Appointment, where the super admin can perform the selection of the applicant. As shown in the modal there is a "Candidate Information" that shows the applicant's complete name, position applied, and rating, Also the "Job Information" which shows the Job details such as the Place of assignment, Salary grade and Monthly salary. There is also

field for assigning of Personnel for reporting of the applicant, then Confirmation of selection button to save and send notification to the applicant that he/she was appointed to the position.

```
Function: Selection of Applicant for Appointment
Require: Applicant application must reach to this stage
Ensure: Applicant has an exam and interview score
   1. Start
   2. View the ranked list of applicants.
   3. For each applicant:
        a. Compute combined rating = (Examination Score × 0.50)
      + (Interview Score × 0.50).
        b. Record the computed rating.
   4. Display applicants based on their combined rating.
   5. If applicant meets the qualification criteria:
          Select or hire the applicant.
        Else
          Do not hire the applicant.
   6. Stop
```

**Figure 25. Pseudocode for Selection of Applicant for Appointment**

The pseudocode for the Selection of Applicant for Appointment illustrates the systematic process of identifying qualified candidates based on their performance. The procedure begins by viewing the ranked list of applicants who have successfully advanced to this stage of the recruitment process. For each applicant, the system computes a combined rating by averaging their examination and interview scores, each weighted equally at 50% (see corresponding code snippet in Appendix Figure 5.2, p. 152). These computed ratings are then recorded and displayed in descending order to show applicants from highest to lowest ranking. If an applicant

meets the established qualification criteria based on the computed rating, they are selected or hired; otherwise, they are not considered for appointment. The process concludes after evaluating all applicants.

**Super Admin module for Viewing System Activity Logs**

The module enables the super admin to view the system activity logs to keep identify and track the process done in the system and ensure the security if there are any unauthorize process or unauthorize account perform specific process without permission to the authority in the system it will be track and identified using the system activity logs report.

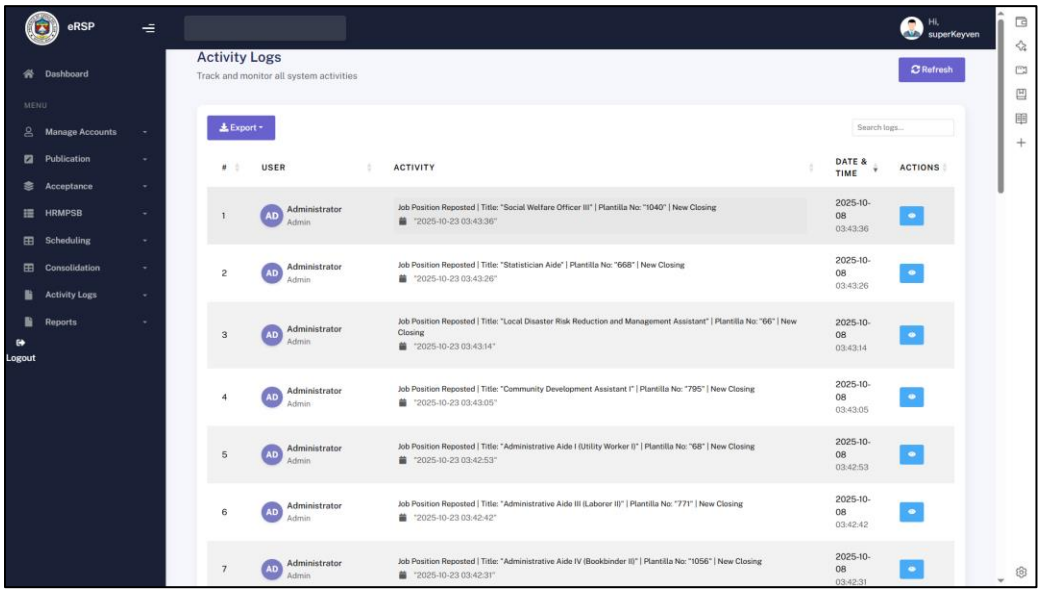



**Figure 26. GUI for Viewing of Activity Logs**

As shown in Figure 26, the Super Admin can access the System Activity Logs, where important processes and actions performed within the system are recorded and displayed in a structured table. The table provides essential information, including the user who performed the action, the type of activity, and the timestamp indicating the exact date and time of the activity. Additionally, the activity logs can be searched, allowing for quick identification and retrieval of specific actions carried out within the system.

```
Function: System Log Monitoring and Investigation
Require: Super admin account Access to system logs
Ensure: Unauthorized activities are detected and investigated
        properly
   1. Start
   2. View the system logs.
   3. Check the system logs for any suspicious or unusual activities.
   4. If there is unauthorized access or illegal activity detected:
          a. Investigate the illegal activity.
          b. Document findings and take necessary security
              actions.
        Else
          a. No action needed; system is secure.
   5. End
```

**Figure 27. Pseudocode for Viewing of Activity Logs**

This pseudocode describes a System Log Monitoring and Investigation process that helps ensure system security. It begins by allowing a super admin to access and review system logs. The system then checks for any suspicious or unauthorized activities within those logs. If such activities are detected, the admin must investigate the incident,

document the findings, and take appropriate security measures to resolve the issue. If no problems are found, no action is needed, and the process ends with the system confirmed as secure (see corresponding code snippet in Appendix Figure 5.3, p. 153).

### Admin module for Job Listing Management

The job listing management module allow the admin to add new job vacancy with the job details and will be available in the job listing of the website so that it will available for the applicant and they can apply for it. The admin can also edit and update the job and when it reaches to its closing date the admin can repost the job.

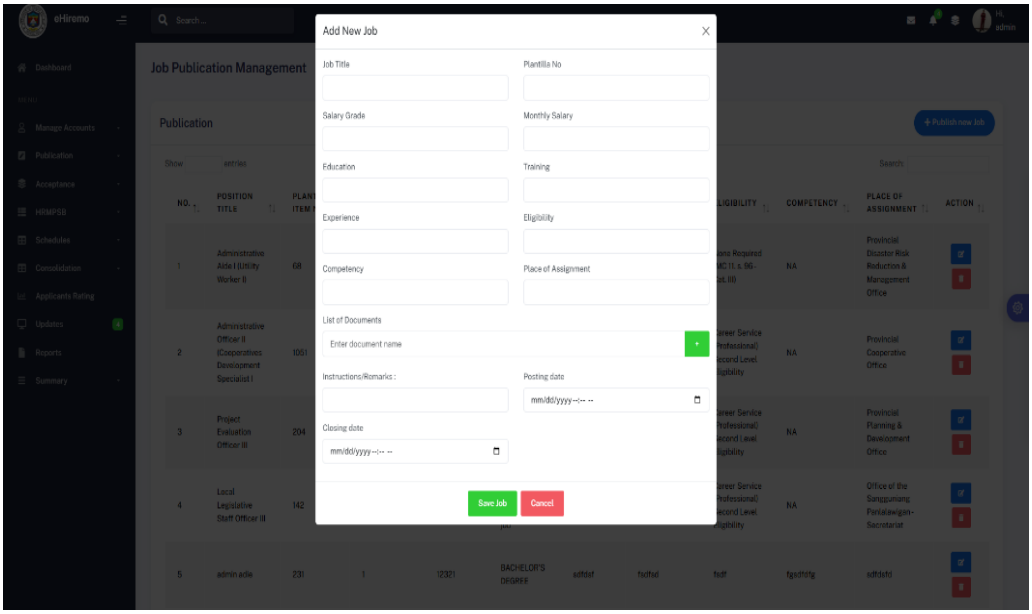



**Figure 28. GUI for adding new job in the publication**

Figure 28 shows the graphical user interface (GUI) for adding new job with its essential field details to be entry such as position title eligibility, salary grade, monthly salary, educational qualifications, training, experience, Competency, posting date, closing date list of documents, instruction, and place of assignment. And button for saving the job in the system and it will now available in the job listing website.

```
Function: Administrator Application and Publication Management
Require: Administrator account access
Ensure: Administrator can publish Job
   1. Administrator logs into the system.
   2. System displays options: Review Application or Add New
      Publication.
   3. If Review Application is selected:
        a. Administrator reviews submitted applications.
        b. Decide whether to Approve or Reject the application.
          i. If approved, return to Dashboard.
          ii. If rejected, notify applicant and return to
      Dashboard.
   4. If Add New Publication is selected:
        a. Administrator inputs Publication Information.
        b. Click Save to store publication details.
        c. Return to Dashboard.
   5. End of Function.
```

**Figure 29. Pseudocode for Job Publication**

As depicted in Figure 29, the pseudocode for the Administrator Application and Publication Management process illustrates how the administrator can efficiently review submitted applications or add new publications within the system. By providing options to approve, reject, or

save entries, the pseudocode ensures that all administrative actions are systematically executed and properly recorded (see corresponding code snippet in Appendix Figure 5.4, p. 153). This structured process helps maintain organized management of both applications and publications, allowing the administrator to update information and return to the dashboard seamlessly.

### Admin module for Assessment and Evaluation of Application

The admin module for assessment and evaluation of application allows the admin to check the submitted application and documents by the applicant, then assess and evaluate if it was correct and legit data and it should also comply or meet the job requirement so that the application will be accepted.

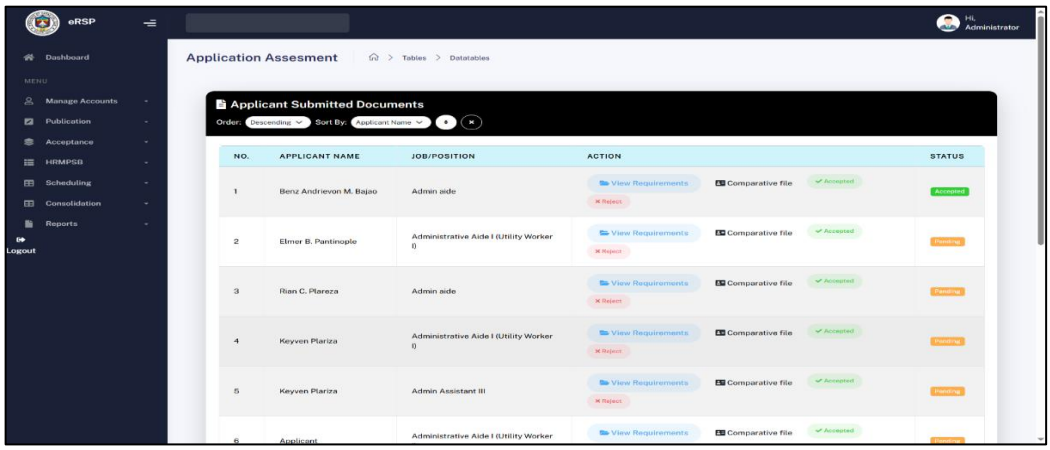



**Figure 30. GUI for the Administrator module for application assessment**

The Figure 30 shows the graphical user interface (GUI) for assessment and evaluation of application it displays the Applicant name job position and status. In the action row there's buttons such as "View Requirements" which enables the admin to check the uploaded document by the applicant, "Comparative file" will generate the comparative report for the applicant to check the data in their PDS entry is correct and meet the job requirement and also check if the uploaded documents are accurate to the data in the comparative file of the applicant. And "Accept" button for accepting the application after checking all the details and if the is something wrong with the data there's a button "Reject" to reject the application it will notify the applicant with the rejection reason so that they can comply the correct and necessary requirement.

```
Function: Application Assessment Process
Require: Submitted applicant documents
Ensure: Proper evaluation and decision on each application
   1. Start
   2. Assess the application.
   3. Review all submitted documents.
   4. If application is accepted:
        a. Include the application in the PSB master list.
   5. Else if application is rejected:
        a. Reject the application and provide the reason.
   6. Stop
   7. End of Function
```

**Figure 31. Pseudocode for Application Assessment**

As depicted in Figure 31, the pseudocode for the Application Assessment Process illustrates how each applicant's submission is systematically evaluated. The process begins with assessing the application and reviewing all required documents to ensure completeness and accuracy. If the application meets the necessary qualifications, it is accepted and added to the PSB master list. Conversely, if it fails to meet the requirements, the application is rejected with a stated reason. This procedure ensures fairness, consistency, and transparency in the evaluation of all applicants (see corresponding code snippet in Appendix Figure 5.5, p. 154).

## Administrator module for generation of Applicant's Comparative assessment

The admin module for generation of applicant's applicants' comparative assessment report allows the admin to generate the comparative assessment report of the applicant the generated report will be the essential information of the applicant based on their Personal Data Sheet (PDS) and this data will help the admin to evaluate if the applicant is eligible for the position base on the comparative assessment report.

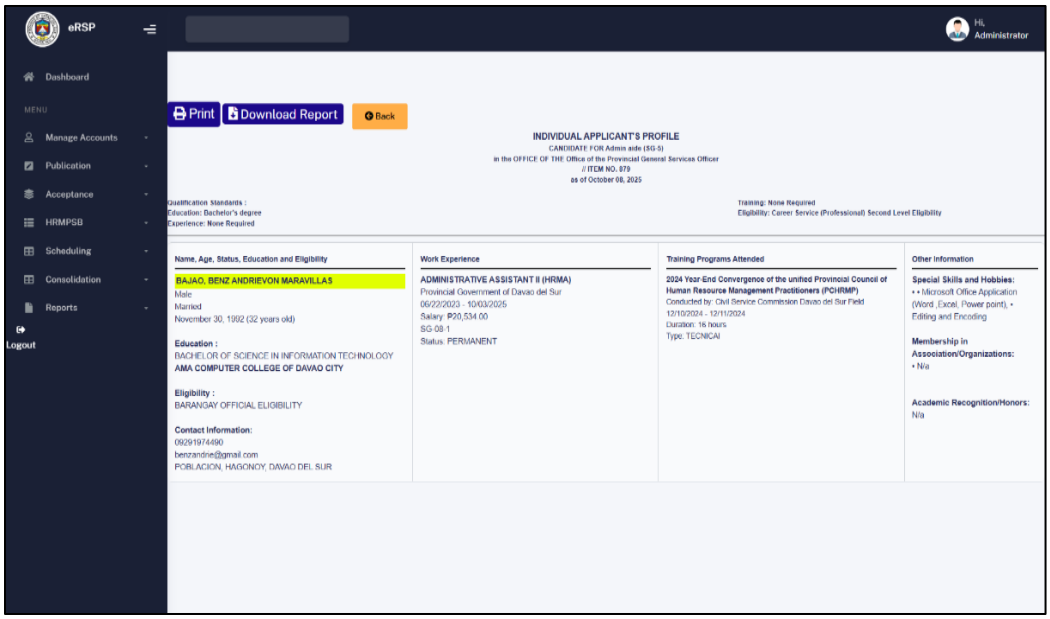



**Figure 32. GUI of Administrator module for generation of Applicant's Comparative assessment**

The figure 32 shows the graphical user interface (GUI) for generation of comparative assessment profile, it holds the information of Applicant their "Personal Information", "Work Experience", "Trainings and Program Attended" and, "Other Information".

```
Function: Comparative Assessment Process
Require: Generated comparative assessment file and applicant
         details
Ensure: Applications are validated and classified accurately
   1. Start
   2. Generate the Comparative Assessment File.
   3. Check the details in the file to determine if they match
      the required criteria.
   4. If details match:
        a. Include the application in the PSB master list.
   5. Else if details do not match:
        a. Reject the application.
   6. Stop
   7. End of Function
```

**Figure 33. Pseudocode for Generation of Comparative**

As depicted in Figure 33, the pseudocode for the Comparative Assessment Process demonstrates how applications are verified and filtered based on their submitted details. The process begins by generating a comparative assessment file, which is then checked to ensure that the applicant's information matches the required criteria. If the details align, the application is accepted and added to the PSB master list. Otherwise, it is rejected. This systematic procedure ensures that only qualified applicants proceed, maintaining accuracy and fairness in the evaluation process (see corresponding code snippet in Appendix Figure 5.6, p. 154).

**Admin module for Sending Push Notification and Email**

The Admin Module for Push Notifications and Email Sending enables administrators to send updates directly to applicants' accounts. These notifications include the status of applications, exam schedules, interview appointments, and final appointment notices, ensuring that applicants remain well-informed throughout the recruitment process.

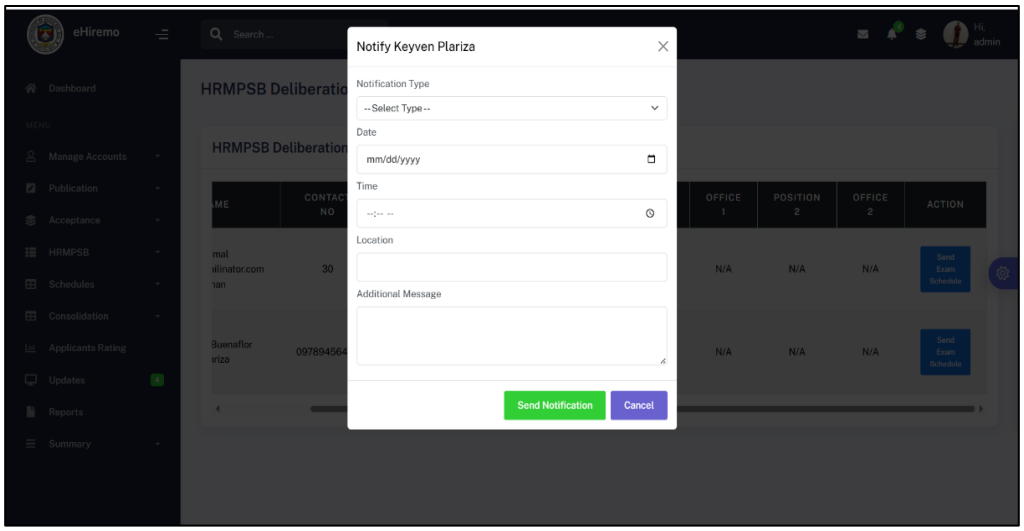



**Figure 34. GUI of Administrator module for sending notifications**

The figure 34 shows the graphical user interface of send notifications the admin will input the necessary details of the notifications such as the notification type, date, time, location, and the specific message depending on what type of notification it is.

```
Function: Notification Sending Process
Require: Notification content and recipient details
Ensure: Notification is successfully created and sent
 1. Start
 2. Create notification details.
 3. Send notification to recipient.
 4. Stop
 5. End of Function
```

**Figure 35. Pseudocode for Sending notifications**

As depicted in the figure 35, the Notification Sending Process outlines how the system creates and delivers notifications to users. By

generating notification details and sending them to the appropriate recipients, the system ensures that users are promptly informed of updates or actions related to their applications, enhancing communication efficiency and user awareness (see corresponding code snippet in Appendix Figure 5.7, p. 155).

**Admin module for Tabulation of Exam Score of the Applicant**

The admin module for tabulation of Exam Score of the Applicant is the module that allow the admin to input the exam score of the applicant after taking the examination. It enables to record and compute applicants' performance in three main components: Written Examination, Computer Proficiency, and Trade/Skill Test. For each component, the administrator inputs the total number of items and the applicant's score obtained. The computation of exam score will be sixty percent of the number of exam item (.6 x Exam item). And if the applicant takes the other exam category which is the "Computer literacy" and "Trade and Skills" the sixty percent of that will be added and get the average exam score.

The figure 36, below shows the Graphical User Interface (GUI) for the Administrator module for tabulation of Examination Score, there are three category which is the Written exam split into two the aptitude and

technical test, other category is the computer literacy, and the Trade and Skill. And the GUI has also a real time calculation and indicator whether it pass the sixty percent or failed.

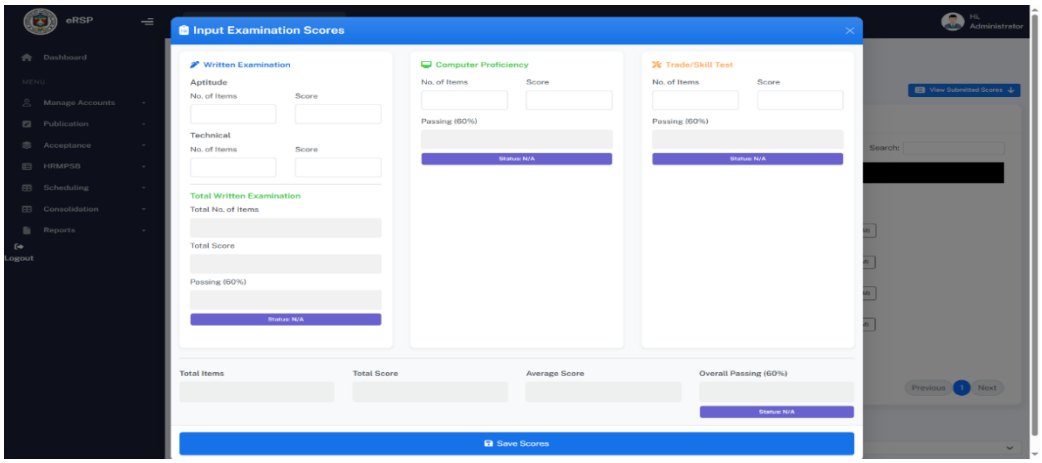



**Figure 36. GUI for the Administrator module for tabulation of Examination Score**

```
Function: Examination Tabulation Process
Require: Applicant's examination score
Ensure: Determine if applicant passes and proceeds to
interview
   1. Start
   2. Input the examination score.
   3. Retrieve total examination items.
   4. Compute 60% of the total examination score as the passing
      mark.
   5. Compare the applicant's score to the 60% passing mark.
   6. If the applicant's score is greater than or equal to 60%:
        a. Mark as "Passed".
        b. Notify the applicant of the result.
        c. Proceed the applicant to the interview stage.
   7. Else:
        a. Mark as "Failed".
        b. Notify the applicant of the result.
   8. Stop
   9. End of Function
```

**Figure 37. Pseudocode for Exam score tabulation**

As shown in the figure, the examination evaluation process allows the system to assess whether an applicant passes based on a 60% cutoff score. If the applicant's score meets or exceeds 60%, they are notified of passing and proceed to the interview stage; otherwise, they are notified of failure. This ensures a fair and consistent evaluation of applicants' examination performance (see corresponding code snippet in Appendix Figure 5.8, p. 155).

### Admin module for Tabulation of Interview Score of the Applicant

The admin module for tabulation of Interview Score of the Applicant enables the admin to input the rating of the applicant in the five categories. And it will be store in the system to use in different module.

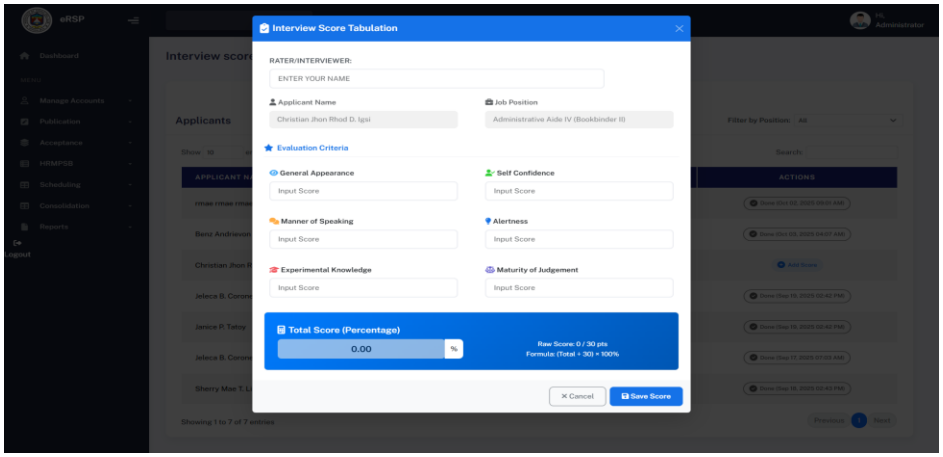



**Figure 38. GUI for Interview Score Tabulation**

The figure 38 shows the graphical user interface of the Interview Score Tabulation, the Interface consist of the six field for entering the score in specific category with the scale of 5, 4, 3, 2, and 1. These categories are the "General appearance", "Manner of speaking", "Experimental knowledge", "Self-confidence", "Alertness", "Maturity of judgement" there is also real time calculation of the average score in the interface and save to store in the system.

```
Function: Interview Rating Process
Require: Interviewer's name and applicant's rating on 6
criteria
Ensure: Applicant's total and percentage score are accurately
recorded
   1. Start
   2. Input interviewer's name.
   3. Rate applicant on 6 criteria (1–5 each).
   4. Compute total raw score (out of 30).
   5. Convert total score to percentage: (Total ÷ 30) × 100.
   6. Record and save the final score.
      Stop
```

**Figure 39. Pseudocode for Interview Score Tabulation**

This process allows the interviewer to evaluate an applicant based on six performance criteria, each rated from 1 to 5. The total score is calculated, converted into a percentage, and then saved to keep a standardized record of the applicant's interview results (see corresponding code snippet in Appendix Figure 5.9, p. 156).

## Admin module for File and Report Generation

The admin module for File and Report Generation allows the admin to generate the file and report for the Comparative, HRMPSD list, Accession report, Assumption, Appointment form, and Oath of office. This module is the most important for PHRMO to have because compare to manual operation the system generated file and report is more efficient.

The figure 40 shows the GUI for file and report generation, in the page there are list of applicants with their name, job, salary grade, and action row consist four buttons that will execute the File and Report Generation. The following button also generate its exact file and report are the Accession report, Assumption, Appointment form, and Oath of office.

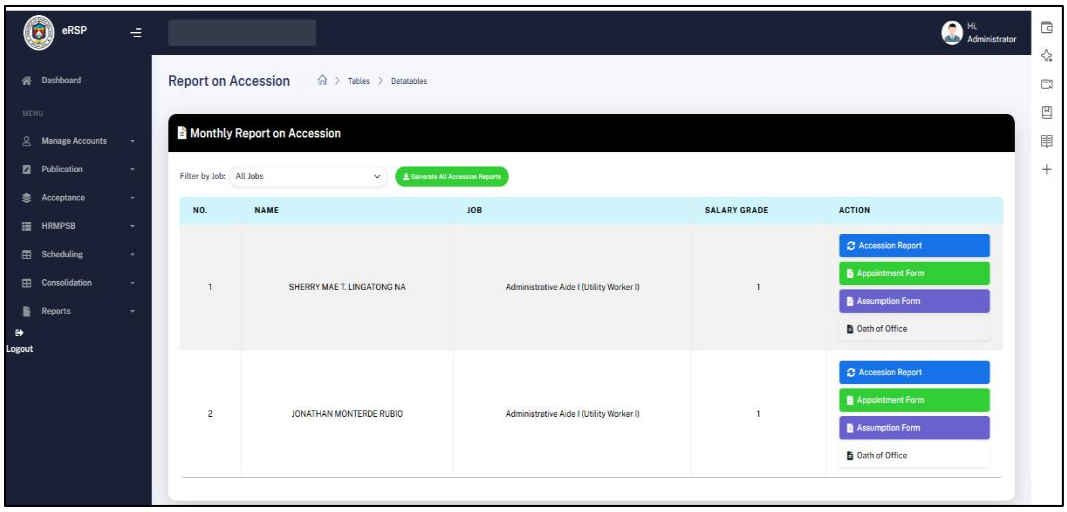



**Figure 40. GUI for File and Report Generation**

```
Function: File and Report Generation
Require: Applicant application
Ensure: Applicant have an application record
   1. Start
   2. Select an Applicant
   3. Choose specific files or document to generate
   4. Ensure all data and information is correct
   6. Stop
```

**Figure 41. Pseudocode for File and Report Generation**

The pseudocode for File and Report Generation outlines the process of producing applicant-related documents within the system. It begins by selecting a specific applicant who has an existing application record. The user then chooses the particular files or reports to generate, such as personal data sheets, certificates, or evaluation summaries. Before generating the documents, the system ensures that all relevant data and information are complete and accurate. Once verified, the process concludes, resulting in properly compiled files or reports ready for use (see corresponding code snippet in Appendix Figure 5.10, p. 156).

### Applicant Job Portal for Job Vacancy Listing

Applicant job portal for job vacancy listing allows the applicant to view the available job and position that are open for application. Applicant

can view the important details for each job listed they can download the job details file in the view details button in the action row.

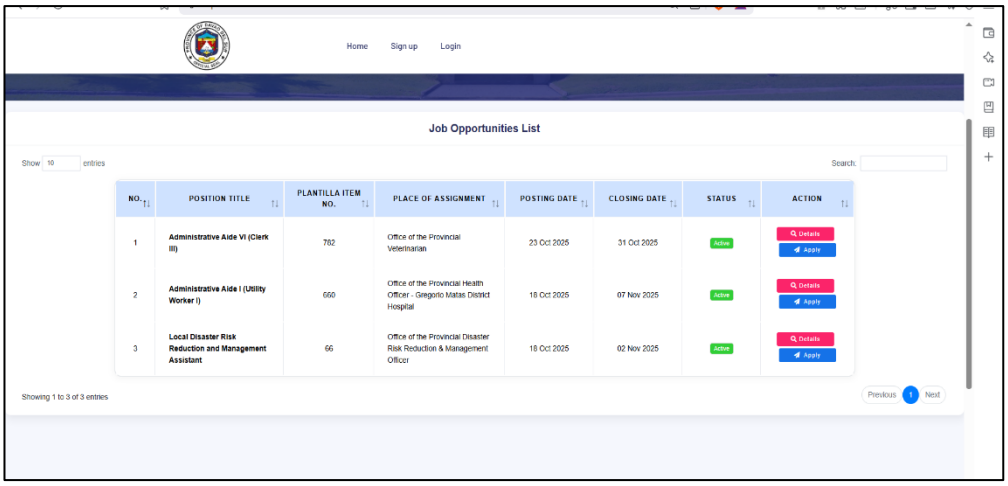



**Figure 42. Job Portal for Job Vacancy Listing**

As shown in the Figure 42, the job portal for job listing where the PHRMO can publish new job vacancy with the complete details about the position. As shown in the figure, a table displays all available job listings**,** including details such as Position Title, Plantilla Number, Posting and Closing Dates, Status**,** and an Action column with two buttons. To view more information about a specific job or position, applicants can click the "Details" button, which generates a PDF file containing all relevant information. The system is accessible across devices, allowing applicants to use a smartphone, laptop, or desktop computer by simply entering the

website's domain name in a web browser. The applicant can apply the specific job with the button in the action column to start the process of their application.

```
Function: Job Portal for Job Vacancy Listing
Require: Complete Job details
Ensure: Job details are accurate
   1. Start
   2. Add new Job listing
   3. Enter all details of the job
   4. Sava and Publish the Job
   5. Stop
```

**Figure 43. Pseudocode for Job Portal for Job Vacancy Listing**

The figure 43, is the Pseudocode for the Job Vacancy Listing, the process starts with the user decide to add new Job vacancy this will trigger the modal to appear to enter all the Job details and when the Job details are enough it will be saved and will Publish in the website of PHRMO (see corresponding code snippet in Appendix Figure 5.12, p. 157).

### Applicant module for Registration of Account

Applicant module for registration of account will allow the applicant to register account in the system the account is used for the application process and submission of application.

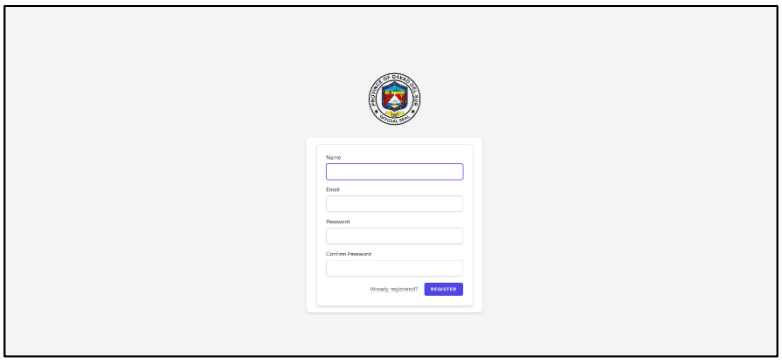


**Figure 44. GUI for Registration Account**

As shown in the Figure 44, it is where the applicant registers their account in the system if they are first time applying for a job applicant will enter their name, email, and password. The user cannot apply if they don't have an account registered in the system because the account is used for processing their application.

```
Function: Registration of Account
Require: User credential
Ensure: User credential is correct
   1. Start
   2. Sign up
   3. Enter all the account credential
   4. Save and Register account
   6. Stop
```

**Figure 45. Pseudocode for Registration of Account**

The Figure 45, is the process of registering an account in the system the user credentials must be unique and not duplicated the system will

prohibit the duplicate email credentials it must original and not used already in the system. After the user already register their account, they can proceed to application (see corresponding code snippet in Appendix Figure 5.11, p. 157).

## Applicant module for Submission of Application and Documents

The Applicant module for submission of application and documents will allow the applicant submit their application and document online and they first fill up the personal data sheet entry and upload the required document so that the application will be completed.

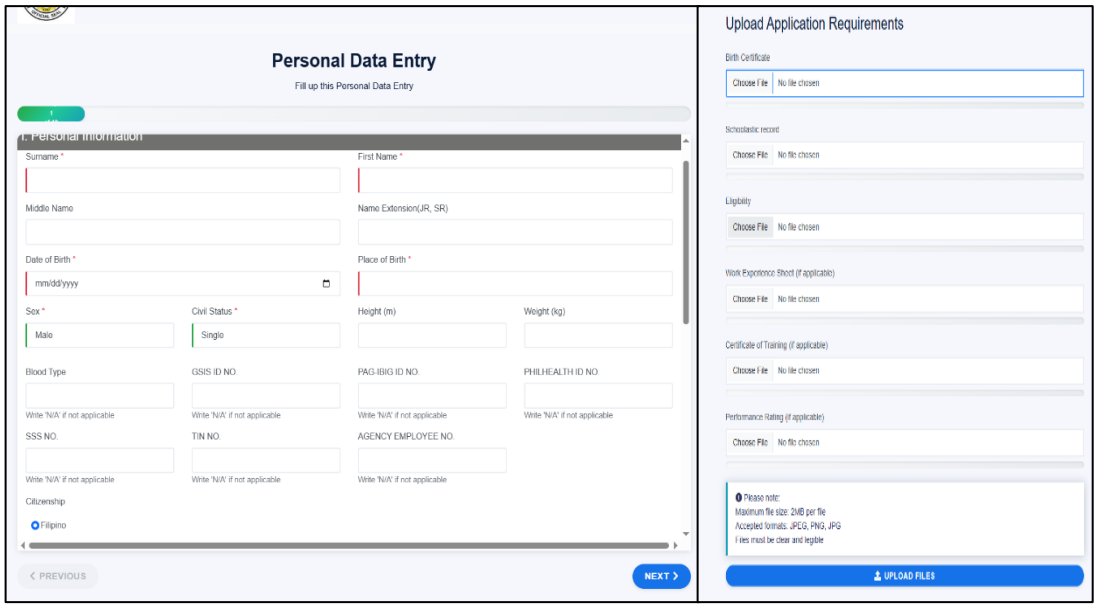



**Figure 46. GUI for Submission of Applicant's Application and Documents**

As depicted in figure 46, a user interface within a Recruitment and Selection Management System for applicant was presented. The interface allows the applicant to fill up the fields in the digital Personal Data Sheet which is one requirement for the application process. The pds entry page was compose of 10 section the user can navigate the section through button next and previous but they cannot proceed to the next section if they did not fill the required fields. The applicant will fill up the PDS entry only once so if they will submit another application for the other position, they will only require to upload the necessary documents according to the job qualification details that can be seen in the Job portal if they will click the see details of that Job Listed in the Website.

```
Function: Applicant Submission Process
Require: Applicant access (New or Existing)
Ensure:
   1. Ask if applicant is new.
   2. If New Applicant:
        a. Fill up PDS entry.
        b. Upload required documents.
        c. Submit specific requirements based on job
      requirements.
        d. Submit application.
   3. If Existing Applicant:
        a. Submit specific requirements.
        b. Submit specific requirements based on job
      requirements.
        c. Submit application.
   4. End of Function.
```

**Figure 47. Pseudocode for Application Submission**

As depicted in Figure 47, the pseudocode for the Applicant Submission Process describes how both new and existing applicants can submit their job applications through the system. New applicants are first required to complete their Personal Data Sheet (PDS), upload necessary documents, and provide specific requirements before submitting their application. Existing applicants, on the other hand, can directly submit the required documents and job-specific requirements. This process ensures that all applicants follow a structured and organized submission workflow, allowing the system to efficiently manage and process applications (see corresponding code snippet in Appendix Figure 5.14, p. 158).

**Applicant module for Tracking Application Status**

Applicant module for tracking application status will allow the applicant to track their application in real time they can view the application status, date and time of the changes of the application status it will display the real time update of the application progress.

The Figure 48 below show the GUI to let the Applicant easily identified the application they submitted and to be aware what is progress

of it. It will display the real-time status and update of the stage of application ang the progress of the application.

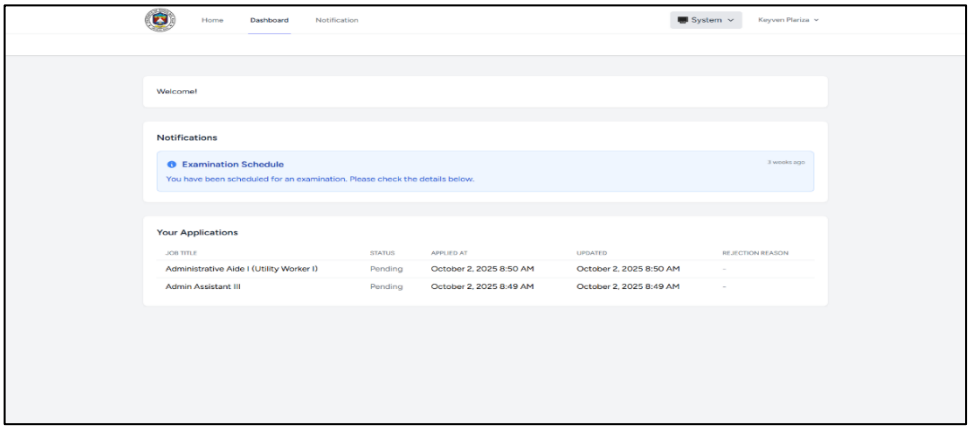



**Figure 48. GUI for viewing and tracking of application status**

```
Function: Application Tracking Process
Require: Applicant account access
Ensure: Applicant can track application progress
   1. Start the application tracking process.
   2. Applicant navigates to their dashboard.
   3. System displays the application tracking interface.
   4. Applicant views and tracks their current application
      status.
   5. End of Function.
```

**Figure 49. Pseudocode for Application Tracking**

As depicted in Figure 49, the pseudocode for the Application Tracking Process outlines how applicants can conveniently monitor the progress of their job applications. By navigating to their dashboard, applicants can access the application tracking interface, where the system

displays real-time updates on their application status. This process ensures transparency and allows applicants to stay informed about their progress without the need for direct administrative assistance.

## Applicant module for Viewing of Notifications

The Applicant module for viewing of notifications enable the applicant to view the notification through their email and dashboard, these notifications serve as an update to their application, schedule, and status which is very important to communicate each end user in the system during the recruitment and selection process.

**Figure 50** illustrates the graphical user interface for viewing notifications**.** The interface displays the notification content along with the time it was received**.** When a user clicks on a highlighted notification, the system displays the full details of that notification for easy reference. The system has two options for viewing notifications and receiving notification the push notification and the email notification which is more convenient and efficient for the user to always keep on updated regarding to its application status, appointment schedules and exam scores.

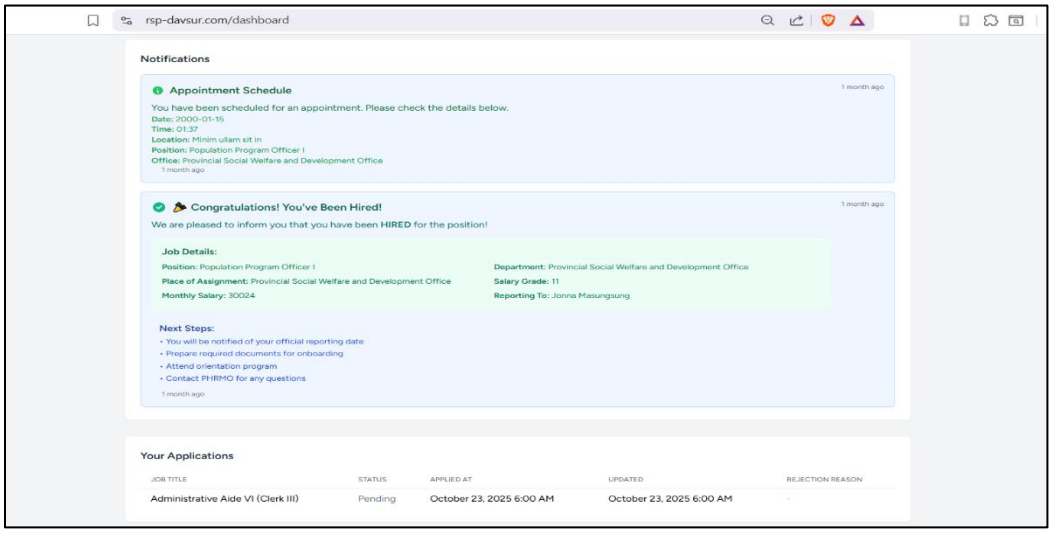



**Figure 50. GUI for Notifications Viewing**

**Function:** Notification Viewing

**Require:** Applicant application and access
**Ensure:** Applicant will be updated and informed

1. **Start**
2. Navigate to the Applicant Account dashboard
3. Check all push notification
4. Check in applicant email
5. Read the Notifications
6. **Stop**

**Figure 51. Pseudocode for User Management**

The pseudocode explains the process of managing user accounts in a system. It starts by going to the User Management section and checking all registered users. If a user is verified, no action is needed. However, if a user is not verified, the system will remove and restrict that account to maintain security and ensure that only verified users have access. The

process ends after all users are checked (see Appendix Figure 5.15, p. 158, for the appropriate code snippet).

## Evaluation Ratings from Respondents in terms of Functional Suitability

Functional suitability corresponds to the way well the features of the system satisfy the expectations of the users and whether it is capable of carrying out essential functions, such as maintaining user accounts, job publication, application submission, application assessment, consolidation of exam and interview scores, setting of schedules, and generates reports and documents. The system which taken the task of the office staff intervention would contributes to the enhancement of Provincial Human Resource Management Office run smoothly and efficiently.

The evaluation of the super admin account's functional fitness is shown in Table 6 below, which shows that the responder thought the system improved the process and was functionally appropriate. Viewing candidate ranks and choosing an applicant for an appointment had the highest mean (4.55), indicating that these aspects had the biggest influence on the system's efficacy. In the meanwhile, activity logs with the lowest mean (4.45) and updating application status require improvement.

**Table 6. Evaluation ratings from the respondents in terms of Functional Suitability (Super Admin Account). October 2025**

| Particulars | Respondents | | Mean | Verbal Interpretation |
|---|---|---|---|---|
| | PHRMO | Advisory Committee | | |
| View the ranking of applicants for a specific position with their ratings. | 4.6 | 4.5 | 4.55 | STRONGLY AGREE |
| Select an Applicant for the appointment of the position; | 4.6 | 4.5 | 4.55 | STRONGLY AGREE |
| View activity logs such as: Posting, updating, or closing of vacant positions**;** | 4.4 | 4.5 | 4.45 | AGREE |
| Updating applicant application status; | 4.4 | 4.5 | 4.45 | AGREE |
| Selection of applicant for appointment; | 4.4 | 4.5 | 4.45 | AGREE |
| **Consolidated Mean** | | | **4.49** | **AGREE** |

The result of the study is consistent with previous research indicating that the use of automated recruitment technologies contributes to greater efficiency and productivity within human resource functions (Kharytonova et al., 2021; Dalsaniya & Patel, 2022).

**Table 7. Evaluation ratings from the respondents in terms of Functional Suitability (Admin Account). October 2025**

| Particulars | Respondents | | Mean | Verbal Interpretation |
|---|---|---|---|---|
| | PHRMO | Advisory Committee | | |
| Post vacant position; | 4.8 | 4.5 | 4.65 | STRONGLY AGREE |
| Assessment and Evaluation of Applications; | 4.4 | 4.5 | 4.45 | AGREE |
| Send Push notification and email for Application Status; | 4.6 | 4.25 | 4.43 | AGREE |
| Send Push notification and email for Schedule for the examination, Panel interview and Appointment; | 4.6 | 4.5 | 4.55 | STRONGLY AGREE |
| Tabulation of the applicant's exam score and interview score; | 4.6 | 4.25 | 4.43 | AGREE |
| Generate file and report for Comparative Assessment Report of Applicants; | 4.6 | 4 | 4.3 | AGREE |
| Generate file and report for HRMPSB list of examinees for examination; | 4.8 | 4.25 | 4.53 | STRONGLY AGREE |
| Generate file and report for Accession report; | 4.8 | 4.25 | 4.53 | STRONGLY AGREE |
| **Consolidated Mean** | | | **4.48** | **AGREE** |

The system's functional adequacy for the Admin Account is evaluated in Table 7, with a consolidated mean of 4.48 (Agree). This suggests that respondents thought the system supported administrative recruitment tasks in a functionally effective manner. The advertising of open positions had the highest mean score (4.65), closely followed by sending alerts for exams, interviews, and appointment schedules (4.55), suggesting that these aspects have the biggest influence on the system's efficiency and usefulness. On the other hand, the lowest mean score (4.30) for producing comparison evaluation reports indicates a small area where the reporting capability of the system needs to be improved. Overall, the results confirm that the system effectively streamlines the hiring and selection process by facilitating job posting, applicant tracking, and report generation. The results of Mohamed et al. (2022), who highlighted that automated recruitment methods enhance HR operational efficiency, are corroborated by this.

Table 8 below, presents the respondents' evaluation of the system in terms of Functional Suitability (Applicant Account). The system obtained a consolidated mean of 4.42 (Agree), suggesting that users generally found the applicant module efficient and functionally reliable.

**Table 8. Evaluation ratings from the respondents in terms of Functional Suitability (Applicant Account). October 2025**

| Particulars | Respondents | | | Mean | Verbal Interpretation |
|---|---|---|---|---|---|
| | Applicants | PHRMO | Advisory Committee | | |
| View published vacant position in the job portal; | 4.70 | 5.00 | 4.25 | 4.65 | STRONGLY AGREE |
| Registration page for new account; | 4.90 | 4.50 | 4.25 | 4.55 | STRONGLY AGREE |
| Personal Data Sheets entry; | 4.80 | 4.50 | 4.25 | 4.52 | STRONGLY AGREE |
| Scanned copy of School Credentials/ Transcript of Records; | 4.70 | 4.50 | 4.25 | 4.48 | AGREE |
| Scanned copy of certificate of Eligibility/ rating (CS, PRC License, LTO License); | 4.60 | 4.50 | 4.25 | 4.45 | AGREE |
| Scanned Work Experience Sheet | 4.80 | 4.00 | 4.25 | 4.35 | AGREE |
| Scanned Certificate of Training; and | 4.90 | 4.50 | 4.25 | 4.55 | STRONGLY AGREE |
| Scanned Performance Rating. | 4.80 | 5.00 | 4.25 | 4.68 | STRONGLY AGREE |

*Continuation from table 8….*

| | | | | | |
|---|---|---|---|---|---|
| Receive Push Notification and Email for Application status; | 3.90 | 4.00 | 3 | 3.63 | AGREE |
| Receive Push Notification and Email for Schedules of Exam, Interview and Appointment including date, time and venue; | 4.80 | 5.00 | 4.25 | 4.68 | STRONGLY AGREE |
| Receive Push Notification and Email for Result of Examination and Selection. | 4.10 | 4.00 | 4 | 4.03 | AGREE |
| **Consolidated Mean** | | | | **4.42** | **AGREE** |

Viewing publicized open positions, scanning performance ratings, and receiving reminders for exam, interview, and appointment dates had the highest mean (4.68). These results indicate that the elements enabling applicant awareness and engagement had the most significant impact on user happiness. The lowest mean (3.63) for application status notifications, on the other hand, indicates that although the functionality works, users believe there is space for improvement in terms of the visibility or timeliness of status updates.

The system's use of automation and notification capabilities, which expedite applicant interactions and guarantee prompt communication throughout the recruiting process, is responsible for this improvement in user experience. These results are consistent with Thawrani's (2021) study, which highlighted how web-based recruitment platforms enhance candidates' familiarity with the hiring process and the organization's reputation.

With an overall mean of 4.39 ("Agree"), Table 9 below presents the key findings, which indicate that respondents view the system's performance as generally positive. This perception is primarily driven by very high satisfaction with the speed of core tasks like viewing job postings, submitting applications, and generating reports (mean 4.56, "Strongly Agree"). However, the system's effectiveness in processing and retrieving massive amounts of data received the lowest, if still favorable, rating (mean 4.22, "Agree"), indicating that users start to perceive performance constraints under greater loads. These findings suggest that while the system already satisfies user expectations for routine interactions, it needs to be optimized for data-intensive operations to prevent emerging delays from undermining satisfaction. This is important because research

consistently demonstrates that slower response times lower user satisfaction and may ultimately discourage continued system use.

**Table 9. Evaluation ratings from the respondents in terms of Performance Efficiency. October 2025**

| Particulars | Respondents | | | Mean | Verbal Interpretation |
|---|---|---|---|---|---|
| | PHRMO | Advisory Committee | Applicant | | |
| The system responds quickly when performing tasks. | 4.67 | 4.5 | 4.50 | 4.56 | STRONGLY AGREE |
| The system Pages and dashboards load within a reasonable time | 4.37 | 4.5 | 4.50 | 4.46 | AGREE |
| The system can handle multiple users performing different actions. | 4.43 | 4 | 4.50 | 4.31 | AGREE |
| The system processes and retrieves data efficiently. | 4.27 | 4 | 4.40 | 4.22 | AGREE |
| **Consolidated Mean** | | | | **4.39** | **AGREE** |

These results imply that the system already meets user expectations for routine interactions but requires optimization for data-intensive operations to prevent emerging delays from eroding satisfaction, which is

crucial because research consistently shows that slower response times reduce user satisfaction and can eventually discourage continued system use.

**Table 10. Evaluation ratings from the respondents in terms of Compatibility. October 2025**

| Particulars | Respondents | | | Mean | Verbal Interpretation |
|---|---|---|---|---|---|
| | PHRMO | Advisory Committee | Applicant | | |
| The system works properly on different devices (desktop, laptop, tablet, mobile). | 4.67 | 4.25 | 4.20 | 4.28 | AGREE |
| The system works consistently across different web browsers. | 4.47 | 4.5 | 4.40 | 4.46 | AGREE |
| The system can access and use the system without encountering compatibility issues. | 4.50 | 4.5 | 4.50 | 4.50 | AGREE |
| The system integrates well with other software or tools when needed. | 4.70 | 4.25 | 4.50 | 4.48 | AGREE |
| **Consolidated Mean** | | | | **4.43** | **AGREE** |

The table indicates that users across PHRMO, Advisory Committee, and applicant groups generally perceive the system as technically dependable and accessible, with a consolidated mean of 4.43 interpreted as "Agree," reflecting high satisfaction with its ability to run on different devices, across browsers, and without major compatibility problems. The system's ability to be accessed and used without experiencing problems with devices or browsers (mean 4.50) and its seamless integration with other software or tools (mean 4.48) received the highest ratings, indicating that it complements users' current workflows rather than posing new technical challenges. Slightly lower, yet still positive, scores for functionality across different devices (mean = 4.28) and consistency among browsers (mean = 4.46) hint that small layout or presentation difficulties may occasionally exist, particularly on certain devices. Nevertheless, overall user happiness is not greatly impacted by these problems. These findings are in line with studies showing that system quality elements—such as accessibility, technical dependability, compatibility, and integration with other tools—are important factors in determining how satisfied users are with information systems because they reduce friction, facilitate ongoing use, and allow for smooth integration into regular work procedures. The table 11 below displays the results reveal

that respondents typically assess the system's usability positively, with a consolidated mean of 4.41 ("Agree"), implying that the interface assists users in accomplishing tasks without excessive effort.

**Table 11. Evaluation ratings from the respondents in terms of Usability. October 2025**

| Particulars | Respondents | | | Mean | Verbal Interpretation |
|---|---|---|---|---|---|
| | PHRMO | Advisory Committee | Applicant | | |
| The system interface is easy to understand and navigate | 4.73 | 4.25 | 4.20 | 4.39 | AGREE |
| The system is easy to learn without extensive training. | 4.37 | 4.5 | 4.80 | 4.56 | STRONGLY AGREE |
| The system provides clear instructions, labels, and feedback during operations. | 4.20 | 4.25 | 4.30 | 4.25 | AGREE |
| The system layout and design of the system are visually appealing and enhance usability. | 4.50 | 4 | 4.80 | 4.43 | AGREE |
| **Consolidated Mean** | | | | **4.41** | **AGREE** |

A key component of usability and a well-known predictor of user satisfaction, the assertion that the system is simple to learn without substantial training received the highest grade (mean 4.56, "Strongly Agree"), suggesting that even new users can become adept fast. Users find the interface intuitive, informative, and aesthetically supportive of their work rather than distracting or perplexing, as evidenced by the means above 4.20 ("Agree") for ease of understanding and navigation, clear instructions and feedback, and a visually pleasing arrangement. These trends are in line with previous studies showing that overall satisfaction and sustained use of information systems are highly and favorably correlated with perceived ease of use, clear instructions, and excellent visual design.

With a consolidated mean of 4.37 ("Agree"), which indicates that the system is typically available, error-free, and consistent during use, table 12's data demonstrate that respondents generally view the system as stable and dependable. System availability and error-free performance receive the highest ratings (both with overall means of 4.46), indicating that users seldom encounter downtime, crashes, or significant disruptions. This is important because availability and reliability are essential components of system quality that have a significant impact on user

satisfaction and trust. Positive ratings for fast recovery without data loss (mean 4.24) and reliable performance under various usage scenarios (mean 4.33) further suggest that, even when issues arise, they are usually fixed without having a significant detrimental impact on users' work.

**Table 12. Evaluation ratings from the respondents in terms of Reliability. October 2025**

| Particulars | Respondents | | | Mean | Verbal Interpretation |
|---|---|---|---|---|---|
| | PHRMO | Advisory Committee | Applicant | | |
| The system is available whenever I need to use it. | 4.53 | 4.25 | 4.60 | 4.46 | AGREE |
| The system performs tasks without unexpected errors or crashes. | 4.27 | 4.5 | 4.60 | 4.46 | AGREE |
| The system recovers quickly without losing data when a problem occurs effectively. | 4.33 | 4 | 4.40 | 4.24 | AGREE |
| The system performs consistently under different usage conditions. | 4.43 | 4.25 | 4.30 | 4.33 | AGREE |
| **Consolidated Mean** | | | | **4.37** | **AGREE** |

Because they lower the danger of disruption, safeguard data integrity, and bolster users' trust in the platform, system dependability, stability, and service continuity are important factors that determine user happiness and continuous use of information systems, according to current research.

With a consolidated mean of 4.30 ("Agree"), which indicates confidence that user accounts and data are properly protected, the results in Table 13 below demonstrate that respondents generally view the system's security as sufficient and reliable. Because perceived security has a significant impact on trust and willingness to use digital systems, ratings for protection from unauthorized access and secure handling of sensitive data (both overall means 4.38) are among the highest, indicating that users feel their passwords and personal information are protected against misuse. Items pertaining to the accuracy of the login procedure and role-based access controls (mean 4.21) and the prevention of unauthorized actions (mean 4.25) are marginally lower but still positive, suggesting that access control mechanisms are mostly operating as intended with only sporadic or minor problems. This pattern is consistent with studies

demonstrating that efficient authorization, access control, and data protection procedures are essential to information security and are

**Table 13. Evaluation ratings from the respondents in terms of Security. October 2025**

| Particulars | Respondents | | | Mean | Verbal Interpretation |
|---|---|---|---|---|---|
| | PHRMO | Advisory Committee | Applicant | | |
| The system user account and data are protected from unauthorized access. | 4.23 | 4.5 | 4.40 | 4.38 | AGREE |
| The system login process and role-based access controls work correctly. | 4.07 | 4.25 | 4.30 | 4.21 | AGREE |
| The system prevents unauthorized actions, such as uploading harmful files or accessing restricted areas. | 4.30 | 4.25 | 4.20 | 4.25 | AGREE |
| The system Sensitive data (such as passwords and personal information) is handled securely by the system. | 4.40 | 4.25 | 4.60 | 4.38 | AGREE |
| **Consolidated Mean** | | | | **4.30** | **AGREE** |

closely linked to user satisfaction and trust, as users are more likely to adopt and continue using systems they perceive as protecting their data and limiting unauthorized behavior.

**Table 14. Evaluation ratings from the respondents in terms of Maintainability. October 2025**

| Particulars | Respondents | | | Mean | Verbal Interpretation |
|---|---|---|---|---|---|
| | PHRMO | Advisory Committee | Applicant | | |
| The system design appears organized and easy to update or improve. | 4.23 | 4 | 4.40 | 4.21 | AGREE |
| The system can be fixed or updated without affecting other features. | 4.23 | 4 | 4.60 | 4.28 | AGREE |
| The system works properly after updates or modifications. | 4.13 | 4.25 | 4.50 | 4.29 | AGREE |
| The system structure makes it easy to diagnose and resolve issues. | 4.37 | 4.25 | 4.30 | 4.31 | AGREE |
| **Consolidated Mean** | | | | **4.27** | **AGREE** |

With a consolidated mean of 4.27 ("Agree"), which indicates confidence that the system can be upgraded, corrected, and debugged without significant disruption, the results indicate that users regard the system as reasonably maintainable and straightforward to enhance. With averages above 4.20, respondents concur that the design is well-organized and simple to update, that updates or repairs typically don't interfere with other features, and that the system typically functions correctly after changes. Additionally, the item on structure making it simple to identify and fix problems (mean 4.31) receives a positive grade, suggesting that error information and underlying architecture are adequate for troubleshooting.

According to software quality standards, a well-structured, modular system that can be altered, tested, and analyzed without unintended side effects is thought to be easier to maintain and more likely to provide stable performance and user satisfaction over time. These findings reflect important aspects of maintainability.

With a consolidated mean of 4.43 ("Agree"), the results in table 15 below reveal that respondents believe the system is very flexible and portable, indicating confidence that it can function well in various locations and be transferred or modified without significant problems. According to

ISO/IEC 25010, portability is defined as the ease of transferring software between hardware, software, or usage environments. High ratings for easy installation or access in various environments and for good adaptation to different configurations or platforms (both overall means around 4.43–4.48) demonstrate that users can reliably use the system across varied setups.

**Table 15. Evaluation ratings from the respondents in terms of Portability. October 2025**

| Particulars | Respondents | | | Mean | Verbal Interpretation |
|---|---|---|---|---|---|
| | PHRMO | Advisory Committee | Applicant | | |
| The system can be easily installed or accessed in different environments. | 4.60 | 4 | 4.70 | 4.43 | AGREE |
| The system adapts well to different configurations or platforms. | 4.60 | 4.25 | 4.60 | 4.48 | AGREE |
| The system could be moved or replaced without major changes or problems. | 4.37 | 4.25 | 4.50 | 4.37 | AGREE |
| The system operates effectively on different | 4.43 | 4 | 4.80 | 4.41 | AGREE |

| | | |
|---|---|---|
| hardware setups. | | |
| **Consolidated Mean** | **4.43** | **AGREE** |

demonstrate that users can dependably operate the system in a variety of configurations, reflecting the ISO/IEC 25010 definition of portability as the ease of software transfer between hardware, software, or usage contexts. Similarly, positive scores for the ability to move or replace the system without major problems and for effective operation on different hardware suggest that the system's architecture supports installability, adaptability, and replaceability, making it easier for the organization to deploy or upgrade the system as needs change. These characteristics are consistent with software quality models that emphasize technology compatibility and portability as crucial aspects of system quality that can lower long-term maintenance and migration costs and improve user satisfaction.

## Overall Ratings from Respondents in terms of Functional Suitability, Performance Efficiency, Maintainability, Compatibility,Security, Reliability, Usability, and Portability

Eight key criteria were used to evaluate the overall performance and efficacy of the Web-Based Recruitment and Selection Management System. Functional appropriateness, performance efficiency, compatibility,

usability, dependability, maintainability, and portability were some of these requirements. These elements were crucial in determining how well the system fulfilled its intended purposes, including its responsiveness and speed, ease of maintenance and updating, compatibility with various platforms and devices, task handling stability, and user-friendliness for various stakeholders. The assessment sought to pinpoint the system's strong points and areas in need of development in order to maximize its overall influence on the PHRMO's recruitment and selection processes.

**Table 16. Summary of User Rating based on ISO 25010 Software Quality Framework. October 2025**

| Particulars | Mean | Descriptive Equivalent |
|---|---|---|
| Functional Suitability | 4.46 | AGREE |
| Performance Efficiency | 4.36 | AGREE |
| Compatibility | 4.46 | AGREE |
| Usability | 4.37 | AGREE |
| Reliability | 4.34 | AGREE |
| Security | 4.28 | AGREE |
| Maintainability | 4.21 | AGREE |
| Portability | 4.34 | AGREE |
| **Consolidated Mean** | **4.35** | **AGREE** |

With an aggregated mean of 4.35 overall ("Agree"), the table demonstrates that respondents rate all eight ISO 25010 quality characteristics favorably, suggesting that users generally believe the system satisfies their needs in terms of functionality, performance, compatibility, usability, reliability, security, maintainability, and portability.

Because these dimensions directly support task completion and seamless interoperability in daily work, the highest mean scores for Functional Suitability and Compatibility (both 4.46) indicate that users believe the system provides the appropriate functions and works well with other systems and environments. This strongly pulls the overall mean upward.

Maintainability, on the other hand, has the lowest mean (4.21), which is still interpreted as "Agree" but is comparatively weaker than the other dimensions. This suggests that users find the system to be a little more difficult to update, modify, or troubleshoot, which somewhat dampens the overall mean and indicates a possible area for long-term improvement. This pattern is in line with research on software quality, which indicates that while lesser maintainability can eventually lead to hidden dangers if left unchecked, robust functional coverage and compatibility are important factors in perceived system success.

Recent studies applying the ISO/IEC 25010 model confirm that functional suitability and compatibility are among the strongest predictors of perceived system quality and user satisfaction, as they directly affect task accomplishment and interoperability in operational settings (Rojas et al., 2025; Gobov & Zuieva, 2025). The system performs effectively, is simple to use, dependable, and adaptable—qualities that recent empirical

evaluations have linked to higher system acceptance and continued use according to the positive ratings for performance efficiency, usability, reliability, security, and portability (Alnanih et al., 2021).

Maintainability, on the other hand, received the lowest mean score (M = 4.21), even though it was still rated as "Agree." This is in line with recent research on software quality, which notes that maintainability is less apparent to end users and is frequently viewed more negatively than externally observable quality attributes (Kumar & Rath, 2023). Overall, the results are consistent with recent ISO/IEC 25010-based research, confirming that while maintainability is still a crucial area for long-term improvement, robust functional coverage and compatibility greatly improve consumers' overall sense of software quality.

# CHAPTER V

# SUMMARY, CONCLUSION, AND RECOMMENDATION

The study is summarized in this chapter, its conclusions, and its ramifications are outlined in this chapter. The conclusion and the significance of the findings in respect to the goals are also presented. Finally, it offers suggestions for future advancements and additional study.

## Summary

In order to automate important parts of the company's hiring procedures, the study set out to create and assess a Web-Based Recruitment and Selection Management System. The system was created to increase administrative efficiency, improve openness in hiring practices, and simplify candidate administration. These goals were effectively supported by the implementation of key features such as the posting of open positions, applicant registration, automatic ranking, online application submission, document uploading, report and file generating, push and email notifications, and activity log monitoring. The system performed well across a number of quality characteristics, including functional suitability, compatibility, usability, reliability, and performance efficiency, all of which

received positive user ratings, according to an evaluation carried out using the ISO/IEC 25010 Software Quality Framework.

The system satisfied the functional and operational criteria of a contemporary recruitment platform while guaranteeing accuracy and fairness in the hiring process, according to findings from the advisory committee, PHRMO administrative staff, and candidates. In order to further maximize system performance and long-term sustainability, feedback also pointed to possible enhancements in maintainability and document handling effectiveness. All things considered, the system that was developed showed that it could facilitate effective and greatly enhance the organization's hiring process.

## Conclusion

The findings of the study lead to the following conclusions can be made:

1. The results indicate that Super Admin account modules was successfully achieved, as all key functionalities received high ratings from respondents. The features for viewing applicant rankings and selecting applicants for appointment obtained mean scores of 4.55, corresponding to a "strongly agree" level on a 5-point Likert scale, showing very strong acceptance of these

functions. Activity log features also performed well, with mean scores of 4.45 and an overall consolidated mean of 4.49 ("agree"), confirming that the page meets the required level of functional suitability for super administrators. However, some Super Admin features were rated only at the "agree" level, indicating that decision-support tools and activity logs can still be strengthened to better support high-stakes appointment decisions.

2. The results indicate that Administrator account objective, was satisfactorily achieved. The different features were consistently rated from Agree to Strongly Agree by both PHRMO and Advisory Committee respondents. The consolidated mean rating of 4.48 (Agree) further affirms the functional suitability of the administrator module, indicating that it effectively supports the work of HR personnel. However, the Administrator account's consolidated "agree" rating suggests that assessment, notification, and reporting functions remain improvable in terms of efficiency, clarity, and depth of information.
3. The results indicate that Applicant account objective, was successfully achieved. With a consolidated mean of 4.42 (Agree), the applicant account page is deemed functionally suitable and

supportive of a more accessible, organized, and transparent application experience. However, several Applicant account features, especially notifications on status and results, received relatively lower "agree" ratings, implying that timeliness, clarity, and overall user experience still need refinement.

4. The findings indicate that the testing and evaluation of the system with regards on the ISO 25010 software quality framework, has been satisfactorily achieved. All eight ISO-based software quality framework obtained mean ratings interpreted as Agree, with a 4.35 consolidated mean. These findings imply that the system generally meets the expected software quality requirements, demonstrating acceptable performance and reliability while remaining usable, secure, and adaptable to future maintenance and deployment needs.

Therefore, the Web-Based Recruitment and Selection Management System successfully addressed the inefficiencies of manual hiring processes by automating applicant evaluation, enhancing transparency, and supporting data-driven recruitment decisions. Based on feedback from PHRMO administrators, the selection committee, and applicants**,** the

system was found to be functionally suitable, compatible, and user-friendly**.** While these results affirm its overall effectiveness, further improvements in document handling, maintainability, and system performance are recommended to ensure its sustainability and scalability for future institutional use.

## Recommendation

Considering the conclusions of the study, the following improvements are proposed:

1. It is recommended that decision-support tools and activity log features be further enhanced by adding more detailed analytics, clearer audit trails, and summarized insights to better support high-stakes appointment and selection decisions.
2. It is recommended to Improvements are recommended for assessment workflows, notification mechanisms, and reporting functions to enhance efficiency, clarity, and the depth of information provided, thereby further supporting HR personnel in managing recruitment and selection processes.
3. It is recommended to improve the notification system should be refined to ensure timely, clear, and transparent updates on application status and

results, with improvements focused on overall usability and user experience.

4. It recommended to have continuous system evaluation and incremental enhancements are recommended, particularly in maintainability and usability, to ensure long-term sustainability, adaptability, and continued compliance with ISO/IEC 25010 software quality standards.

The recommendations above aim to strengthen the functionality, security, and performance of the Web-Based Recruitment and Selection System. Implementing these enhancements will ensure smoother recruitment operations, improve user satisfaction, and promote transparency and efficiency within FACT Provincial Human Resource Management Office.

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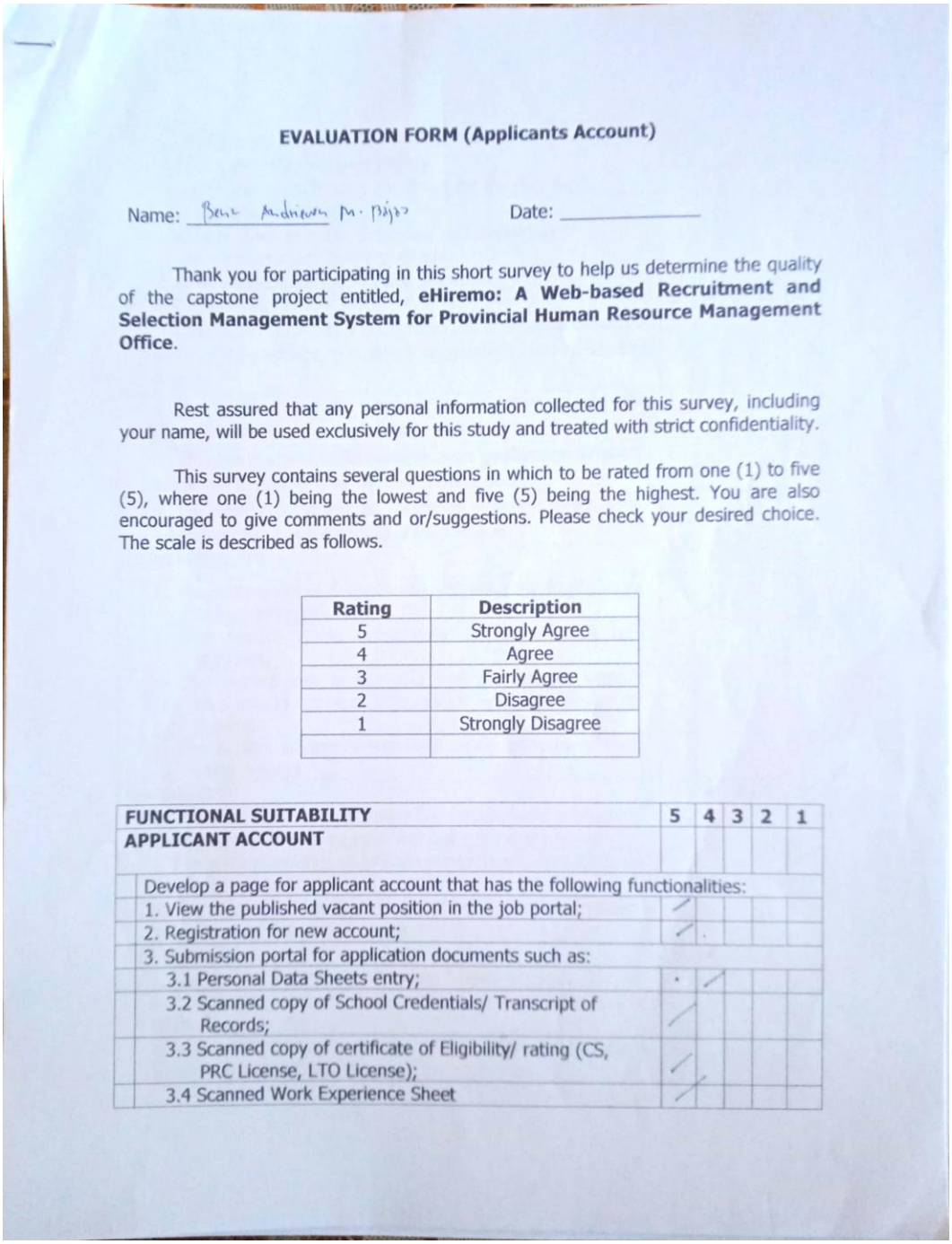
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# APPENDICES

**Appendix 1.** Filled-up Survey Questionnaire



**EVALUATION FORM (Applicants Account)**

Name: Bea Andrew M. Pajes Date: _____

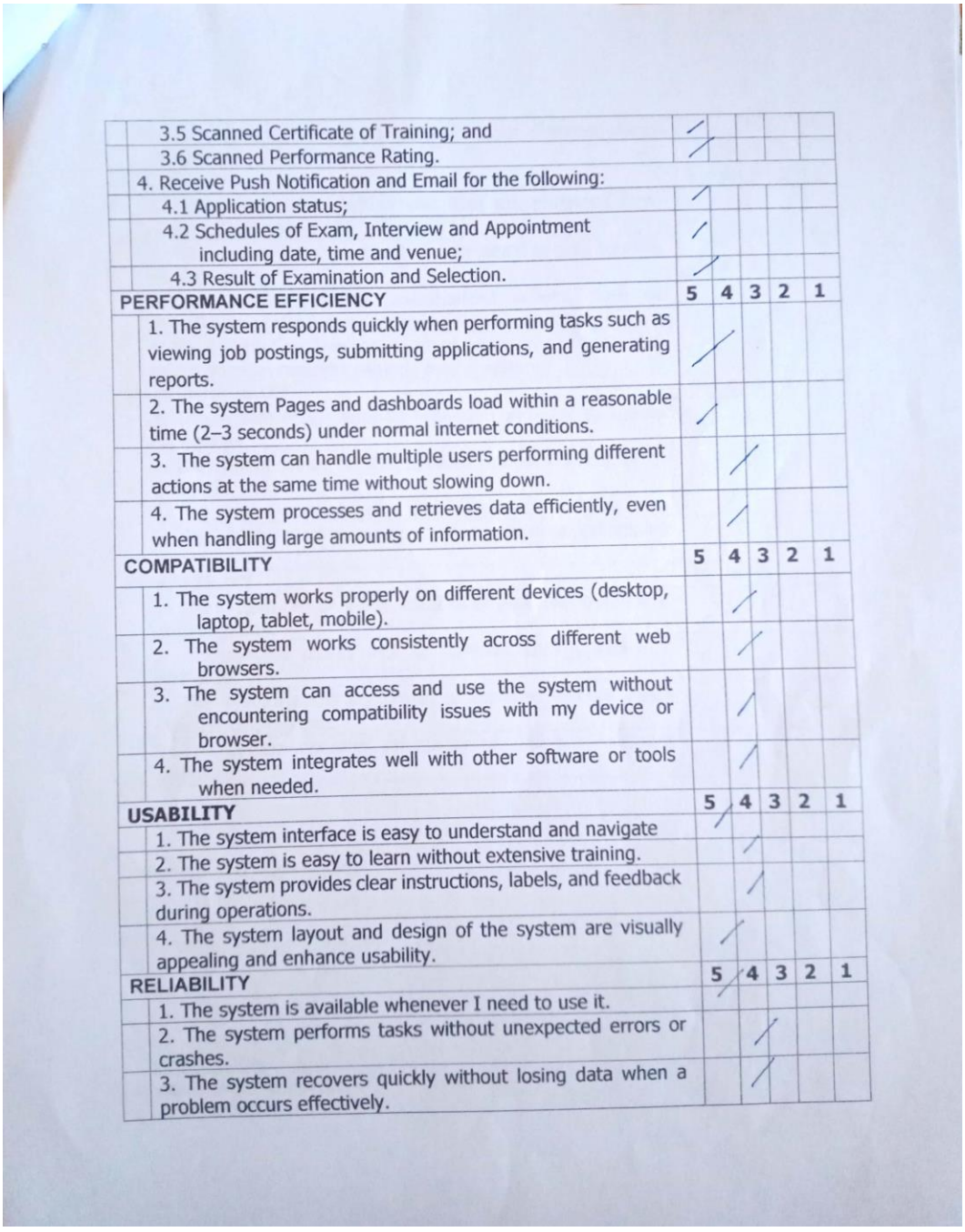
Thank you for participating in this short survey to help us determine the quality of the capstone project entitled, **eHiremo: A Web-based Recruitment and Selection Management System for Provincial Human Resource Management Office**.

Rest assured that any personal information collected for this survey, including your name, will be used exclusively for this study and treated with strict confidentiality.

This survey contains several questions in which to be rated from one (1) to five (5), where one (1) being the lowest and five (5) being the highest. You are also encouraged to give comments and or/suggestions. Please check your desired choice. The scale is described as follows.

| Rating | Description |
|---|---|
| 5 | Strongly Agree |
| 4 | Agree |
| 3 | Fairly Agree |
| 2 | Disagree |
| 1 | Strongly Disagree |
| | |

| FUNCTIONAL SUITABILITY | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|
| **APPLICANT ACCOUNT** | | | | | |
| Develop a page for applicant account that has the following functionalities: | | | | | |
| 1. View the published vacant position in the job portal; | ✓ | | | | |
| 2. Registration for new account; | ✓ | | | | |
| 3. Submission portal for application documents such as: | | | | | |
| 3.1 Personal Data Sheets entry; | | ✓ | | | |
| 3.2 Scanned copy of School Credentials/ Transcript of Records; | ✓ | | | | |
| 3.3 Scanned copy of certificate of Eligibility/ rating (CS, PRC License, LTO License); | ✓ | | | | |
| 3.4 Scanned Work Experience Sheet | ✓ | | | | |



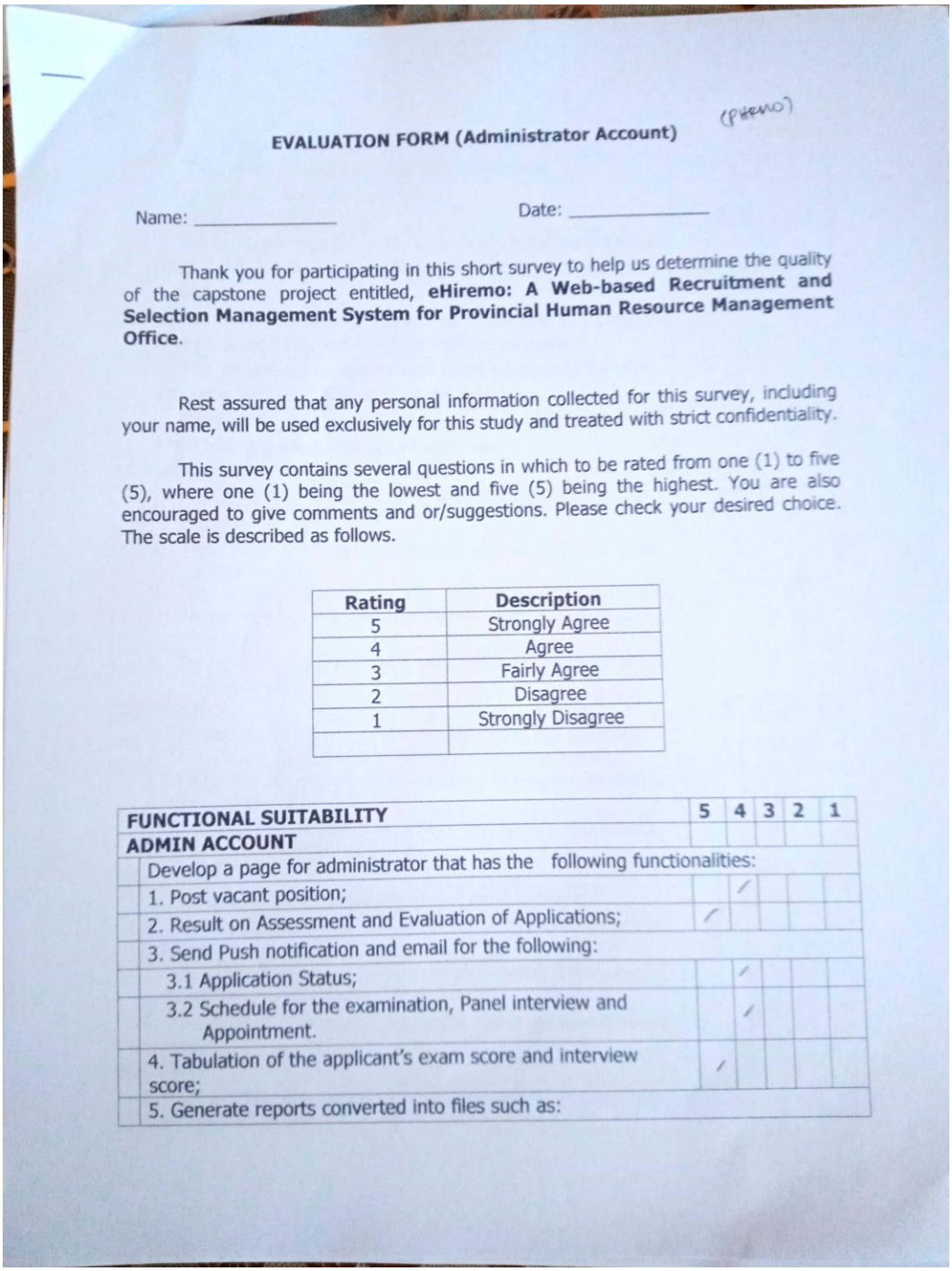
| | | | | | | |
|---|---|---|---|---|---|---|
| | 3.5 Scanned Certificate of Training; and | ✓ | | | | |
| | 3.6 Scanned Performance Rating. | ✓ | | | | |
| | 4. Receive Push Notification and Email for the following: | | | | | |
| | 4.1 Application status; | ✓ | | | | |
| | 4.2 Schedules of Exam, Interview and Appointment including date, time and venue; | ✓ | | | | |
| | 4.3 Result of Examination and Selection. | ✓ | | | | |
| **PERFORMANCE EFFICIENCY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system responds quickly when performing tasks such as viewing job postings, submitting applications, and generating reports. | | ✓ | | | |
| | 2. The system Pages and dashboards load within a reasonable time (2–3 seconds) under normal internet conditions. | ✓ | | | | |
| | 3. The system can handle multiple users performing different actions at the same time without slowing down. | | | ✓ | | |
| | 4. The system processes and retrieves data efficiently, even when handling large amounts of information. | | | ✓ | | |
| **COMPATIBILITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system works properly on different devices (desktop, laptop, tablet, mobile). | | | ✓ | | |
| | 2. The system works consistently across different web browsers. | | | ✓ | | |
| | 3. The system can access and use the system without encountering compatibility issues with my device or browser. | | | ✓ | | |
| | 4. The system integrates well with other software or tools when needed. | | | ✓ | | |
| **USABILITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system interface is easy to understand and navigate | ✓ | | | | |
| | 2. The system is easy to learn without extensive training. | | ✓ | | | |
| | 3. The system provides clear instructions, labels, and feedback during operations. | | | ✓ | | |
| | 4. The system layout and design of the system are visually appealing and enhance usability. | ✓ | | | | |
| **RELIABILITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system is available whenever I need to use it. | ✓ | | | | |
| | 2. The system performs tasks without unexpected errors or crashes. | | | ✓ | | |
| | 3. The system recovers quickly without losing data when a problem occurs effectively. | | | ✓ | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| | 4. The system performs consistently under different usage conditions. | | ✓ | | | |
| **SECURITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system user account and data are protected from unauthorized access. | ✓ | | | | |
| | 2. The system login process and role-based access controls work correctly. | | ✓ | | | |
| | 3. The system prevents unauthorized actions, such as uploading harmful files or accessing restricted areas. | ✓ | | | | |
| | 4. The system Sensitive data (such as passwords and personal information) is handled securely by the system. | ✓ | | | | |
| **MAINTAINABILITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system design appears organized and easy to update or improve. | ✓ | | | | |
| | 2. The system can be fixed or updated without affecting other features. | ✓ | | | | |
| | 3. The system works properly after updates or modifications. | | ✓ | | | |
| | 4. The system structure makes it easy to diagnose and resolve issues. | ✓ | | | | |
| **PORTABILITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system can be easily installed or accessed in different environments. | ✓ | | | | |
| | 2. The system adapts well to different configurations or platforms. | ✓ | | | | |
| | 3. The system could be moved or replaced without major changes or problems. | ✓ | | | | |
| | 4. The system operates effectively on different hardware setups. | ✓ | | | | |

Comments and Suggestions:

Bene Austin Jon M. Bajo

Signature over printed name



**EVALUATION FORM (Administrator Account)** (pitemo)

Name: _____ Date: _____

Thank you for participating in this short survey to help us determine the quality of the capstone project entitled, **eHiremo: A Web-based Recruitment and Selection Management System for Provincial Human Resource Management Office**.

Rest assured that any personal information collected for this survey, including your name, will be used exclusively for this study and treated with strict confidentiality.

This survey contains several questions in which to be rated from one (1) to five (5), where one (1) being the lowest and five (5) being the highest. You are also encouraged to give comments and or/suggestions. Please check your desired choice. The scale is described as follows.

| Rating | Description |
|---|---|
| 5 | Strongly Agree |
| 4 | Agree |
| 3 | Fairly Agree |
| 2 | Disagree |
| 1 | Strongly Disagree |
| | |

| FUNCTIONAL SUITABILITY | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|
| **ADMIN ACCOUNT** | | | | | |
| Develop a page for administrator that has the following functionalities: | | | | | |
| 1. Post vacant position; | | ✓ | | | |
| 2. Result on Assessment and Evaluation of Applications; | ✓ | | | | |
| 3. Send Push notification and email for the following: | | | | | |
| 3.1 Application Status; | | ✓ | | | |
| 3.2 Schedule for the examination, Panel interview and Appointment. | | ✓ | | | |
| 4. Tabulation of the applicant's exam score and interview score; | ✓ | | | | |
| 5. Generate reports converted into files such as: | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| | 5.1 Comparative Assessment Report of Applicants; | ✓ | | | | |
| | 5.2 HRMPSB list of examinees for examination; | | ✓ | | | |
| | 5.3 Monthly accession report. | | ✓ | | | |
| **PERFORMANCE EFFICIENCY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system responds quickly when performing tasks such as viewing job postings, submitting applications, and generating reports. | ✓ | | | | |
| | 2. The system Pages and dashboards load within a reasonable time (2–3 seconds) under normal internet conditions. | ✓ | | | | |
| | 3. The system can handle multiple users performing different actions at the same time without slowing down. | | ✓ | | | |
| | 4. The system processes and retrieves data efficiently, even when handling large amounts of information. | ✓ | | | | |
| **COMPATIBILITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system works properly on different devices (desktop, laptop, tablet, mobile). | ✓ | | | | |
| | 2. The system works consistently across different web browsers. | ✓ | | | | |
| | 3. The system can access and use the system without encountering compatibility issues with my device or browser. | ✓ | | | | |
| | 4. The system integrates well with other software or tools when needed. | ✓ | | | | |
| **USABILITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system interface is easy to understand and navigate | ✓ | | | | |
| | 2. The system is easy to learn without extensive training. | | ✓ | | | |
| | 3. The system provides clear instructions, labels, and feedback during operations. | ✓ | | | | |
| | 4. The system layout and design of the system are visually appealing and enhance usability. | ✓ | | | | |
| **RELIABILITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system is available whenever I need to use it. | ✓ | | | | |
| | 2. The system performs tasks without unexpected errors or crashes. | | ✓ | | | |
| | 3. The system recovers quickly without losing data when a problem occurs effectively. | ✓ | | | | |
| | 4. The system performs consistently under different usage conditions. | ✓ | | | | |
| **SECURITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system user account and data are protected from unauthorized access. | | ✓ | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| | 2. The system login process and role-based access controls work correctly. | | | ✓ | | |
| | 3. The system prevents unauthorized actions, such as uploading harmful files or accessing restricted areas. | | ✓ | | | |
| | 4. The system Sensitive data (such as passwords and personal information) is handled securely by the system. | | ✓ | | | |
| **MAINTAINABILITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system design appears organized and easy to update or improve. | | ✓ | | | |
| | 2. The system can be fixed or updated without affecting other features. | | ✓ | | | |
| | 3. The system works properly after updates or modifications. | ✓ | | | | |
| | 4. The system structure makes it easy to diagnose and resolve issues. | | ✓ | | | |
| **PORTABILITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system can be easily installed or accessed in different environments. | ✓ | | | | |
| | 2. The system adapts well to different configurations or platforms. | ✓ | | | | |
| | 3. The system could be moved or replaced without major changes or problems. | ✓ | | | | |
| | 4. The system operates effectively on different hardware setups. | ✓ | | | | |

Comments and Suggestions:

(PHREMO)

## EVALUATION FORM (Super Administrator Account)

Name: _____ Date: _____

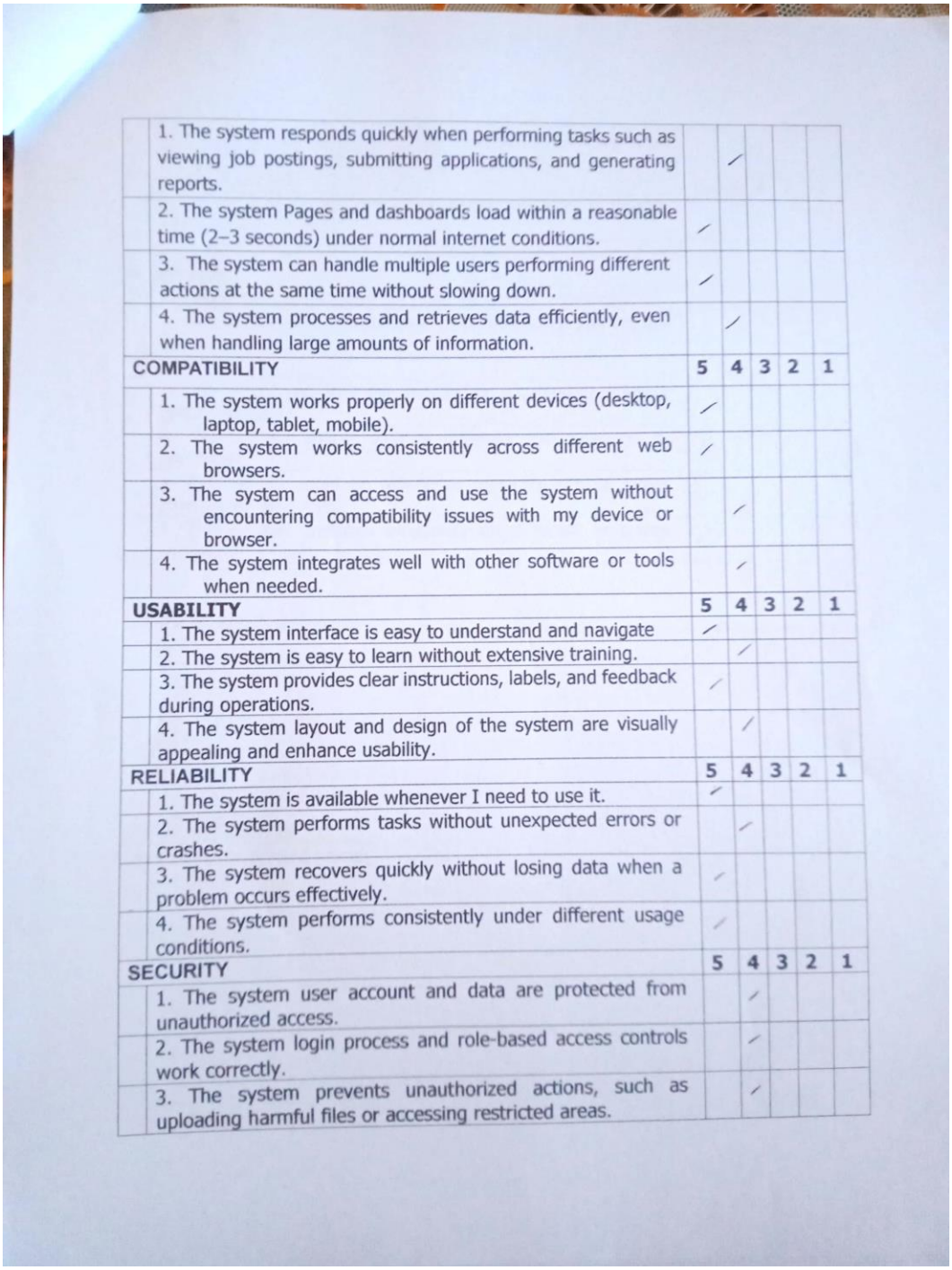
Thank you for participating in this short survey to help us determine the quality of the capstone project entitled, **eHiremo: A Web-based Recruitment and Selection Management System for Provincial Human Resource Management Office**.

Rest assured that any personal information collected for this survey, including your name, will be used exclusively for this study and treated with strict confidentiality.

This survey contains several questions in which to be rated from one (1) to five (5), where one (1) being the lowest and five (5) being the highest. You are also encouraged to give comments and or/suggestions. Please check your desired choice. The scale is described as follows.

| Rating | Description |
|---|---|
| 5 | Strongly Agree |
| 4 | Agree |
| 3 | Fairly Agree |
| 2 | Disagree |
| 1 | Strongly Disagree |
| | |

| FUNCTIONAL SUITABILITY | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|
| **SUPER ADMIN ACCOUNT** | | | | | |
| Develop a page for super administrator that has the following functionalities: | | | | | |
| 1. View the ranking of applicants for a specific position with their ratings. | ✓ | | | | |
| 2. Select an Applicant for the appointment of the position; | ✓ | | | | |
| 3. View activity logs such as: | | | | | |
| 3.1 Posting, updating, or closing of vacant positions; | ✓ | | | | |
| 3.2 Updating applicant application status; | | ✓ | | | |
| 3.3 Selection of applicant for appointment; | ✓ | | | | |
| **PERFORMANCE EFFICIENCY** | **5** | **4** | **3** | **2** | **1** |



| | | | | | | |
|---|---|---|---|---|---|---|
| | 1. The system responds quickly when performing tasks such as viewing job postings, submitting applications, and generating reports. | | ✓ | | | |
| | 2. The system Pages and dashboards load within a reasonable time (2–3 seconds) under normal internet conditions. | ✓ | | | | |
| | 3. The system can handle multiple users performing different actions at the same time without slowing down. | ✓ | | | | |
| | 4. The system processes and retrieves data efficiently, even when handling large amounts of information. | | ✓ | | | |
| **COMPATIBILITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system works properly on different devices (desktop, laptop, tablet, mobile). | ✓ | | | | |
| | 2. The system works consistently across different web browsers. | ✓ | | | | |
| | 3. The system can access and use the system without encountering compatibility issues with my device or browser. | | ✓ | | | |
| | 4. The system integrates well with other software or tools when needed. | | ✓ | | | |
| **USABILITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system interface is easy to understand and navigate | ✓ | | | | |
| | 2. The system is easy to learn without extensive training. | | ✓ | | | |
| | 3. The system provides clear instructions, labels, and feedback during operations. | ✓ | | | | |
| | 4. The system layout and design of the system are visually appealing and enhance usability. | | ✓ | | | |
| **RELIABILITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system is available whenever I need to use it. | ✓ | | | | |
| | 2. The system performs tasks without unexpected errors or crashes. | | ✓ | | | |
| | 3. The system recovers quickly without losing data when a problem occurs effectively. | ✓ | | | | |
| | 4. The system performs consistently under different usage conditions. | ✓ | | | | |
| **SECURITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system user account and data are protected from unauthorized access. | | ✓ | | | |
| | 2. The system login process and role-based access controls work correctly. | | ✓ | | | |
| | 3. The system prevents unauthorized actions, such as uploading harmful files or accessing restricted areas. | | ✓ | | | |

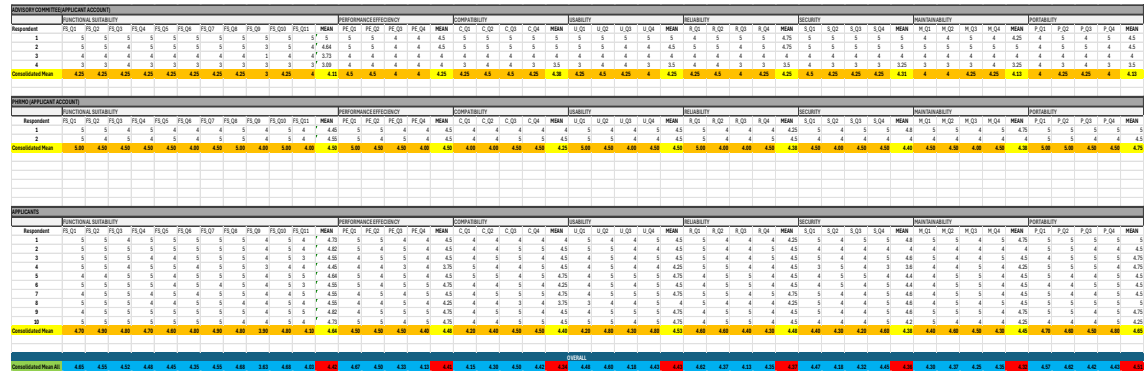
| | | | | | | |
|---|---|---|---|---|---|---|
| | 4. The system Sensitive data (such as passwords and personal information) is handled securely by the system. | | ✓ | | | |
| **MAINTAINABILITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system design appears organized and easy to update or improve. | | ✓ | | | |
| | 2. The system can be fixed or updated without affecting other features. | | ✓ | | | |
| | 3. The system works properly after updates or modifications. | ✓ | | | | |
| | 4. The system structure makes it easy to diagnose and resolve issues. | ✓ | | | | |
| **PORTABILITY** | | **5** | **4** | **3** | **2** | **1** |
| | 1. The system can be easily installed or accessed in different environments. | | ✓ | | | |
| | 2. The system adapts well to different configurations or platforms. | | ✓ | | | |
| | 3. The system could be moved or replaced without major changes or problems. | ✓ | | | | |
| | 4. The system operates effectively on different hardware setups. | ✓ | | | | |

Comments and Suggestions: _____

LEONOR M. ALQUISAR

Signature over printed name

## **Appendix 2.** Raw/Tabulated data



ADVISORY COMMITTEE(ADMIN ACCOUNT)

| | FUNCTIONAL SUITABILITY | | | | | | | | | PERFORMANCE EFFECIENCY | | | | COMPATIBILITY | | | | USABILITY | | | | RELIABILITY | | | | SECURITY | | | | MAINTAINABILITY | | | | PORTABILITY | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Respondent | FS_Q1 | FS_Q2 | FS_Q3 | FS_Q4 | FS_Q5 | FS_Q6 | FS_Q7 | FS_Q8 | MEAN | PE_Q1 | PE_Q2 | PE_Q3 | PE_Q4 | MEAN | C_Q1 | C_Q2 | C_Q3 | C_Q4 | MEAN | U_Q1 | U_Q2 | U_Q3 | U_Q4 | MEAN | R_Q1 | R_Q2 | R_Q3 | R_Q4 | MEAN | S_Q1 | S_Q2 | S_Q3 | S_Q4 | MEAN | M_Q1 | M_Q2 | M_Q3 | M_Q4 | MEAN | P_Q1 | P_Q2 | P_Q3 | P_Q4 | MEAN |
| 1 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 4 | 4.5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 5 | 5 | 5 | 4.75 | 5 | 5 | 5 | 5 | 5 | 4 | 4 | 5 | 4 | 4.25 | 4 | 5 | 4 | 5 | 4.5 |
| 2 | 5 | 5 | 4 | 5 | 5 | 5 | 5 | 5 | 4.75 | 5 | 5 | 4 | 4 | 4.5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 4 | 4.5 | 5 | 5 | 4 | 4.75 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 5 | 5 | 4 | 4.5 |
| 3 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 4 | 4 | 4 | 4 | 4 | 0 | 0 | 0 | 0 | 3.5 | 4 | 4 | 4 | 4 | 4 | 0 | 4 | 4 | 0 | 3.5 | 0 | 4 | 4 | 0 | 3.5 | 4 | 4 | 0 | 0 | 3.5 | 4 | 0 | 0 | 0 | 3.25 | 0 | 0 | 0 | 4 | 3.25 | 4 | 0 | 4 | 0 | 3.5 |
| Consolidated M | 4.5 | 4.5 | 4.25 | 4.5 | 4.25 | 4 | 4.25 | 4.25 | 4.31 | 4.5 | 4.5 | 4 | 4 | 4.25 | 4.25 | 4.5 | 4.5 | 4.25 | 4.38 | 4.25 | 4.5 | 4.25 | 4 | 4.25 | 4.25 | 4.5 | 4 | 4.25 | 4.25 | 4.5 | 4.25 | 4.25 | 4.31 | 4 | 4 | 4.25 | 4.25 | 4.13 | 4 | 4.25 | 4.25 | 4 | 4.12 |

PHRMO(ADMIN ACCOUNT)

| | FUNCTIONAL SUITABILITY | | | | | | | | | PERFORMANCE EFFECIENCY | | | | COMPATIBILITY | | | | USABILITY | | | | RELIABILITY | | | | SECURITY | | | | MAINTAINABILITY | | | | PORTABILITY | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Respondent | FS_Q1 | FS_Q2 | FS_Q3 | FS_Q4 | FS_Q5 | FS_Q6 | FS_Q7 | FS_Q8 | MEAN | PE_Q1 | PE_Q2 | PE_Q3 | PE_Q4 | MEAN | C_Q1 | C_Q2 | C_Q3 | C_Q4 | MEAN | U_Q1 | U_Q2 | U_Q3 | U_Q4 | MEAN | R_Q1 | R_Q2 | R_Q3 | R_Q4 | MEAN | S_Q1 | S_Q2 | S_Q3 | S_Q4 | MEAN | M_Q1 | M_Q2 | M_Q3 | M_Q4 | MEAN | P_Q1 | P_Q2 | P_Q3 | P_Q4 | MEAN |
| 1 | 4 | 5 | 4 | 4 | 5 | 5 | 4 | 4 | 4.38 | 5 | 5 | 4 | 5 | 4.75 | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 5 | 5 | 4.75 | 5 | 5 | 4 | 5 | 4.75 | 4 | 3 | 4 | 4 | 3.75 | 4 | 4 | 5 | 4 | 4.25 | 5 | 5 | 5 | 5 | 5 |
| 2 | 5 | 4 | 5 | 4 | 4 | 5 | 5 | 5 | 4.63 | 5 | 4 | 5 | 5 | 4.75 | 5 | 4 | 5 | 5 | 4.75 | 4 | 4 | 4 | 5 | 4.25 | 5 | 4 | 4 | 4 | 4.25 | 4 | 4 | 3 | 4 | 3.75 | 5 | 4 | 4 | 4 | 4 | 5 | 4 | 4 | 4 | 4.25 |
| 3 | 5 | 5 | 4 | 5 | 5 | 4 | 5 | 5 | 4.75 | 4 | 4 | 4 | 4 | 4 | 5 | 4 | 5 | 5 | 4.5 | 5 | 4 | 4 | 4 | 4.25 | 5 | 4 | 5 | 4 | 4.5 | 4 | 5 | 3 | 3 | 3.75 | 4 | 4 | 4 | 4 | 4 | 5 | 4 | 4 | 4 | 4.25 |
| 4 | 5 | 4 | 5 | 5 | 4 | 5 | 5 | 5 | 4.75 | 5 | 4 | 5 | 5 | 4.75 | 5 | 4 | 5 | 5 | 4.75 | 5 | 4 | 4 | 4 | 4.25 | 4 | 3 | 5 | 3 | 3.75 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 5 | 4 | 4 | 4 | 4.25 |
| 5 | 5 | 4 | 5 | 5 | 4 | 5 | 5 | 5 | 4.75 | 5 | 4 | 4 | 4 | 4.25 | 4 | 5 | 5 | 5 | 4.75 | 5 | 4 | 4 | 4 | 4.25 | 3 | 4 | 4 | 3 | 3.5 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| Consolidated M | 4.8 | 4.4 | 4.6 | 4.6 | 4.6 | 4.6 | 4.8 | 4.8 | 4.65 | 4.8 | 4.2 | 4.4 | 4.6 | 4.5 | 4.6 | 4.6 | 4.8 | 5 | 4.75 | 4.8 | 4.2 | 4.2 | 4.4 | 4.4 | 4.4 | 4 | 4.4 | 3.8 | 4.15 | 4 | 4 | 3.8 | 3.8 | 3.9 | 3.8 | 3.8 | 4.2 | 4.2 | 4 | 4.6 | 4.4 | 4.2 | 4.2 | 4.35 |

OVERALL

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Consolidated M | 4.65 | 4.45 | 4.43 | 4.55 | 4.43 | 4.3 | 4.53 | 4.48 | 4.65 | 4.35 | 4.28 | 4.38 | 4.38 | 4.43 | 4.55 | 4.65 | 4.62 | 4.57 | 4.53 | 4.25 | 4.23 | 4.28 | 4.28 | 4.33 | 4.25 | 4.28 | 4.03 | 4.28 | 4.25 | 4.13 | 4.03 | 4.03 | 4.11 | 3.98 | 3.98 | 4.23 | 4.23 | 4.02 | 4.38 | 4.33 | 4.23 | 4.38 | 4.24 |

ADVISORY COMMITTEE (SUPER ADMIN ACCOUNT)

| | FUNTIONAL SUITABILITY | | | | | | PERFORMANCE EFFECIENCY | | | | | COMPATIBILITY | | | | | USABILITY | | | | | RELIABILITY | | | | | SECURITY | | | | | MAINTAINABILITY | | | | | PORTABILITY | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Respondent | FS_Q1 | FS_Q2 | FS_Q3 | FS_Q4 | FS_Q5 | MEAN | PE_Q1 | PE_Q2 | PE_Q3 | PE_Q4 | MEAN | C_Q1 | C_Q2 | C_Q3 | C_Q4 | MEAN | U_Q1 | U_Q2 | U_Q3 | U_Q4 | MEAN | R_Q1 | R_Q2 | R_Q3 | R_Q4 | MEAN | S_Q1 | S_Q2 | S_Q3 | S_Q4 | MEAN | M_Q1 | M_Q2 | M_Q3 | M_Q4 | MEAN | P_Q1 | P_Q2 | P_Q3 | P_Q4 |
| 1 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 4 | 4.5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 5 | 5 | 5 | 4.75 | 5 | 5 | 5 | 5 | 5 | 4 | 4 | 5 | 4 | 4.25 | 4 | 5 | 4 | 5 |
| 2 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 4 | 4.5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 4 | 4.5 | 5 | 5 | 4 | 5 | 4.75 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 5 | 5 | 4 |
| 3 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 3 | 4 | 4 | 3 | 3.5 | 3 | 4 | 4 | 3 | 3.5 | 4 | 4 | 3 | 3.5 | 4 | 3 | 3 | 3 | 3.25 | 3 | 3 | 3 | 4 | 3.25 | 4 | 3 | 4 | 3 |
| onsolidated Mear | 4.5 | 4.5 | 4.5 | 4.5 | 4.5 | 4.5 | 4.5 | 4.5 | 4 | 4 | 4.25 | 4.25 | 4.5 | 4.5 | 4.25 | 4.38 | 4.25 | 4.5 | 4.25 | 4 | 4.25 | 4.25 | 4.5 | 4 | 4.25 | 4.25 | 4.5 | 4.25 | 4.25 | 4.31 | 4 | 4 | 4.25 | 4.25 | 4.13 | 4 | 4.25 | 4.25 | 4 |

PHRMO (SUPER ADMIN ACCOUNT)

| | FUNTIONAL SUITABILITY | | | | | | PERFORMANCE EFFECIENCY | | | | | COMPATIBILITY | | | | | USABILITY | | | | | RELIABILITY | | | | | SECURITY | | | | | MAINTAINABILITY | | | | | PORTABILITY | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Respondent | FS_Q1 | FS_Q2 | FS_Q3 | FS_Q4 | FS_Q5 | MEAN | PE_Q1 | PE_Q2 | PE_Q3 | PE_Q4 | MEAN | C_Q1 | C_Q2 | C_Q3 | C_Q4 | MEAN | U_Q1 | U_Q2 | U_Q3 | U_Q4 | MEAN | R_Q1 | R_Q2 | R_Q3 | R_Q4 | MEAN | S_Q1 | S_Q2 | S_Q3 | S_Q4 | MEAN | M_Q1 | M_Q2 | M_Q3 | M_Q4 | MEAN | P_Q1 | P_Q2 | P_Q3 | P_Q4 |
| 1 | 5 | 5 | 5 | 4 | 5 | 4.8 | 4 | 5 | 5 | 4 | 4.5 | 5 | 5 | 4 | 4 | 4.5 | 5 | 4 | 5 | 5 | 4.75 | 5 | 4 | 5 | 5 | 4.75 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 5 | 4.25 | 4 | 4 | 5 | 5 |
| 2 | 5 | 5 | 4 | 5 | 5 | 4.8 | 4 | 4 | 5 | 4 | 4.25 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 4 | 4.5 | 4 | 5 | 4 | 5 | 4.5 | 4 | 4 | 5 | 5 | 4.5 | 4 | 4 | 5 | 5 | 4.5 | 5 | 5 | 4 | 4 |
| 3 | 5 | 5 | 4 | 5 | 4 | 4.6 | 5 | 5 | 4 | 5 | 4.75 | 4 | 5 | 4 | 5 | 4.5 | 4 | 4 | 4 | 5 | 4.25 | 4 | 5 | 5 | 5 | 4.75 | 4 | 4 | 5 | 4 | 4.25 | 5 | 4 | 4 | 4 | 4.25 | 4 | 5 | 4 | 4 |
| 4 | 4 | 4 | 5 | 4 | 4 | 4.2 | 4 | 4 | 4 | 4 | 4 | 4 | 5 | 5 | 5 | 4.75 | 4 | 5 | 4 | 5 | 4.5 | 4 | 5 | 4 | 5 | 4.5 | 5 | 4 | 5 | 5 | 4.75 | 5 | 5 | 4 | 4 | 4.5 | 4 | 4 | 4 | 5 |
| 5 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 5 | 4 | 3 | 4 | 4 | 4 | 5 | 4 | 4 | 4.25 | 4 | 5 | 5 | 5 | 4.75 | 4 | 5 | 4 | 5 | 4.5 | 4 | 5 | 4 | 4 | 4.25 | 4 | 4 | 5 | 5 |
| Consolidated Mea | 4.6 | 4.6 | 4.4 | 4.4 | 4.4 | 4.48 | 4.2 | 4.4 | 4.4 | 4.2 | 4.3 | 4.6 | 4.8 | 4.2 | 4.6 | 4.55 | 4.4 | 4.4 | 4.4 | 4.6 | 4.45 | 4.2 | 4.8 | 4.6 | 5 | 4.65 | 4.2 | 4.2 | 4.6 | 4.6 | 4.4 | 4.4 | 4.4 | 4.2 | 4.4 | 4.35 | 4.2 | 4.4 | 4.4 | 4.6 |

OVERALL

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Consolidated Mea | 4.55 | 4.55 | 4.45 | 4.45 | 4.45 | 4.49 | 4.35 | 4.45 | 4.20 | 4.10 | 4.28 | 4.43 | 4.65 | 4.35 | 4.43 | 4.46 | 4.33 | 4.45 | 4.33 | 4.30 | 4.35 | 4.23 | 4.65 | 4.30 | 4.63 | 4.45 | 4.35 | 4.23 | 4.43 | 4.43 | 4.36 | 4.20 | 4.20 | 4.23 | 4.33 | 4.24 | 4.10 | 4.33 | 4.33 | 4.30 |

**Appendix 3.** Acceptance form

Republic of the Philippines
Institute of Computing, Engineering and Technology
Davao Del Sur State College
**DEPARTMENT OF COMPUTING**
Matti, Digos City, Davao Del Sur

**CAPSTONE PROJECT OUTLINE – ACCEPTANCE FORM**

This letter confirm is that the Specific Objectives, Scope, and Limitations, and the hundred percent (100%) User Interface with eighty percent (80%) Functionality of the Project entitled eHiremo: A WEB-BASED RECRUITMENT AND SELECTION MANAGEMENT SYSTEM FOR THE PROVINCIAL HUMAN RESOURCE MANAGEMENT OFFICE has been accepted and approved by the respective technic's panel committee members before the specifications set with their conditional pass.

The following specific objectives, scope, and limitations, functional and non-functional requirements are as follow:

**Specific Objectives:**

1. Develop a page for applicant account that has the following functionalities:
   - 1.1 View published vacant position in the job portal;
   - 1.2 Registration page for new account;
   - 1.3 Submission portal for application documents such as:
     - 1.3.1 Personal Data Sheets entry;
     - 1.3.2 Scanned copy of School Credentials/ Transcript of Records;
     - 1.3.3 Scanned copy of certificate of Eligibility (CS, PRC License, LTO License);
     - 1.3.4 Scanned Work Experience Sheet;
     - 1.3.5 Scanned Certificate of Training; and
     - 1.3.6 Scanned Performance Rating.
   - 1.4 View and Track application status;
   - 1.5 Receive Push Notification and Email for the following:
     - 1.5.1 Application status;
     - 1.5.2 Schedules of Exam, Interview and Appointment including date, time and venue;
     - 1.5.3 Result of Examination and Selection.
2. Develop a page for administrator that has the following functionalities:
   - 2.1 Post vacant position;
   - 2.2 Result on Assessment and Evaluation of Applications;
   - 2.3 Send Push notification and email for the following:
     - 2.3.1 Application Status;
     - 2.3.2 Schedule for the examination, Panel interview and Appointment;
   - 2.4 Tabulation of the applicant's exam score and interview score;
   - 2.5 Generate reports converted into pdf files such as:

Republic of the Philippines
Institute of Computing, Engineering and Technology
Davao Del Sur State College
**DEPARTMENT OF COMPUTING**
Matti, Digos City, Davao Del Sur

2.5.1 Comparative Assessment Report of Applicants;
2.5.2 HRMPSB list of examinees for examination;
2.5.3 Monthly accession report;

3. Develop a page for super administrator that has the following functionalities:
   3.1 View the ranking of applicants for a specific position with their ratings.
   3.2 Select an Applicant for the appointment of the position;
   3.3 View activity logs such as:
      3.3.1 Applicant-related actions on registrations and submission;
      3.3.3 Job posting activities.

**Scope and Limitation**

The study aims to develop a comprehensive web-based system designed to optimize and automate the job application and assessment processes for applicants, administrators, and super administrators. The system will allow applicants to view published vacant jobs and positions, register for new accounts, and submit necessary application documents, including personal data sheets, school credentials, eligibility certificates, work experience sheets, certificates of training, and performance ratings.

Additionally, applicants will be able to track their application status and receive notifications regarding assessments, selection schedules, and results. For administrators, the system will enable posting of vacant jobs, assessment of applications based on qualifications (such as education, experience, and eligibility), and sending notifications about application statuses, exam schedules, panel interviews, and appointment notices. The administrators will also be able to calculate exam and interview scores, generate comparative assessments, and prepare monthly reports. For super administrators, the system will facilitate the viewing of top applicants, applicant ratings, and the selection of applicants for appointment, as well as the monitoring of activity logs related to job postings, applicant registrations, and application reviews.

Despite the anticipated contributions of the study, there are several limitations to consider. First, the development of the system will be constrained by the current capabilities of web technologies, which may not fully accommodate future advancements or integrations with emerging technologies. Additionally, the accuracy and reliability of the system's outputs, such as applicant status and scores, will depend heavily on the quality of the data provided by both applicants and administrators. Any discrepancies or incomplete information could impact the system's overall performance.

Furthermore, the study will focus on a limited user base, which means that the system's scalability to handle larger numbers of users or higher traffic volumes may require additional optimization beyond the scope of this research. While the system will undergo controlled testing to meet various criteria, real-world scenarios involving unpredictable user behavior or varying internet conditions may not be fully accounted for. Security measures will be implemented, but the study's scope will not include exhaustive security assessments, which could require further improvements to safeguard sensitive data.

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**Non-functional Requirements**

1. **Performance Efficiently:** The system should remain fast, responsive, and stable, delivering quick access to all features even under heavy use and growing data.
2. **Compatibility:** The system should work seamlessly across different devices, operating systems, and browsers, ensuring consistent functionality and user experience for all users.
3. **Usability:** The system should be easy to learn, navigate, and operate, providing a clear and intuitive interface that allows users to accomplish tasks efficiently with minimal errors.
4. **Reliability:** The system should consistently perform its intended functions without errors, remain available when needed, and recover quickly from any failures or interruptions.
5. **Security:** The system should protect user data and system resources through secure authentication, access control, and data encryption, preventing unauthorized access or malicious activities.
6. **Maintainability:** The system should be easy to update, fix, and enhance, with a clear structure that allows changes without affecting other functions.
7. **Portability:** The system should be adaptable to different environments, allowing installation, configuration, and operation on various platforms with minimal effort.

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**SURVEY QUESTIONNAIRE VALIDATORS**

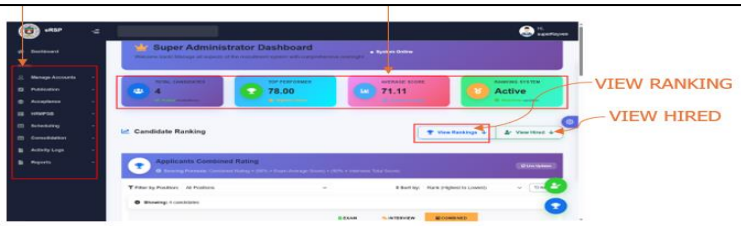
KRIS C. BAHAYA, MIT
Internal Validator

MICHELLE ALCEBAR, MBA
External Validator

**Appendix 4.** User's Manual per User Level

**Appendix 4.1.** Super Administrator Manual

**Step 1.**




**Step 2.**



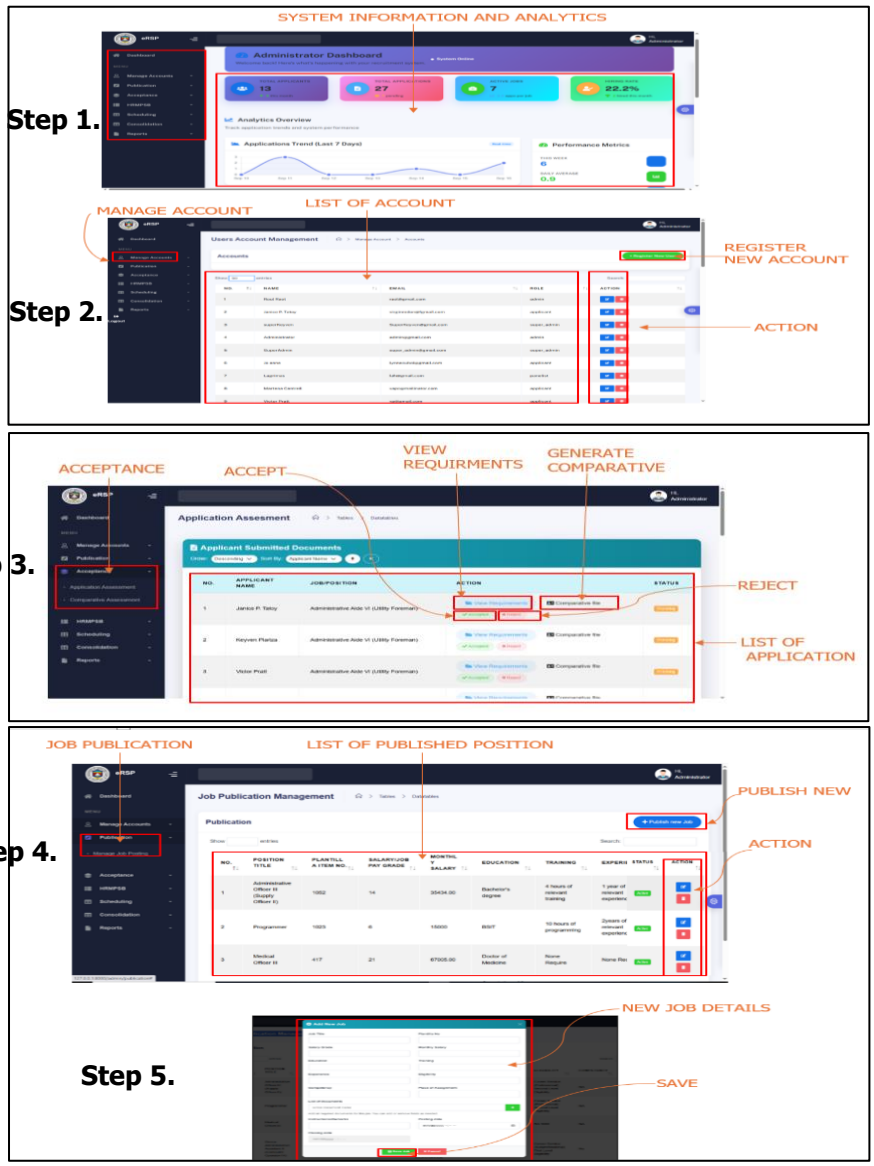

**Step 3.**




**Step 4.**



**Appendix 4.2.** Administrator User's Manual



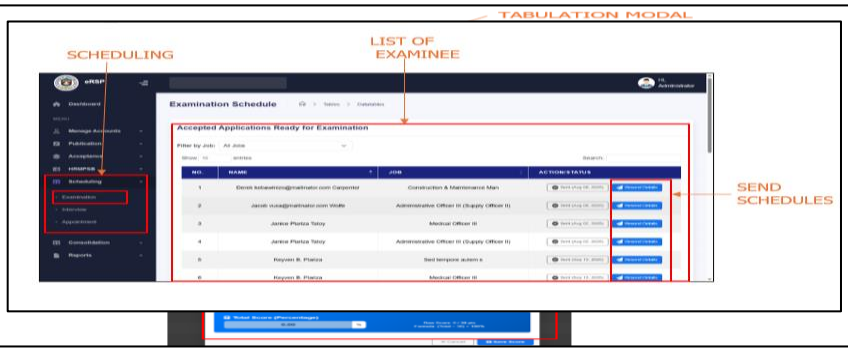
**Step 6.**



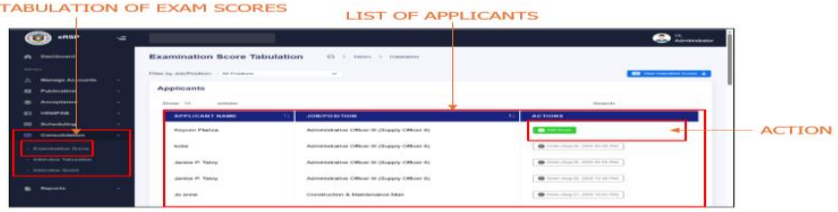

**Step 7.**



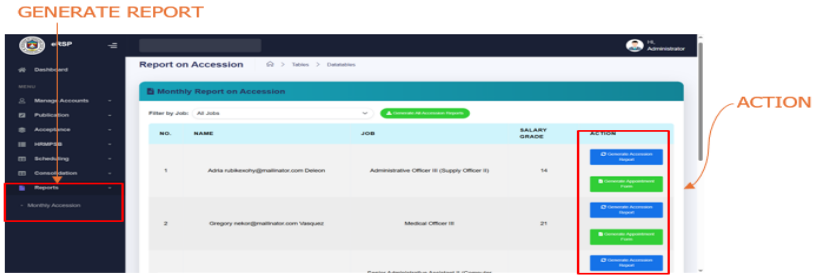

**Step 8.**



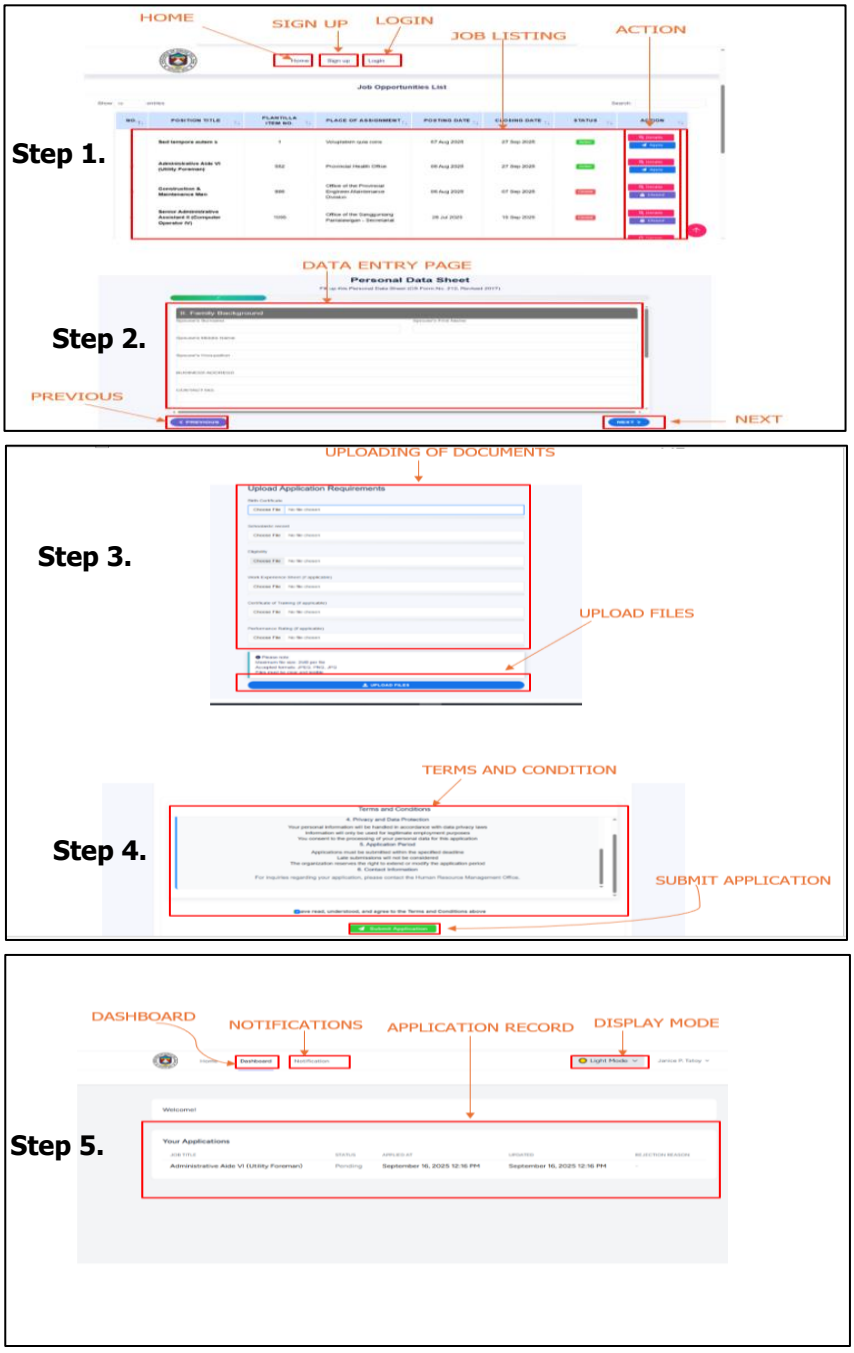

**Step 9.**




**Step 10.**



**Appendix 4.3.** Applicant User's Manual



# **Appendix 5.** Relevant Source Code

## **Appendix 5.1.** Code Snippet for Applicant Ranking and Rating Viewing

```
public function index()
{
    $applications = Application::with(['applicant.user', 'job', 'scoreExam', 'interviewRating'])->get();

    $ratings = $applications->map(function ($application, $index) {
        $examScore = $application->examScore;
        $interview = $application->interviewRating;

        $averageScore = $examScore?->average_score ?? 0;
        $interviewScore = $interview?->total_score ?? 0;
        $overallRating = ($averageScore + $interviewScore) / 2;

        return (object) [
            'no' => $index + 1,
            'name' => $application->applicant->user->name ?? 'N/A',
            'position' => $application->job->title ?? 'N/A',
            'average_score' => number_format($averageScore, 2),
            'total_score' => number_format($interviewScore, 2),
            'overall_rating' => number_format($overallRating, 2),
        ];
    });

    return view('SuperAdmin.index', compact('ratings'));
}


}
```

## **Appendix 5.2.** Code Snippet for Selection of Applicant for Appointment

```
public function updateStatus(Request $request, $hashid)
{
    $request->validate([
        'status' => 'required|in:pending,accepted,shortlisted,rejected,hired',
        'reason' => 'nullable|string|max:1000',
        'reporting_to' => 'required_if:status,hired|string|max:255',
    ]);

    // Decode the hashid
    $decoded = \Vinkla\Hashids\Facades\Hashids::decode($hashid);
    if (empty($decoded)) {
        abort(404, 'Invalid application id');
    }
    $realId = $decoded[0];

    $application = Application::with(['applicant.user', 'job'])->findOrFail($realId);
    $application->status = $request->status;
    if ($request->status === 'rejected') {
        $application->rejection_reason = $request->reason;
    }
    $application->save();

    // Log the status change
    // Debug log to check the data
    \Log::info('Application data for logging:', [
        'applicant' => $application->applicant,
        'user' => $application->applicant->user ?? null,
        'job' => $application->job,
        'old_status' => $application->getOriginal('status'),
        'new_status' => $request->status
    ]);

    // Ensure we have all required data
    $applicantName = $application->applicant->user->name ?? 'Unknown Applicant';
    $jobTitle = $application->job->title ?? 'Unknown Position';
    $oldStatus = ucfirst($application->getOriginal('status') ?? 'None');
```

**Appendix 5.3.** Code Snippet for System Activity Logs view

```php

namespace App\Http\Controllers;

use App\Models\ActivityLog;
use Illuminate\Http\Request;

class ActivityLogController extends Controller
{
    public function index()
    {
        $activityLogs = ActivityLog::with('user')
            ->orderBy('created_at', 'desc')
            ->paginate(50);

        return view('Admin.tables.Logs', compact('activityLogs'));
    }
}

```

**Appendix 5.4.** Code Snippet for Job Listing management

```php
use LogsActivity;
    public function addJob(Request $request)
{
$validated = $request->validate([
    'title' => 'required|string|max:255',
    'plantilla_no' => 'required|string',
    'salary_grade' => 'required|string',
    'monthly_salary' => 'required|string',
    'education' => 'required|string',
    'training' => 'required|string',
    'experience' => 'nullable|string',
    'eligibility' => 'required|string',
    'competency' => 'required|string',
    'place_assignment' => 'required|string',
    'reporting_to' => 'nullable|string',
    'documents' => 'nullable|array',
    'Instructions' => 'nullable|string',
    'closing_date' => 'required|date',
    'posted_at' => 'nullable|date',
]);

// ✅ Encode documents field to JSON if it's present
if (isset($validated['documents'])) {
    $validated['documents'] = json_encode($validated['documents']);
}

$job = Job::create($validated);

$this->logActivity(
    'job_created',
    'New Job Position Created | Title: ":title" | Plantilla No: ":plantilla" | Salary Grade: ":salary_grade" | Monthly Salary: ":salary" | Place of Assignment: ":pl
    $job,
    [
        'title' => $job->title,
        'plantilla' => $job->plantilla_no,
        'salary_grade' => $job->salary_grade,
        'salary' => $job->monthly_salary,
```

**Appendix 5.5.** Code Snippet for Application Assessment and Evaluation

```php
/**
 * Application Assessment and Evaluation.
 */
public function updateStatus(Request $request, $hashid)
{
    $request->validate([
        'status' => 'required|in:pending,accepted,shortlisted,rejected,hired',
        'reason' => 'nullable|string|max:1000',
        'reporting_to' => 'required_if:status,hired|string|max:255',
    ]);

    // Decode the hashid
    $decoded = \Vinkla\Hashids\Facades\Hashids::decode($hashid);
    if (empty($decoded)) {
        abort(404, 'Invalid application id');
    }
    $realId = $decoded[0];

    $application = Application::with(['applicant.user', 'job'])->findOrFail($realId);
    $application->status = $request->status;
    if ($request->status === 'rejected') {
        $application->rejection_reason = $request->reason;
    }
    $application->save();

    // Log the status change
    // Debug log to check the data
    \Log::info('Application data for logging:', [
        'applicant' => $application->applicant,
        'user' => $application->applicant->user ?? null,
        'job' => $application->job,
        'old_status' => $application->getOriginal('status'),
        'new_status' => $request->status
    ]);

    // Ensure we have all required data
    $applicantName = $application->applicant->user->name ?? 'Unknown Applicant';
    $jobTitle = $application->job->title ?? 'Unknown Position';
```

**Appendix 5.6.** Code Snippet for Comparative Assessment File Generation of the Applicants

```php
public function generateComparativeReport($hashid)
{
    // Decode Hashid to get the real ID
    $decoded = Hashids::decode($hashid);
    if (empty($decoded)) {
        abort(404, 'Invalid ID');
    }
    $id = $decoded[0];

    // Get the application and related data with the most recent PDS
    $application = Application::with([
        'job',
        'applicant.pds' => function($query) {
            $query->orderBy('created_at', 'desc'); // Get the most recent PDS
        },
        'applicant.pds.educationalBackgrounds',
        'applicant.pds.eligibilities',
        'applicant.pds.workExperiences',
        'applicant.pds.trainings',
        'applicant.pds.otherInfo'
    ])->findOrFail($id);

    // Load Word template
    $templateProcessor = new TemplateProcessor(storage_path('app/templates/comparative.docx'));

    // Get the job data
    $job = $application->job;

    // Get the applicant and their most recent PDS at the time of application
    $applicant = $application->applicant;
    $pds = $applicant->pds()->where('created_at', '<=', $application->created_at)
                            ->orderBy('created_at', 'desc')
                            ->first();

    if (!$pds) {
        abort(404, 'No Personal Data Sheet found for this application.');
```

**Appendix 5.7.** Code Snippet for Push Notification and Email Sending Module

```php
public function toMail($notifiable)
{
    if (($this->details['type'] ?? null) === 'exam_details') {
        return (new \Illuminate\Notifications\Messages\MailMessage)
            ->subject('Examination Invitation')
            ->markdown('emails.exam_invitation', [
                'date' => now()->format('F d, Y'),
                'recipient_name' => $this->details['recipient_name'],
                'address' => $this->details['address'],
                'contact' => $this->details['contact'],
                'last_name' => $this->details['last_name'],
                'exam_status' => 'passed the Application assessment  qualifying examination',
                'position' => $this->details['position'],
                'office' => $this->details['office'],
                'exam_date' => \Carbon\Carbon::parse($this->details['date'])->format('F d, Y'),
                'exam_day' => \Carbon\Carbon::parse($this->details['date'])->format('l'),
                'exam_time' => $this->details['time'],
                'exam_location' => $this->details['location'],
                'salutation' => $this->details['salutation'],
            ]);
    }
    if (($this->details['type'] ?? null) === 'interview_details') {
        return (new \Illuminate\Notifications\Messages\MailMessage)
            ->subject('Interview Invitation')
            ->markdown('emails.interview_invitation', [
                'date' => now()->format('F d, Y'),
                'recipient_name' => $this->details['recipient_name'],
                'address' => $this->details['address'],
                'contact' => $this->details['contact'],
                'last_name' => $this->details['last_name'],
                'interview_status' => 'passed the qualifying examination and are now scheduled for an interview',
                'position' => $this->details['position'],
                'office' => $this->details['office'],
                'interview_date' => \Carbon\Carbon::parse($this->details['date'])->format('F d, Y'),
                'interview_day' => \Carbon\Carbon::parse($this->details['date'])->format('l'),
```

**Appendix 5.8.** Code Snippet for Tabulation of Exam score of the Applicant

```php
class ExaminationScoreController extends Controller
{
    public function store(Request $request)
    {
        $validated = $request->validate([
            'application_id' => 'required|exists:applications,id',
            'written_items' => 'integer|nullable',
            'written_score' => 'numeric|nullable',
            'computer_items' => 'integer|nullable',
            'computer_score' => 'numeric|nullable',
            'trade_items' => 'integer|nullable',
            'trade_score' => 'numeric|nullable',
            'average_score' => 'numeric|nullable',
            'remarks' => 'string|nullable',
        ]);

        $score = ExaminationScore::updateOrCreate(
            ['application_id' => $request->application_id],
            [
                'written_exam' => $request->written_score ?? null,
                'computer_proficiency' => $request->computer_score ?? null,
                'trade_skill' => $request->trade_score ?? null,
                'average_score' => $request->average_score,
                'written_items' => $request->written_items,
                'written_score' => $request->written_score,
                'computer_items' => $request->computer_items,
                'computer_score' => $request->computer_score,
                'trade_items' => $request->trade_items,
                'trade_score' => $request->trade_score,
                'remarks' => $request->remarks,
            ]
        );

        // Send notification if exam is failed
        if ($request->remarks === 'Failed') {
            $application = Application::with(['applicant.user', 'job'])->find($request->application_id);
            if ($application && $application->applicant && $application->applicant->user) {
```

**Appendix 5.9.** Code Snippet for Tabulation of Interview score of the Applicant

```php
public function store(Request $request)
    {
        $validated = $request->validate([
            'application_id' => 'required|exists:applications,id',
            'panelist_id' => 'required|exists:users,id', // panelist must exist in users
            'interviewer' => 'required|string|max:255',
            'general_appearance' => 'required|numeric|min:1|max:5',
            'self_confidence' => 'required|numeric|min:1|max:5',
            'manner_of_speaking' => 'required|numeric|min:1|max:5',
            'alertness' => 'required|numeric|min:1|max:5',
            'experimental_knowledge' => 'required|numeric|min:1|max:5',
            'maturity_of_judgement' => 'required|numeric|min:1|max:5',
        ]);

        $totalScore = array_sum([
            $validated['general_appearance'],
            $validated['self_confidence'],
            $validated['manner_of_speaking'],
            $validated['alertness'],
            $validated['experimental_knowledge'],
            $validated['maturity_of_judgement']
        ]);

        $totalScorePercentage = ($totalScore /30) * 100;

        InterviewRating::updateOrCreate(
            [
                'application_id' => $validated['application_id'],
                'panelist_id' => $validated['panelist_id'], // unique to panelist + applicant
            ],
            array_merge($validated, [
                'total_score' => $totalScorePercentage
            ])
        );
```

**Appendix 5.10.** Snippet code for File and Report Generation

```php
public function generateReport()
{
    // 1. Kunin ang lahat ng empleyado/applicants
    $applications = Application::with(['applicant.user', 'applicant.pds', 'job'])
    ->where('status', 'Hired')
    ->get();

            // 2. Get manual inputs (or use default if wala)
            $statusInput    = $request->input('status_of_appointment', 'Permanent');
            $accessionInput = $request->input('mode_of_accession', 'Original');
            $remarksInput   = $request->input('remarks', '—');

            // 3. Prepare data for template
            $rows = $applications->map(function ($a, $index) use ($statusInput, $accessionInput, $remarksInput) {
        return [
            'no'         => $index + 1,
            'name'       => $e->surname . ', ' . $e->first_name . ' ' . ($e->middle_name ?? '') . ' ' . ($e->extension ?? ''),
            'dob'        => optional($e->date_of_birth)->format('m/d/Y'),
            'position'   => $e->title,
            'status'    => $statusInput,      // Permanent / Temp / Casual
            'salary'     => $e->salary_grade,
            'accession' => $accessionInput,        // Original / Transfer / Promotion
            'effective'  => optional($e->effective_date)->format('m/d/Y'),
            'remarks'   => $remarksInput,
        ];
    })->all();

    // 3. Load the DOCX template
    $templatePath = Storage::path('templates/Accession_report.docx');
    $template = new TemplateProcessor($templatePath);

    // 4. Clone the row and fill values
    $template->cloneRowAndSetValues('no', $rows);

    // 5. Save DOCX first
```

**Appendix 5.11.** Snippet code for Registration of Account



```
/**
 * Handle an incoming registration request.
 *
 * @throws \Illuminate\Validation\ValidationException
 */ // For Register Accounts
public function store(Request $request)
{
    // Validate the incoming request data
    $request->validate([
        'name' => 'required|string|max:255',
        'email' => 'required|email|unique:users,email',
        'password' => 'required|string|min:6|confirmed',

    ]);

    // Create the user with default role as 'applicant'
    $user = User::create([
        'name' => $request->name,
        'email' => $request->email,
        'role' => 'applicant', // Default role
        'password' => Hash::make($request->password),
    ]);

    // Automatically create associated applicant record
    $user->applicant()->create([
        'contact_number' => 'N/A',
        'address' => 'N/A',
    ]);

    return redirect()->route('home')->with('success', 'Account created successfully!');
}
```

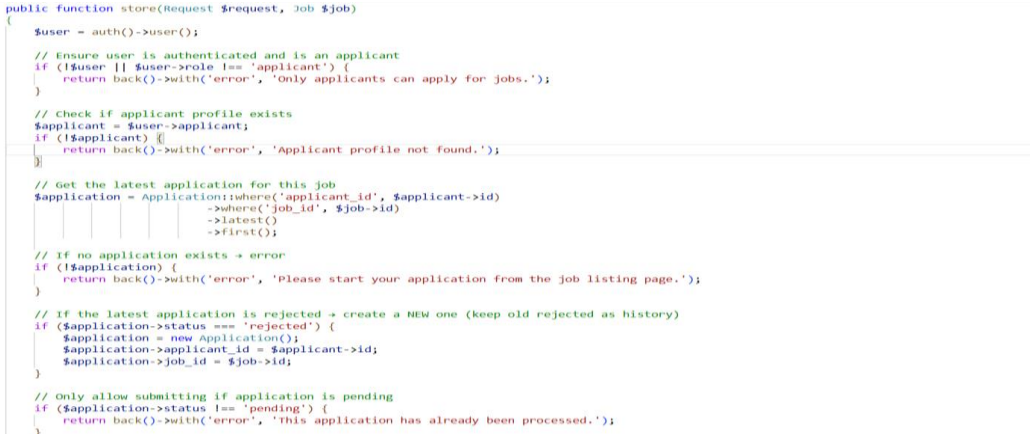
**Appendix 5.12.** Snippet code for Job Vacancy Listing



```
  <h4 class="card-title text-center">Job Opportunities List</h4>
</div>
<div class="card-body">
  <div class="table-responsive px-10 py-10"> <!-- Increased horizontal and vertical padding here -->
    <div class="table-responsive">
      <table id="add-row" class="table table-bordered table-hover align-middle" style="background: #fff; border-radius: 12px; box-shadow: 0 2px 12px
      <thead class="table-primary text-center align-middle">
        <tr>
        </tr>
      </thead>
      <tbody>
        @forelse ($jobs as $job)
          <tr>
            <td class="text-center">{{ $loop->iteration }}</td>
            <td>
              <strong>{{ $job->title }}</strong>

            </td>
            <td class="text-center">{{ $job->plantilla_no }}</td>
            <td>{{ $job->place_assignment }}</td>
            <td class="text-center">{{ \Carbon\Carbon::parse($job->posted_at)->format('d M Y') }}</td>
            <td class="text-center">{{ \Carbon\Carbon::parse($job->closing_date)->format('d M Y') }}</td>
            <td class="text-center">
              @if($job->isClosed())
                <span class="badge bg-danger">Closed</span>
              @else
                <span class="badge bg-success">Active</span>
              @endif
            </td>
            <td class="text-center">
              <a href="{{ route('jobs.pdf', $job->hashid) }}" target="_blank" class="btn btn-outline-secondary btn-sm me-2" style="min-width: 110px; mar
```

**Appendix 5.13.** Snippet code for Submission of Applicant's Application and Documents



```
public function store(Request $request, Job $job)
{
    $user = auth()->user();

    // Ensure user is authenticated and is an applicant
    if (!$user || $user->role !== 'applicant') {
        return back()->with('error', 'Only applicants can apply for jobs.');
    }

    // Check if applicant profile exists
    $applicant = $user->applicant;
    if (!$applicant) {
        return back()->with('error', 'Applicant profile not found.');
    }

    // Get the latest application for this job
    $application = Application::where('applicant_id', $applicant->id)
                              ->where('job_id', $job->id)
                              ->latest()
                              ->first();

    // If no application exists → error
    if (!$application) {
        return back()->with('error', 'Please start your application from the job listing page.');
    }

    // If the latest application is rejected → create a NEW one (keep old rejected as history)
    if ($application->status === 'rejected') {
        $application = new Application();
        $application->applicant_id = $applicant->id;
        $application->job_id = $job->id;
    }

    // Only allow submitting if application is pending
    if ($application->status !== 'pending') {
        return back()->with('error', 'This application has already been processed.');
    }
```

**Appendix 5.14.** Snippet code for Application Tracking



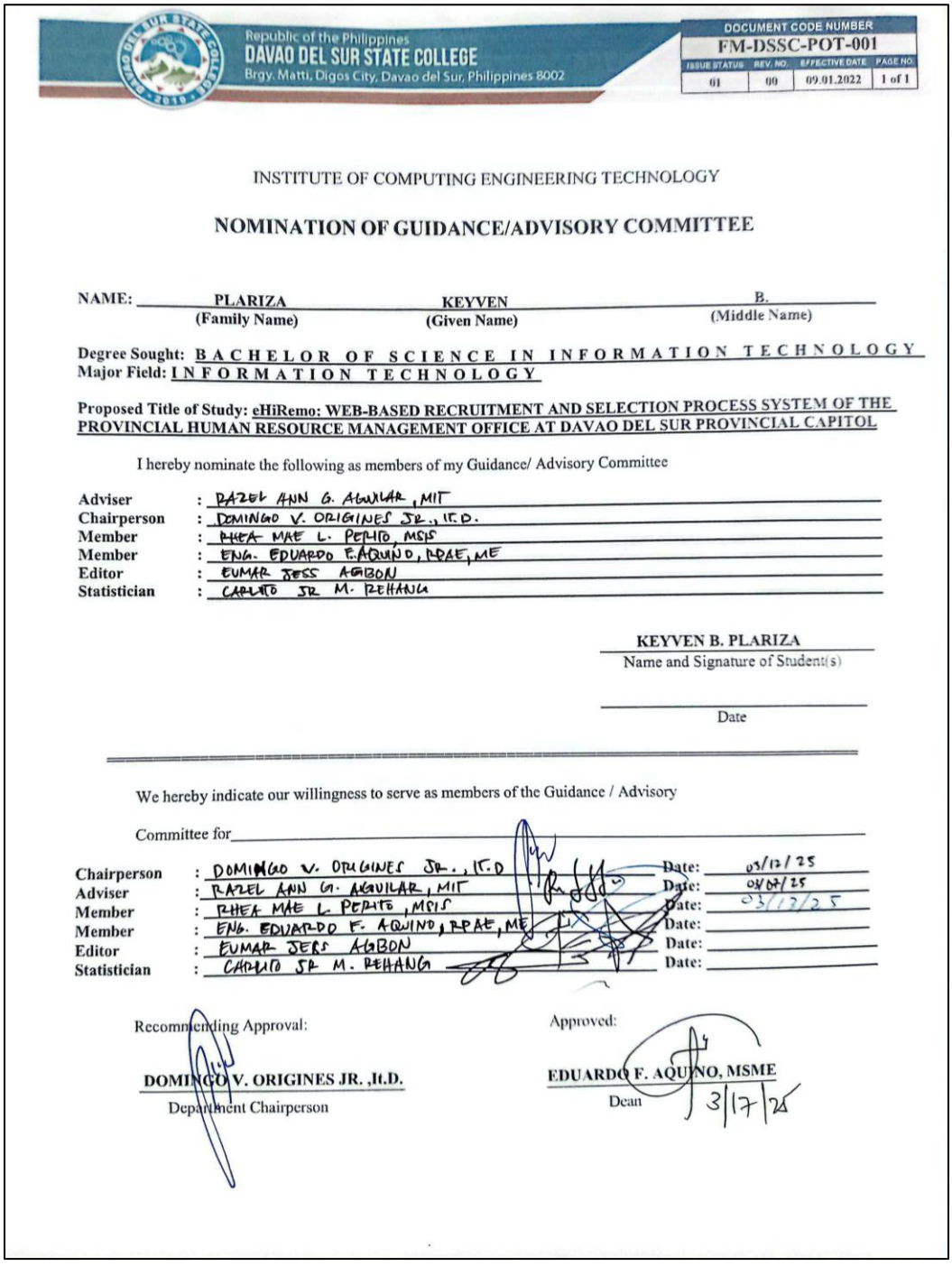
```
<h3 class="text-lg font-semibold text-gray-800 dark:text-gray-200">
    {{ __('Your Applications') }}
</h3>
<div class="mt-4">
    <table class="min-w-full divide-y divide-gray-200 dark:divide-gray-700">
        <thead>
            <tr>
                <th class="px-4 py-2 text-left text-xs font-medium text-gray-500 uppercase">Job Title</th>
                <th class="px-4 py-2 text-left text-xs font-medium text-gray-500 uppercase">Status</th>
                <th class="px-4 py-2 text-left text-xs font-medium text-gray-500 uppercase">Applied At</th>
                <th class="px-4 py-2 text-left text-xs font-medium text-gray-500 uppercase">Updated</th>
                <th class="px-4 py-2 text-left text-xs font-medium text-gray-500 uppercase">Rejection Reason</th>
            </tr>
        </thead>
        <tbody class="bg-white dark:bg-gray-800 divide-y divide-gray-200 dark:divide-gray-700">
            @foreach ($applications as $application)
                <tr>
                    <td class="px-4 py-2">{{ $application->job->title ?? 'N/A' }}</td>
                    <td class="px-4 py-2">
                        <span class="capitalize text-{{ match($application->status) {
                            'pending' => 'gray-500',
                            'accepted' => 'green-600',
                            'shortlisted' => 'yellow-500',
                            'rejected' => 'red-600',
                            'hired' => 'blue-600',
                            default => 'gray-400',
                        } }}">
                            {{ $application->status }}
                        </span>
                    </td>
                    <td class="px-4 py-2">{{ \Carbon\Carbon::parse($application->applied_at)->format('F j, Y g:i A') }}</td>
                    <td class="px-4 py-2">{{ \Carbon\Carbon::parse($application->updated_at)->format('F j, Y g:i A') }}</td>
```

**Appendix 5.15.** Snippet code for Notifications Viewing



```
{{-- Notifications Section --}}
@if ($notifications && $notifications->count())
    <div class="mt-8 bg-white dark:bg-gray-800 overflow-hidden shadow-sm sm:rounded-lg">
        <div class="p-6">
            <h3 class="text-lg font-semibold text-gray-800 dark:text-gray-200 mb-4">
                {{ __('Notifications') }}
            </h3>
            <div class="space-y-4">
                @foreach ($notifications as $notification)
                    <a href="{{ route('notifications.show', $notification->id) }}" class="block">
                    <div class="p-4 border rounded-lg {{ $notification->type === 'App\\Notifications\\ExamFailureNotification' ? 'bg-red-50 border-red-200 dar
                        <div class="flex items-start justify-between">
                            <div class="flex-1">
                                @if ($notification->type === 'App\\Notifications\\ExamFailureNotification')
                                    <div class="flex items-center mb-2">
                                        <svg class="w-4 h-4 text-red-500 mr-2" fill="currentColor" viewBox="0 0 20 20">
                                            <path fill-rule="evenodd" d="M10 18a8 8 0 100-16 8 8 0 000 16zM8.707 7.293a1 1 0 00-1.414 1.414L8.586 10l-1.293 1.
                                        </svg>
                                        <h4 class="text-lg font-semibold text-red-800 dark:text-red-200">
                                            {{ $notification->data['title'] ?? 'Examination Failed' }}
                                        </h4>
                                    </div>
                                    <p class="text-red-700 dark:text-red-300 mb-2">
                                        {{ $notification->data['message'] ?? 'You did not pass the examination.' }}
                                    </p>
                                    <div class="text-sm text-red-600 dark:text-red-400">
                                        <p><strong>Position:</strong> {{ $notification->data['position'] ?? 'N/A' }}</p>
                                        <p><strong>Your Score:</strong> {{ $notification->data['total_score'] ?? 'N/A' }}</p>
                                        <p><strong>Passing Score:</strong> {{ $notification->data['passing_score'] ?? 'N/A' }}</p>
                                    </div>
                                @elseif ($notification->type === 'App\\Notifications\\ApplicantStatusUpdate' && ($notification->data['type'] ?? null) === 'int
                                    <div class="flex items-center mb-2">
                                        <svg class="w-5 h-5 text-green-500 mr-2" fill="currentColor" viewBox="0 0 20 20">
                                            <path fill-rule="evenodd" d="M18 10a8 8 0 11-16 0 8 8 0 0116 0zm-7-4a1 1 0 11-2 0 1 1 0 012 0zM9 9a1 1 0 000 2v3a1
```

**Appendix 6.** Nomination form (Outline)



Republic of the Philippines
DAVAO DEL SUR STATE COLLEGE
Brgy. Matti, Digos City, Davao del Sur, Philippines 8002

| DOCUMENT CODE NUMBER | | | |
|---|---|---|---|
| FM-DSSC-POT-001 | | | |
| ISSUE STATUS | REV. NO. | EFFECTIVE DATE | PAGE NO. |
| 01 | 00 | 09.01.2022 | 1 of 1 |

INSTITUTE OF COMPUTING ENGINEERING TECHNOLOGY

**NOMINATION OF GUIDANCE/ADVISORY COMMITTEE**

NAME: PLARIZA (Family Name) KEYVEN (Given Name) B. (Middle Name)

Degree Sought: BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
Major Field: INFORMATION TECHNOLOGY

Proposed Title of Study: eHiRemo: WEB-BASED RECRUITMENT AND SELECTION PROCESS SYSTEM OF THE PROVINCIAL HUMAN RESOURCE MANAGEMENT OFFICE AT DAVAO DEL SUR PROVINCIAL CAPITOL

I hereby nominate the following as members of my Guidance/ Advisory Committee

Adviser : RAZEL ANN G. AGUILAR, MIT
Chairperson : DOMINGO V. ORIGINES JR., IT.D.
Member : RHEA MAE L. PERITO, MSIS
Member : ENG. EDUARDO F. AQUINO, RPAE, ME
Editor : EUMAR JESS AGBON
Statistician : CARLITO JR M. REHANG

KEYVEN B. PLARIZA
Name and Signature of Student(s)

Date

We hereby indicate our willingness to serve as members of the Guidance / Advisory Committee for

Chairperson : DOMINGO V. ORIGINES JR., IT.D Date: 03/12/25
Adviser : RAZEL ANN G. AGUILAR, MIT Date: 03/07/25
Member : RHEA MAE L. PERITO, MSIS Date: 03/12/25
Member : ENG. EDUARDO F. AQUINO, RPAE, ME Date:
Editor : EUMAR JESS AGBON Date:
Statistician : CARLITO JR M. REHANG Date:

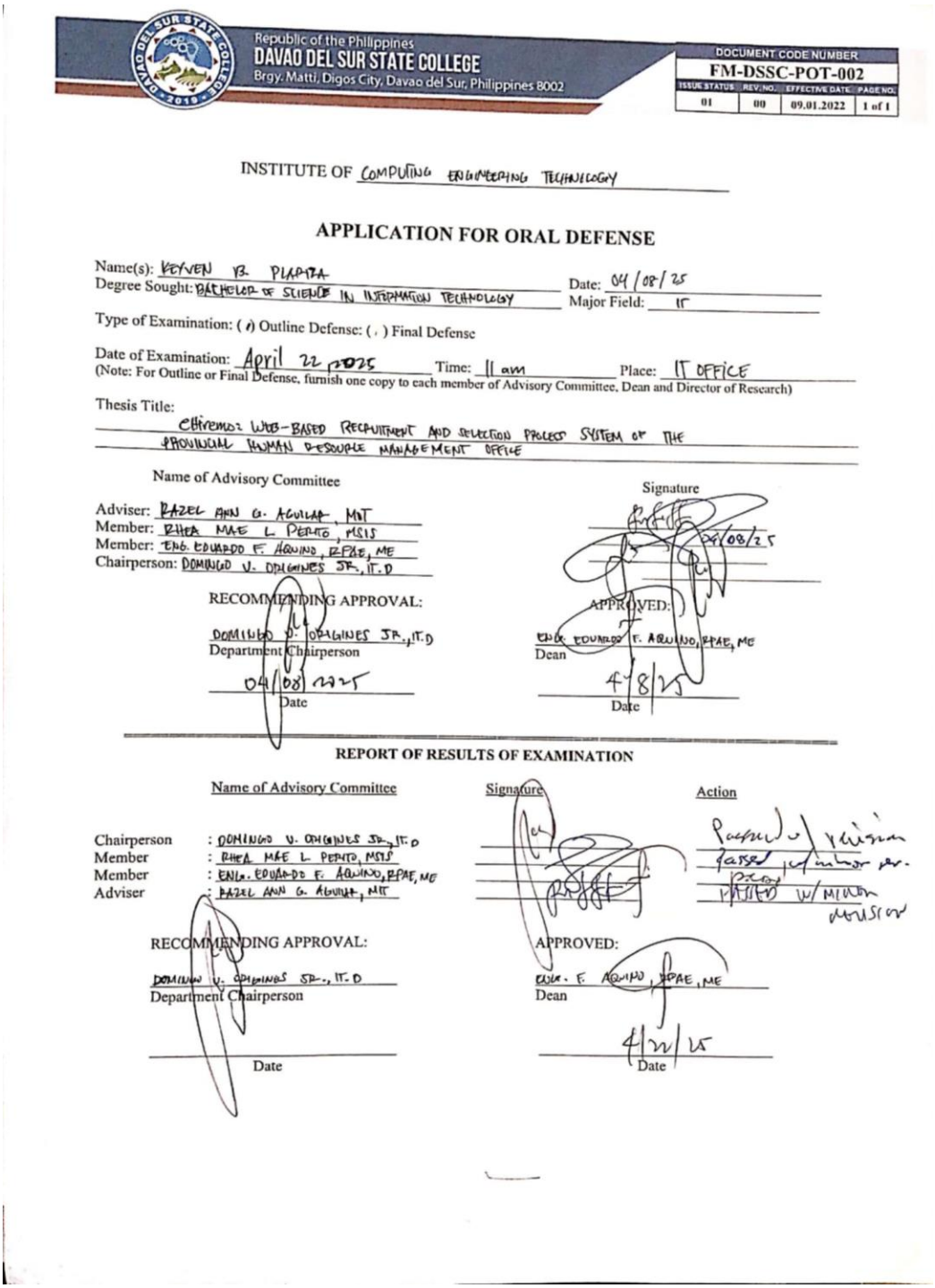
Recommending Approval:

DOMINGO V. ORIGINES JR., IT.D.
Department Chairperson

Approved:

EDUARDO F. AQUINO, MSME
Dean
3/17/25

**Appendix 7.** Application for Oral Outline Defense



Republic of the Philippines
DAVAO DEL SUR STATE COLLEGE
Brgy. Matti, Digos City, Davao del Sur, Philippines 8002

| DOCUMENT CODE NUMBER | | | |
|---|---|---|---|
| FM-DSSC-POT-002 | | | |
| ISSUE STATUS | REV. NO. | EFFECTIVE DATE | PAGE NO. |
| 01 | 00 | 09.01.2022 | 1 of 1 |

INSTITUTE OF COMPUTING ENGINEERING TECHNOLOGY

## APPLICATION FOR ORAL DEFENSE

Name(s): KEYVEN B. PLAZA
Degree Sought: BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
Date: 04/08/25
Major Field: IT

Type of Examination: ( / ) Outline Defense: ( ) Final Defense

Date of Examination: April 22 2025 Time: 11 am Place: IT OFFICE
(Note: For Outline or Final Defense, furnish one copy to each member of Advisory Committee, Dean and Director of Research)

Thesis Title:
CHIVEMOZ: WEB-BASED RECRUITMENT AND SELECTION PROCESS SYSTEM OF THE PROVINCIAL HUMAN RESOURCE MANAGEMENT OFFICE

| Name of Advisory Committee | Signature |
|---|---|
| Adviser: HAZEL ANN G. AGUILAR, MIT | |
| Member: RHEA MAE L. PERITO, MSIS | |
| Member: ENG. EDUARDO F. AQUINO, RPAE, ME | 04/08/25 |
| Chairperson: DOMINGO V. OBLINES JR., IT.D | |

RECOMMENDING APPROVAL:

DOMINGO V. OBLINES JR., IT.D
Department Chairperson

04/08/2025
Date

APPROVED:

ENG. EDUARDO F. AQUINO, RPAE, ME
Dean

4/8/25
Date

### REPORT OF RESULTS OF EXAMINATION

| Name of Advisory Committee | Signature | Action |
|---|---|---|
| Chairperson : DOMINGO V. OBLINES JR., IT.D | | Passed with minor revision |
| Member : RHEA MAE L. PERITO, MSIS | | Passed with minor revision |
| Member : ENG. EDUARDO F. AQUINO, RPAE, ME | | PASSED W/ MINOR REVISION |
| Adviser : HAZEL ANN G. AGUILAR, MIT | | |

RECOMMENDING APPROVAL:

DOMINGO V. OBLINES JR., IT.D
Department Chairperson

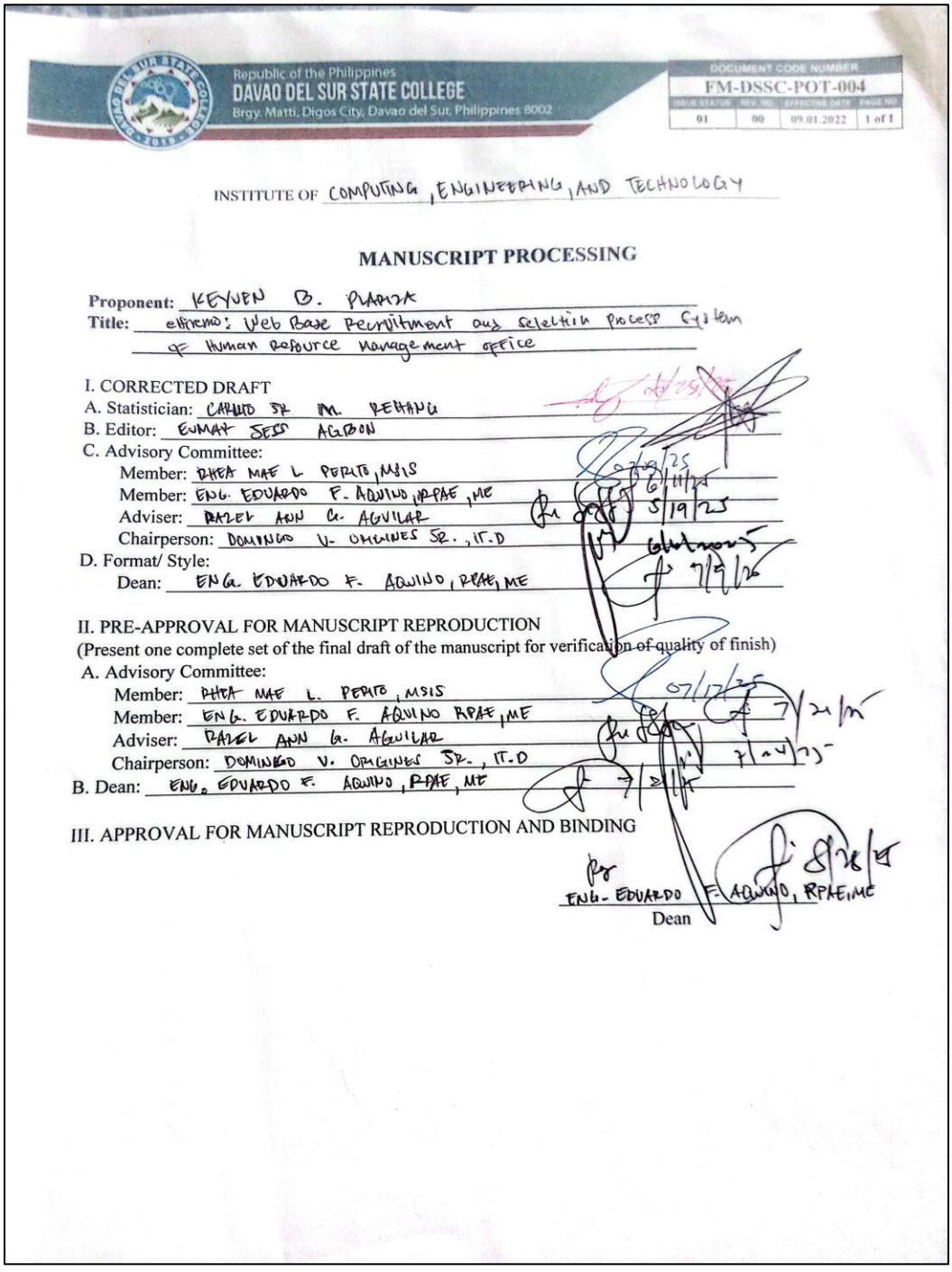
Date

APPROVED:

ENG. F. AQUINO, RPAE, ME
Dean

4/22/25
Date

**Appendix 8.** Capstone Outline Processing Form



Republic of the Philippines
DAVAO DEL SUR STATE COLLEGE
Brgy. Matti, Digos City, Davao del Sur, Philippines 8002

| DOCUMENT CODE NUMBER | | | |
|---|---|---|---|
| FM-DSSC-POT-004 | | | |
| ISSUE STATUS | REV. NO. | EFFECTIVE DATE | PAGE NO. |
| 01 | 00 | 09.01.2022 | 1 of 1 |

INSTITUTE OF COMPUTING, ENGINEERING, AND TECHNOLOGY

**MANUSCRIPT PROCESSING**

**Proponent:** KEYDEN B. PLANZA
**Title:** efficient Web Base Recruitment and Selection Process System of Human Resource Management Office

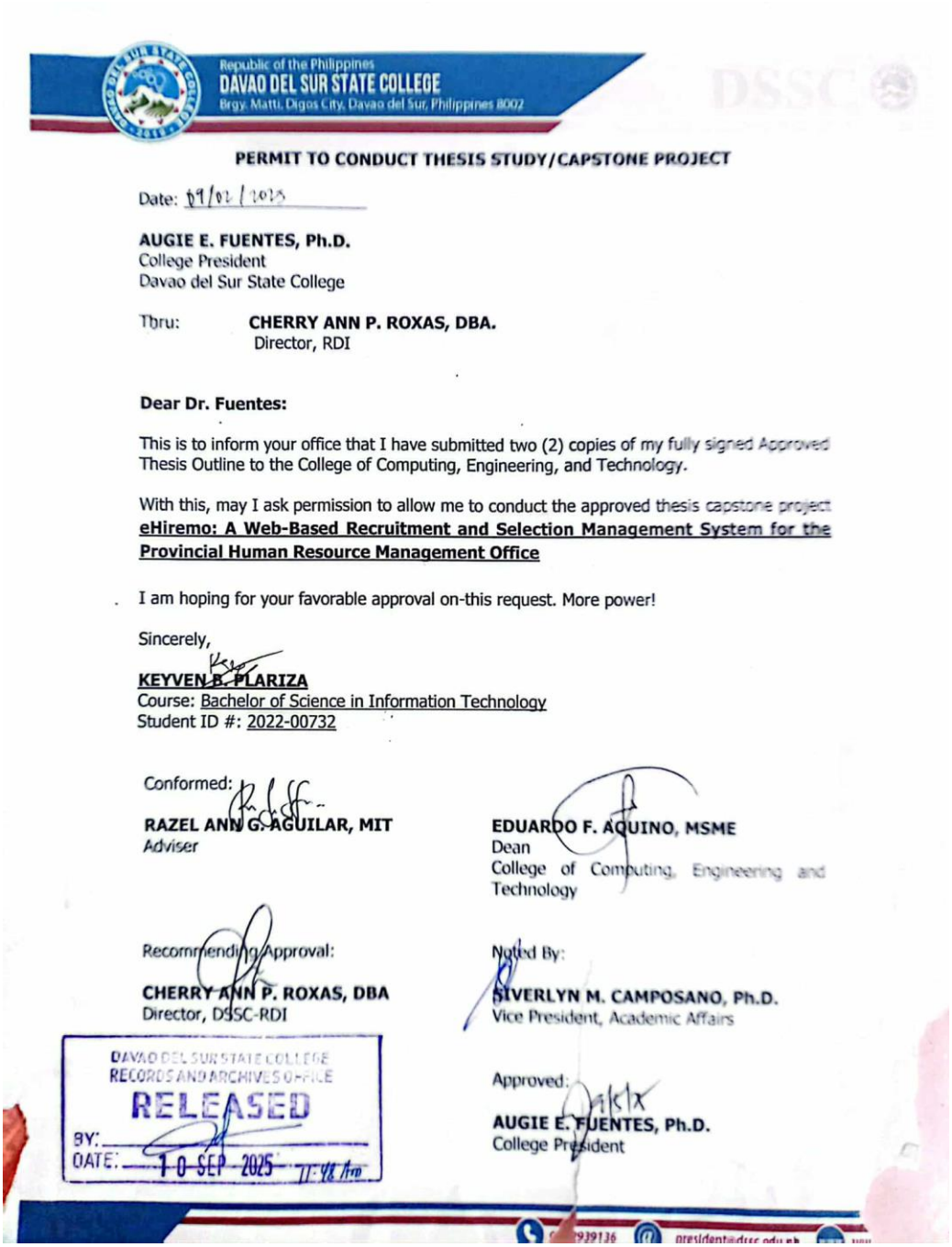
I. CORRECTED DRAFT
A. Statistician: CARLITO JR M. PEÑANO
B. Editor: EUMAR JESS AGUBON
C. Advisory Committee:
Member: RHEA MAE L PERETO, MSIS
Member: ENG. EDUARDO F. AQUINO, RPAE, ME
Adviser: RAZEL ANN G. AGUILAR
Chairperson: DOMINGO V. OMGINES SR., IT-D
D. Format/ Style:
Dean: ENG. EDUARDO F. AQUINO, RPAE, ME

II. PRE-APPROVAL FOR MANUSCRIPT REPRODUCTION
(Present one complete set of the final draft of the manuscript for verification of quality of finish)
A. Advisory Committee:
Member: RHEA MAE L. PERETO, MSIS
Member: ENG. EDUARDO F. AQUINO RPAE, ME
Adviser: RAZEL ANN G. AGUILAR
Chairperson: DOMINGO V. OMGINES SR., IT-D
B. Dean: ENG. EDUARDO F. AQUINO, RPAE, ME

III. APPROVAL FOR MANUSCRIPT REPRODUCTION AND BINDING

By
ENG. EDUARDO F. AQUINO, RPAE, ME
Dean

**Appendix 9.** Permit to Conduct



Republic of the Philippines
DAVAO DEL SUR STATE COLLEGE
Brgy. Matti, Digos City, Davao del Sur, Philippines 8002

**PERMIT TO CONDUCT THESIS STUDY/CAPSTONE PROJECT**

Date: 09/02/2025

**AUGIE E. FUENTES, Ph.D.**
College President
Davao del Sur State College

Thru: **CHERRY ANN P. ROXAS, DBA.**
Director, RDI

**Dear Dr. Fuentes:**

This is to inform your office that I have submitted two (2) copies of my fully signed Approved Thesis Outline to the College of Computing, Engineering, and Technology.

With this, may I ask permission to allow me to conduct the approved thesis capstone project **eHiremo: A Web-Based Recruitment and Selection Management System for the Provincial Human Resource Management Office**

I am hoping for your favorable approval on-this request. More power!

Sincerely,

**KEYVEN B. PLARIZA**
Course: Bachelor of Science in Information Technology
Student ID #: 2022-00732

Conformed:

**RAZEL ANN G. AGUILAR, MIT**
Adviser

**EDUARDO F. AQUINO, MSME**
Dean
College of Computing, Engineering and Technology

Recommending Approval:

**CHERRY ANN P. ROXAS, DBA**
Director, DSSC-RDI

Noted By:

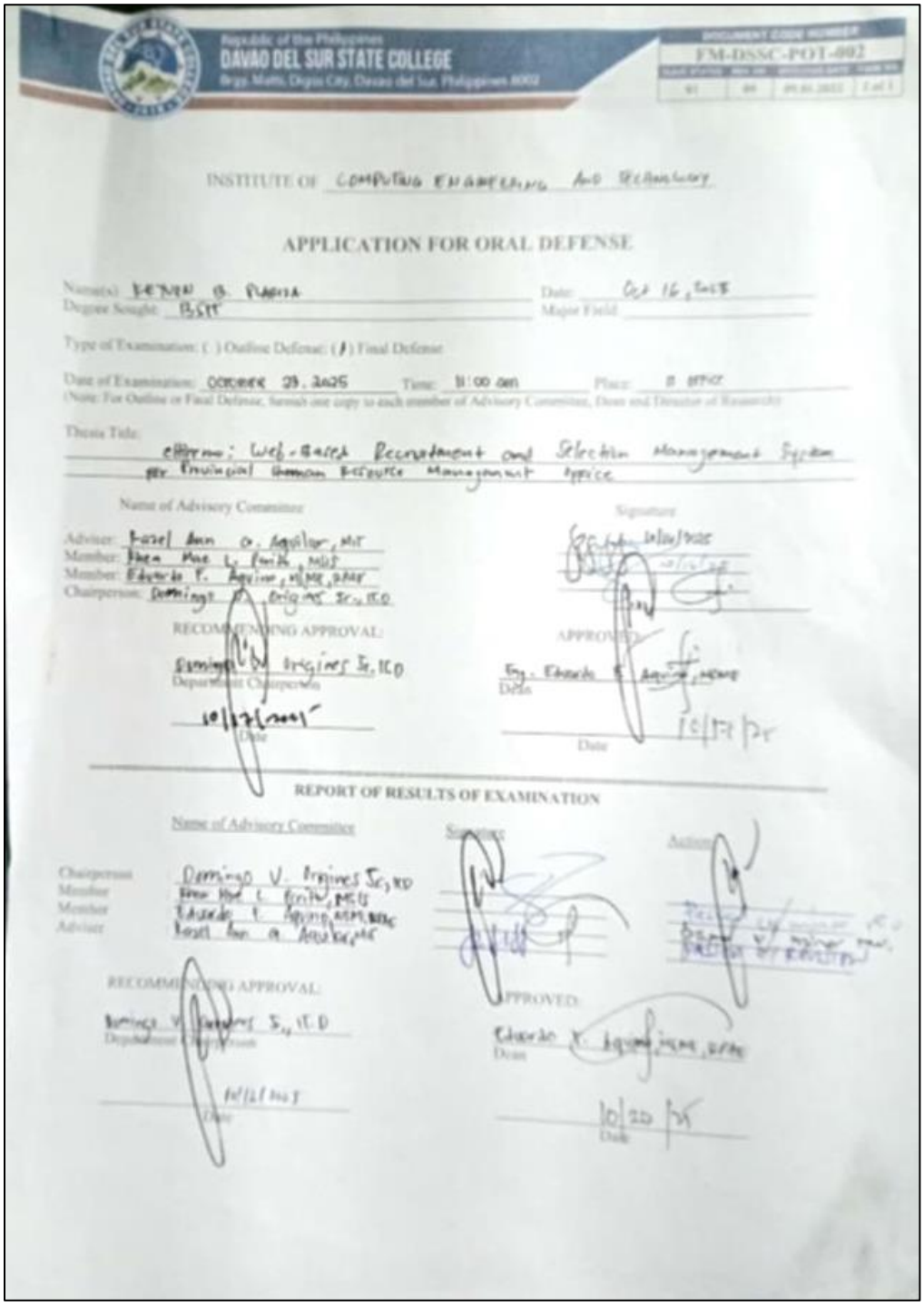
**SILVERLYN M. CAMPOSANO, Ph.D.**
Vice President, Academic Affairs

DAVAO DEL SUR STATE COLLEGE
RECORDS AND ARCHIVES OFFICE
RELEASED
BY:
DATE: 10 SEP 2025 11:48 AM

Approved:

**AUGIE E. FUENTES, Ph.D.**
College President

**Appendix 10.** Application for Oral Defense



Republic of the Philippines
DAVAO DEL SUR STATE COLLEGE
Brgy. Matti, Digos City, Davao del Sur, Philippines 8002

DOCUMENT CODE NUMBER
FM-DSSC-POT-002

INSTITUTE OF COMPUTING ENGINEERING AND TECHNOLOGY

APPLICATION FOR ORAL DEFENSE

Name(s): KENNY B. PLARISA
Degree Sought: BSIT
Date: Oct 16, 2025
Major Field:

Type of Examination: ( ) Outline Defense; (/) Final Defense

Date of Examination: OCTOBER 29, 2025 Time: 11:00 am Place: D OFFICE
(Note: For Outline or Final Defense, furnish one copy to each member of Advisory Committee, Dean and Director of Research)

Thesis Title:
eHRMO: Web-Based Recruitment and Selection Management System
for Provincial Human Resource Management Office

Name of Advisory Committee — Signature

Adviser: Hazel Ann O. Aguilar, MIT
Member: Rhea Mae L. Ponce, MSIS
Member: Edwardo F. Aquino, MEME, DPAP
Chairperson: Domingo V. Briones Jr., II.D

RECOMMENDING APPROVAL:
Domingo V. Briones Jr., II.D
Department Chairperson
10/17/2025
Date

APPROVED:
Eng. Edwardo F. Aquino, MEME
Dean
10/17/25
Date

REPORT OF RESULTS OF EXAMINATION

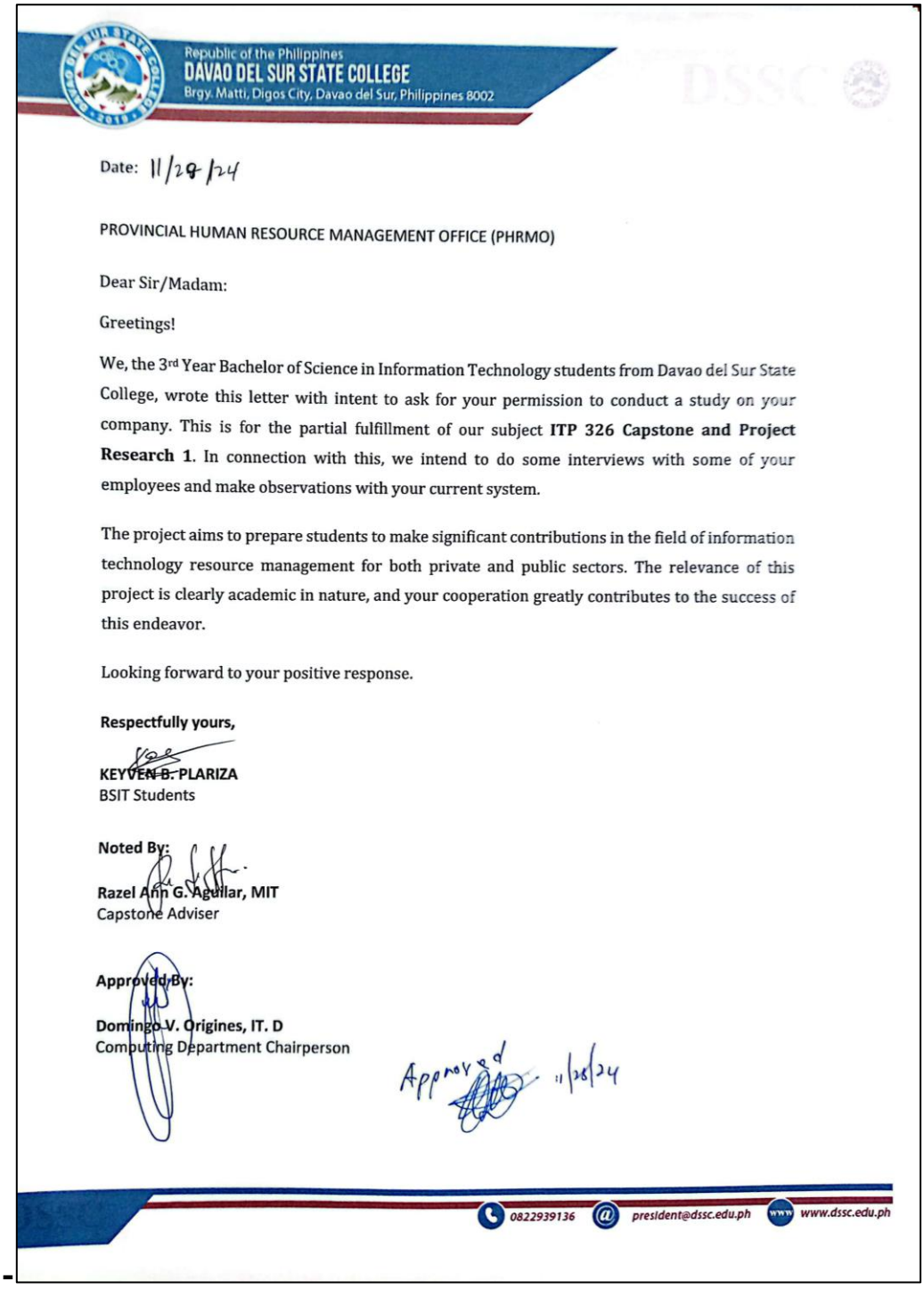
Name of Advisory Committee — Signature — Action

Chairperson: Domingo V. Briones Jr., II.D
Member: Rhea Mae L. Ponce, MSIS
Member: Edwardo F. Aquino, MEME, DPAP
Adviser: Hazel Ann O. Aguilar, MIT

RECOMMENDING APPROVAL:
Domingo V. Briones Jr., II.D
Department Chairperson
10/21/2025
Date

APPROVED:
Edwardo F. Aquino, MEME, DPAP
Dean
10/20/25
Date

**Appendix 11.** Communication letter



Republic of the Philippines
DAVAO DEL SUR STATE COLLEGE
Brgy. Matti, Digos City, Davao del Sur, Philippines 8002

Date: 11/28/24

PROVINCIAL HUMAN RESOURCE MANAGEMENT OFFICE (PHRMO)

Dear Sir/Madam:

Greetings!

We, the 3rd Year Bachelor of Science in Information Technology students from Davao del Sur State College, wrote this letter with intent to ask for your permission to conduct a study on your company. This is for the partial fulfillment of our subject **ITP 326 Capstone and Project Research 1**. In connection with this, we intend to do some interviews with some of your employees and make observations with your current system.

The project aims to prepare students to make significant contributions in the field of information technology resource management for both private and public sectors. The relevance of this project is clearly academic in nature, and your cooperation greatly contributes to the success of this endeavor.

Looking forward to your positive response.

**Respectfully yours,**

**KEYVEN B. PLARIZA**
BSIT Students

**Noted By:**

**Razel Ann G. Aguilar, MIT**
Capstone Adviser

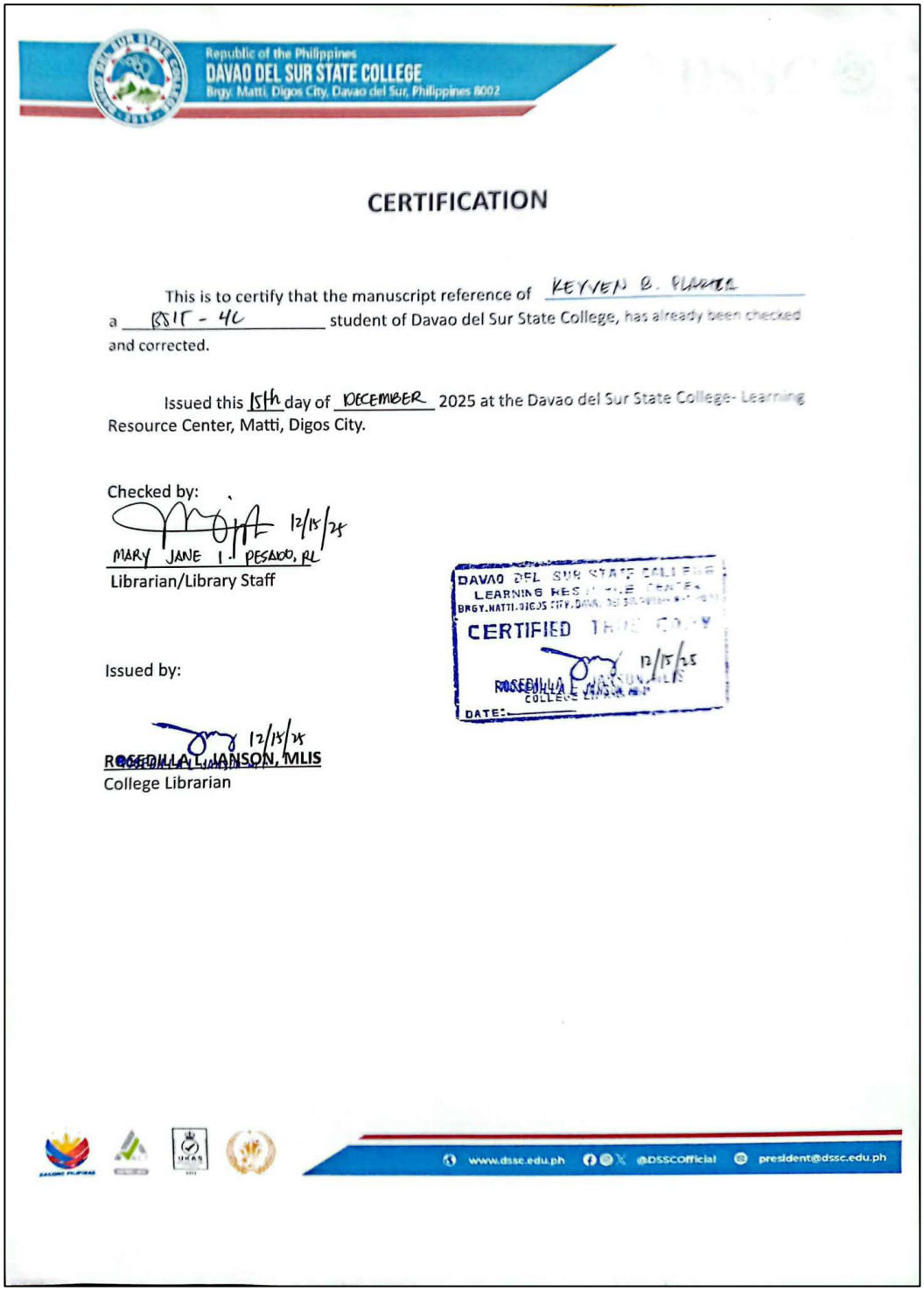
**Approved By:**

**Domingo V. Origines, IT. D**
Computing Department Chairperson

Approved 11/28/24

0822939136 president@dssc.edu.ph www.dssc.edu.ph

**Appendix 12.** Library Reference Certificate



Republic of the Philippines
DAVAO DEL SUR STATE COLLEGE
Brgy. Matti, Digos City, Davao del Sur, Philippines 8002

## CERTIFICATION

This is to certify that the manuscript reference of KEYVEN B. [illegible] a BSIT - 4C student of Davao del Sur State College, has already been checked and corrected.

Issued this 15th day of DECEMBER 2025 at the Davao del Sur State College- Learning Resource Center, Matti, Digos City.

Checked by:

12/15/25

MARY JANE I. PESADO, RL
Librarian/Library Staff

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COLLEGE LIBRARIAN
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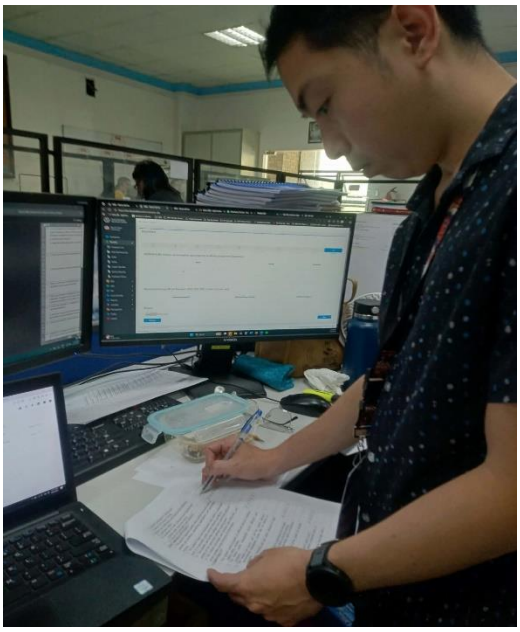
12/15/25

ROSEDILLA L. JANSON, MLIS
College Librarian

www.dssc.edu.ph @DSSCofficial president@dssc.edu.ph

**Appendix 18.** Photo Documentation







## CURRICULUM VITAE



Address: Simon, San Isidro, Magsaysay Davao del Sur
Cellphone No. 09973529511
E-mail: KeyvenPlariza@gmail.com

### EDUCATION BACKGROUND

**Tertiary Education**

| | |
|---|---|
| Field of Education: | Batchelor of Science in Information Technology |
| Name of School: | Davao del Sur State College |
| Address: | Digos Campus, Matti, Digos City, Davao del Sur |
| Year: | 2025-2026 |

**Secondary Education**

| | |
|---|---|
| Name of School: | Barayong National High School |
| Address: | Barayong, Magsaysay, Davao del Sur |
| Year: | 2016-2022 |

**Elementary Education**

| | |
|---|---|
| Name of School: | San Isidro Elementary School |
| Address: | San Isidro, Magsaysay Davao del Sur |
| Year: | 2010-2016 |

### PERSONAL PARTICULARS

| | |
|---|---|
| Full name: | Keyven B. Plariza |
| Date of Birth: | February, 14, 2003 |
| Place of Birth: | Digos City |
| Civil Status: | Single |
| Gender: | Male |
| Religion: | United Church of Christ in the Philippines |
| Father: | Ireneo P. Plariza |
| Mother: | Josephine B. Plariza |